



Annual Health-Sector Performance Report 2022



MINISTRY OF HEALTH

ANNUAL HEALTH-SECTOR PERFORMANCE REPORT 2022

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Foreword

Amid the emerging or re-emerging challenges our society faces, such as climate change, incessant threats to our health security, substance abuse and a whole host of other challenges, in 2022 the Ministry of Health, and the health sector as a whole, endeavoured, not just to maintain the successes of our forebears but also to vigorously push and pull ourselves much further forward.

Bruised but not defeated by the COVID-19 pandemic, as a Nation, we sought, collectively, to upgrade the quantity and quality of services we provide, bringing them closer and closer to where people live, work and play, keeping the targets of the sustainable development goals and other commitments our country has made, nationally and internationally, clearly in focus.

This 2022 Annual Health-Sector Performance Report is an accurate enough reflection of the amount of gruelling work that the more than 2,000 health professionals in the state-run health services, the private sector and the country as a whole have undertaken, individually and collectively, to keep on promoting, protecting and restoring the health of our nation.

The sine qua non for good report writing is complete and accurate data. Our electronic health information system is on the way but long overdue. Its potential to generate a huge difference in the accuracy and timeliness of our annual health performance report cannot be underestimated. Similarly, more health service data from the rapidly sprouting private sector would certainly provide a much more complete picture of the state of health and of health services of our Nation.

Nevertheless, I am satisfied that the document you now hold in your hand serves its purpose of informing all stakeholders about where we are currently, in relation to where we want to be in the future.

Going forward, more success in all areas of health will depend on all of us working, assiduously and methodically, to bolster specific areas of weakness, in spite of the vicissitudes of our time. The government is committed to strengthen any mal-aligned structure and process to usher in the change we all want to see.

Writing this report has again been a painstaking undertaking but a true labour of love. All staff deserve commendation for the strenuous health work they have accomplished for the year 2022 and for the passion they have manifested in writing this awe-inspiring report.

I therefore say thank you to all and sundry.

Peggy Vidot
Minister

Executive Summary

The Ministry of Health (MOH) launched the **National Health-Sector Strategic Plan (NHSP) 2022-2026** in April 2022, articulating four goals for the sector: (1) Increase life expectancy and healthy life expectancy; (2) Achieve and sustain all dimensions of Universal Health Coverage (UHC); (3) Prevent, prepare for, detect and respond to health emergencies; and (4) Promote healthy populations. The NHSP outlines milestones, indicators and targets for the next five years, with responsible lead entities for implementation.

The other important strategic document launched in 2022 was the **Primary Health Care (PHC) Package**, defining the essential health services the sector commits to provide, and sets out five new programmes.

Substance use services were integrated into the MOH as from January 2022, as the Division for Substance Abuse Prevention, Treatment and Rehabilitation (**DSAPTR**). The Minister for Health appeared before the National Assembly to answer 14 questions. MOH launched the **inter-sectoral policy network** in June 2022, to improve collaboration in healthy public policy-making.

The public health sector received significant resources from Government, **SCR 1.3 billion** allocated in 2022 (equivalent to 13% of total Government budget and 4.4% of GDP), allowing the MOH to deliver primary, secondary and specialised health services. The private health sector, where payments are out-of-pocket, continued to expand.

Health care providers are an essential resource for the sector to perform its function. **Health workforce density is high, and comparable to other high-income countries** (SDG 3.8 and 3.c). Seychelles had 241 practising doctors (79% in public sector) and 44 practising dentists (68% in public sector) in 2022. There were 56 medical students but *no* dental students undergoing training as at end-2022.

There were 623 practising nurses on the nurses' roll in 2022 (73% in public sector); however, attrition was relatively high in 2022 (17 retirements, 10 resignations), exceeding new recruits (18). There were 481 allied health professionals on the register; and only eight of 81 registered pharmacists worked in the public sector.

The public health system consists of one tertiary hospital, with four annexes, and a network of 17 primary care centres, organised into six regions. Key infrastructure projects in 2022 were completion of a new NIHSS building, and start of construction of the new La Digue Hospital. Amongst other important health system functions, spending on medicines was SCR 63 million, an increase of 6% from 2021. Implementation of an electronic Health Information System (eHIS) started late 2021, with slow progress in 2022.

An increase in health service utilisation was seen in 2022 compared to 2021, but still below pre-pandemic level. In line with NHSP, the MOH has started work to improve quality-of-care (QoC) through set-up of a **QoC committee** in 2022. Addressing equity in access and health outcomes remain a challenge.

Community health services reported 271,000 doctors' consultations, 145,877 nursing visits and 8,972 home visits. **Specialist clinic visits were 39,068** in 2022, of which 73% were at Yellow Roof and 27% in regional health centres, including Praslin and La Digue.

In 2022, there were **9,868 hospitalisations**; average bed occupancy (acute beds) was 63% (up from 53% in 2021). The highest bed occupancies were in ICU (91%) and Male Medical Ward (90%). **Respiratory conditions were the leading cause of admissions.** Half of admissions to psychiatric ward were related to substance use (cannabis, alcohol, opioid and poly-drug use). Over 5,200 surgical procedures were performed, an increase to above pre-pandemic levels. The Clinical laboratory performed over 811,000 tests, a 65% increase over 2021. MOH saw an increase in units of blood collected in 2022 (2,355 units, a 23% increase over 2021).

DSAPTR had 3,166 active clients on treatment register in 2022, **80% for low-threshold methadone maintenance programme**. Utilisation of services for alcohol abuse were relatively low. Coverage of other harm reduction care, including needle-syringe programme and HIV testing, were low.

Traditionally **Seychelles achieves high coverage for key services such as Childhood vaccination (>98% SDG 3.8.1 and 3.b.1), skilled birth attendance (>99%, SDG 3.1.2) and antenatal care (>99%, SDG 3.8.1), with Universal Health Coverage (UHC) index (SDG 3.8) of 0.85, almost unchanged from 2021.** Child developmental assessment and school health coverage increased in 2022. However coverage of **screening for cancers remained low**, and new programmes outlined in the PHC package are pending implementation (youth health, men's health, healthy ageing, palliative care and home-based care programmes).

The patterns of births remained steady, with **1,569 births** registered in 2022 (52% M, 48% F). **Deaths** numbered **929** (58% M, 42% F) and were above pre-pandemic levels, with 167 **(+21.8%) excess deaths** in 2022 compared to 2015-2019 average. **No maternal death** was registered in 2022 (SDG 3.1.1), and infant mortality was at 15.3 per 1,000 live births (NHSP target <10). **Total fertility rate was 2.01 births per woman** aged 15-48y. **Adolescent fertility rate was 46 per 1,000 girls aged 15-19y** (SDG 3.7.2, NHSP target <30).

Life expectancy (LE) was at 74.2y (69.8y M, 77.8y F). After a reduction of LE in 2021, this is a partial recovery for women (NHSP target >79y for both sexes, >74y M, >83y F); but male LE deteriorated further in 2022. While MOH reports on LE, it is an indicator of societal development influenced by all sectors.

Cardiovascular diseases (CVD) attributed to 280 deaths (30% of total), followed by cancer (159, 17%), respiratory diseases (117, 13%), COVID-19 (97, 10%), and accidents and injuries (71, 8%).

The trends and analyses of the mortality and available morbidity data have demonstrated the **continued increase in non-communicable disease (NCDs)** burden (especially cancer deaths amongst men). We note **re-emergence of infectious and communicable diseases** (e.g. Syphilis); and an **increase in deaths from external causes**: 13 from suicide (SDG 3.4.2), six homicides (SDG 16.1), 18 from RTAs (SDG 3.6.1), nine from falls (typically occupational) and 16 from drowning.

Cancer Registry reported **290 (167M, 123F) new cancer cases**. For men, the most common new cancers diagnosed were prostate (57), bowel (15), and head and neck (22). For women, breast (39), bowel (16), uterus (13) and cervix (11).

CDCU reported **72 new cases of HIV** (64% M, 36% F), including one case of mother-to-child-transmission of HIV; and 27 deaths occurred amongst PLHIV in 2022, well-above pre-pandemic average. Testing challenges remain for diagnosis of sexually transmitted infections.

According to the Global AIDS Monitoring report, 988 people were known to be living with HIV in Seychelles (630 M 15y+, 327 F 15y+ and 11 aged <15y), of whom 893 were on treatment (90%), and 709 (79%) virally suppressed. This implies **Seychelles achieved one of the UNAIDS 90-90-90 targets in 2022.**

The school health screening showed **14.5% of boys and 16.3% of girls were obese, and 15.2% and 18.7% overweight** in 2022. Amongst **S4 students, self-reported use of alcohol (62% M, 67% F), tobacco (17.7% M, 13.6% F) and substances of abuse (9.1% M, 5.8% F)** in past month were high.

The country continued to face major **health security** threats in 2022. Seychelles' IHR core capacities index was 52 in 2022; an improvement from 2021, but significantly below global averages.

The PHA registered **25,069 new COVID-19 infections** in 2022; the primary variant in circulation was Omicron. The MOH administered over **66,000 COVID-19 vaccine doses**, including 1,600 for adolescents aged 12-17y. A **surge in Syphilis infections** were reported in 2022, 71 cases (63% M, 37% F), predominantly in the 15-39y age group. CDCU and Maternity services reported one case of congenital Syphilis.

Considerable resources are invested in health and the MOH has a large number of outputs, with service utilisation reaching pre-pandemic levels. **MOH achieves good service coverage** (see Summary Indicator Table next). While MOH has key efforts to make, to achieve improved results, we **need improvements across health system functions**, and the country also needs to work on the **social determinants of health**.

Summary Indicator Table

The table shows 2022 values of select NHSP indicators, to assess performance towards key goals and strategic directions (SD), with linkages to respective SDG goals.

SD	SDG	Indicator	Baseline (2020)	Value 2022	NHSP End-Term Target	Rating
1/5		Govt. Health expenditure as % of GDP	4%	4.4%	>6%	■
1/5		Allocation to Health as % of Total Budget	11.75%	13%	>15%	■
4		Life Expectancy at Birth (Male)§	72.7	69.8	74	■
4		Life Expectancy at Birth (Female)§	82	77.8	83	■
4		Life Expectancy at Birth (Both Sexes)§	77.3	74.2	79	■
2	3.8.1	Universal Health Coverage Index (WHO Estimate)	71	75	>80	■
3	3.8.1	IHR Core Capacities Index	88	52	-	■
2/4	3.1.1	Maternal Mortality Ratio	64.4	0	<70	■
2/4		Infant Mortality Rate	14.2	15.3	≤10	■
3	3.6.1	Mortality from RTAs/100,000 pop.	11.2	15	<10	■
3	3.4.2	Mortality from Suicides/100,000 pop.	6.1	10.8	<5	■
3	16.1	Mortality from Homicide/100,000 pop.	1	5	-	■
4	3.7.2	Adolescent birth rate/1,000 girls 15-19y	60	46	<30	■
2		Family Planning Coverage	47%	43%	>60%	■
2	3.8.1	Antenatal Care Coverage	99%	>99%	>99%	■
2	3.1.2	Skilled birth attendance	99%	>99%	>99%	■
2		Post-natal Care (Home-visit at 6 wks)	<70%	52.9%	>99%	■
2/3	3.b.1	Pentavalent Vaccine Coverage (Dose 3)	97%	96.8%	>95%	■
2/3		HPV Vaccine Coverage (Girls)	99%	82%	>95%	■
2/3		COVID-19 Vaccine Coverage (12y+)	80%	83.3%	>80%	■
5	3.c	Practising doctors/10,000 pop.	26	24.3	-	■
5	3.c	Practising nurses/10,000 pop.	66	62.7	-	■
5		NIHSS Graduates/year	71	71	-	■
5/6		eHIS patient registration coverage estimate (SOPD)	0%	81%	100%	■
5/6		eHIS patient registration coverage estimate (PHCs)	0%	21%	100%	■
2		Clinic Doctors' Consultations	262,430	271,437	-	■
2		Emergency Visits (Seychelles Hospital)	41,456	41,153	-	■
2		SOPD Specialist Consultations	35,601	39,068	-	■
2		Admissions to Seychelles Hospital	9,749	9,869	-	■
2/5		Prescriptions filled	425,208	440,484	-	■
2		Surgeries at Seychelles Hospital	4,484	5,211	-	■
2		Haemodialysis (# of patients on chronic dialysis)	193	189	-	■
2		Overseas Treatment (# cases referred)	162	201	-	■
2		Mammograms	773	1,138	>2,500/y	■
2		Clinical Lab Tests	677,873	811,145	-	■
2/3		Cervical Cancer Screening (Pap smears done)	6,096	5,925	>20,143	■
2/3		# Active Clients on Methadone Programme	1,884	2,778	>2,100	■
2/3		# Needles-Syringes distributed (by DSPATR)	37,516	11,038	-	■
3/4		Overweight & Obesity among School Boys	28.6%	29.7%	≤28.6%	■
3/4		Overweight & Obesity among School Girls	28.6%	35%	≤28.6%	■
3	3.3.1	HIV New Cases	84	72‡	<50	■
3	3.3.1	TB Incidence (per 100,000 pop.)	9	16	≤5	■
2/3		Anti-retroviral treatment coverage	82%	90%	>90%	■
2/3	3.8.1	TB treatment coverage	100%	87%	100%	■
3	3.3.4	New cases of Hepatitis B‡	34	13‡	-	■
3		New cases of Hepatitis C‡	91	13‡	<45	■
3		COVID-19 new cases	0.8/day	0.7/day*	<20/day	■
2/3		COVID-19 deaths	-	97	-	■

§ from Population and Vital Statistics Bulletin 2022 (NBS, 2023); Ratings: **Good**; **Average**; **Poor** and **Not rateable**.
 * For Dec 2022; ‡ CDCU-reported only;

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Abbreviations

A&E – Accident and Emergency	MTCT – Mother-To-Child-Transmission of HIV/AIDS
AEFI – Adverse Events Following Immunisation	NAC – National AIDS Council
AIDS – Acquired Immunodeficiency Syndrome	NBS – National Bureau of Statistics
ANC – Ante-Natal Clinic/Care	NCD – Non-Communicable Disease(s)
ARTs – Anti-Retroviral Treatment(s)	NEC/NES – Not Elsewhere Classified/Specified
AU – African Union	NGOs – Non-Governmental Organisations
CKD – Chronic Kidney (Renal) Disease	NIHSS – National Institute for Health and Social Studies
CMO – Chief Medical Officer	NITAG – National Immunisation Technical Advisory Group
CNO – Chief Nursing Officer	NMR – Neonatal Mortality Rate
CRVS – Civil Registration and Vital Statistics	NPHC – National Population and Housing Census
CVA – Cerebrovascular Disease	OECD – Organization for Economic Cooperation and Development
CVD – Cardiovascular Disease	PCR – Polymerase Chain Reaction
DoCS – Department of Civil Status	PE – Pulmonary Embolism
DSAPTR – Division for Substance Abuse Prevention, Treatment and Response.	PHA – Public Health Authority
DSRU – Disease Surveillance and Response Unit	PHC – Primary Health Care
DVT – Deep Vein Thrombosis	PHEOC – Public Health Emergency Operations Committee/Centre
DW – D’Offay (Male Surgical) Ward	PLHIV – Person/people living with HIV
ECCE – Early Childhood Care and Education	PMTCT – Prevention of Mother-to-Child Transmission of HIV/AIDS
eHIS – electronic Health Information System	Pop. – Population
ENDS – Electronic Nicotine Delivery System	PPBB – Programme Performance Based Budgeting
EPI – Expanded Programme on Immunisation	PSA – Prostate Specific Antigen
FMW – Female Medical Ward	PVSB – Population and Vital Statistics Bulletin
GAM – Global AIDS Monitoring (report)	PW – Paediatric Ward
GDP – Gross Domestic Product	PWID – Person/people who inject drugs
GPW – Global Programme of Work	QoC – Quality of Care
HCA – Health Care Agency	RMNCAH – Reproductive, Maternal, Child and Adolescent Health
HCW – Health Care Worker(s)/ Health Worker(s)	RTA – Road Traffic Accident(s)
HIS – Health Information System	SD – Strategic Direction (of NHSP 2022-2026)
HIV – Human Immuno-deficiency Virus	SDG – Sustainable Development Goals
HPC – Health Professionals Council	SDH – Social Determinants of Health.
HPV – Human Papilloma Virus	SEIS – Surveillance and Epidemic Intelligence Section
HR(H) – Human Resources (for Health)	SEY-PEN – Seychelles Package of Essential NCD interventions
HSPA – Health System Performance Assessment	SMDC – Seychelles Medical and Dental Council
HW – Hermitte (Female Surgical) Ward	SNMC – Seychelles Nurses and Midwives Council
IDSR – Integrated Disease Surveillance and Response	SOPD – Specialist Outpatient Department
IECD – Institute for Early Childhood Development	SPHL –Seychelles Public Health Laboratory
IHR – International Health Regulations	U5MR – Under-five Mortality Rate
IMR – Infant Mortality Rate	VTE – Venous Thromboembolism
IPC – Infection Prevention and Control	WASH – Water Access, Sanitation and Hygiene.
KPI – Key Performance Indicator(s)	WHO – World Health Organization
LE – Life Expectancy	
MCH – Maternal-Child Health	
MDROs – Multi-Drug Resistant Organisms	
MI – Myocardial Infarction	
MMW – Male Medical Ward	
MOH – Ministry of Health	

Glossary

Excess mortality: an unusual mortality increase during a specific period, in a given population.

Healthy adjusted life expectancy: Health-adjusted life expectancy (HALE) or healthy life expectancy is the number of years that a person at a given age can expect to live in good health, taking into account mortality and disability.

Health system: comprises all the organizations, institutions and resources that are devoted to producing health actions (WHO, 2000).

Health workforce density: Number of core medical professionals, including physicians, non-physicians, clinicians, registered nurses and midwives per 10,000 populations.

Incidence: The number of cases of disease that have their onset during a prescribed period of time. It is often expressed as a rate. Incidence is a measure of morbidity or other events that occur within a specified period of time. See related prevalence.

Infant mortality rate: The probability that a child born in a specific year or period will die before reaching the age of 1 year, if subjected to the age-specific mortality rates of that period, expressed per 1000 live births.

Isolation: The separation, or the period of communicability, of known infected people in such places and under such condition as to prevent or limit the transmission of the infectious agent.

Life expectancy at birth: The average number of years that a new-born could expect to live if he or she were to pass through life exposed to the sex- and age-specific death rates prevailing at the time of his or her birth, for a specific year, in a given country, territory or geographical area.

Maternal mortality ratio: The annual number of female deaths from any cause related to or aggravated by pregnancy or its management (excluding accidental or incidental causes) during pregnancy and childbirth or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy, expressed per 100 000 live births, for a specified time period.

Perinatal Mortality rate: Combined number of deaths of babies aged less than 7 days and the number of stillbirths in a year per 1,000 total births during the year.

Morbidity: A measure of disease incidence or prevalence in a given population, location or other grouping of interest.

Mortality: A measure of deaths in a given population, location or other grouping of interest.

Neonatal mortality rate: Probability that a child born in a specific year or period will die during the first 28 completed days of life if subjected to the age-specific mortality rates of that period, expressed per 1000 live births.

Premature mortality: Death that occurs before the average age of death in a certain population.

Quarantine: The restriction of the activities of healthy people who have been exposed to a communicable disease, during its period of communicability, to prevent disease transmission during the incubation period should infection occur.

Under- five mortality rate: The probability of a child born in a specific year or period dying before reaching the age of 5 years, if subjected to the age-specific mortality rates of that period, expressed per 1000 live births.

1 Introduction

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1.1 Background

Health is defined as “a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity”¹. The Republic guarantees the constitutional right² to health through state provision of essential health services. The Government’s vision for health³ is “the attainment, by all people in Seychelles, of the highest level of physical, social, mental and spiritual health and living in harmony with nature”.

The health sector consists of all organisations, institutions, resources and people in the country, whose primary purpose is to improve health of the people in Seychelles. This includes Government Ministries, Departments and Agencies (MDAs), and civil society and private sector actors. The Ministry of Health (MOH) acts to lead coordinate and steward the activities of the health sector.

Implementation of the current **National Health-sector Strategic Plan (NHSP) 2022-2026** began in April 2022, with new goals, indicators and targets for progress. The new strategic framework is outlined in Chapter 2 Leadership and Governance.

1.2 What is the Annual Health-Sector Performance Report?

The Annual Health-Sector Performance Report (APR) 2022 serves as a key **sector accountability mechanism** to demonstrate **progress of the health sector towards achieving sector goals** and targets set-out in the NHSP 2022-2026, national health priorities as outlined the National Development Strategy, and global commitments, including Agenda 2063 and the Sustainable Development Agenda (2030).

The report provides an account of vital statistics, and through a set of core indicators, population-level health trends, health service utilisation, the state of the health system, and progress towards the global goals. For more detailed information, refer to relevant entity, unit or programme reports (see 1.2.4 Key Data Sources).

The MOH has consistently published APRs from 2018 (for year 2017) to-date⁴ and they have served as a key source of vital information on health and the public health sector for local and international stakeholders.

1.2.1 Organisation of the Report

The report has the following core content by chapter:

- **Chapter 1 Introduction** – introduces the report, its aim, intended audience and development process.
- **Chapter 2 Leadership and Governance** – outlines sector organisation, stewardship, new policies, legislation, and health information systems.
- **Chapter 3 Health Status** – covers vital statistics, life expectancy, excess mortality, fertility, causes of death, and selected morbidity data.
- **Chapter 4 Health Security** – key outbreaks and functions of preventive public health services.
- **Chapter 5 Health System** – health financing and resource generation (focusing on human resources for health, and infrastructure and supplies), and details of service utilisation.
- **Chapter 6 Progress towards NHSP and SDG3 Goals** – describes current trends SDG and NHSP goals, and outlines key coverage indicators for health programmatic services.
- **Chapter 7 Conclusions** – a synthesis of key conclusions.

¹ World Health Organisation (WHO). Constitution of the World Health Organisation; 1946.

² Constitution of the Republic of Seychelles (1993): Article 29, Right to Health care.

³ National Health Policy, 2015 (MOH, 2015).

⁴ Available in public domain at www.health.gov.sc; can also request by e-mail to the MOH;

Although performance is compared against NHSP 2022-2026 goals and objectives, and global (SDG) targets, we note that target setting for the NHSP was difficult during the COVID-19 pandemic. In addition, with large changes in population data evident from the 2022 census, targets may be revisited in future, possible as part of a mid-term review exercise.

1.2.2 Intended Audience

The primary audience are **leaders in health, health service providers and policy makers**. Other stakeholders include Non-Governmental Organisations (NGOs), legislators and the public.

1.2.3 Development Process

A technical working group, led by the Policy and Planning Unit of the MOH, developed the APR 2022. Groundwork for development of the APR 2022 began in December 2022, with initial meetings with data processing teams from January 2023. Following preparation of a report outline, working groups were set-up as from February 2023, with small teams to lead write-up of each chapter (or part thereof).

The full drafts of each chapter were compiled as from May 2023; shared with reviewers towards early July 2023, and the report finalised mid-July 2023. The primary dissemination meetings were scheduled for end-July 2023 onwards. See Acknowledgement for the lead writers of chapters and supporting teams.

1.2.4 Key Data Sources

The main sources of data are from the Statistics Unit; additional data from Disease Surveillance and Response Unit (DSRU), Seychelles National Cancer Registry (SNCR), programmes and annual reports of many units, were used. All Data Sources are acknowledged under all Figures and Tables, and as footnotes for in-text references.

The key Data Sources were:

- Monthly reports of births from maternity services, and deaths from Department of Civil Status (DoCS).
- Routine compilation of inpatient admissions data (Statistics Unit, MOH).
- Unit/service annual reports (for specific diseases and programmes).
- Notifiable disease incidence (including COVID-19) from Annual report from Surveillance and Epidemic Intelligence Section (SEIS), 2022 (PHA, 2023).
- Cancer incidence and deaths (Cancer Registry, PHA).
- Routine service utilisation (Statistics Unit, MOH).
- HR data: reports of the CMO and CNO; Regulatory Councils (SMDC, SNMC, HPC);
- Finance data: Financial Controllers; NHA; International Cooperation Unit.
- Through direct communications to relevant officials within MOH (e.g. donations, infrastructure).
- World Health Statistics Report 2023 (WHO, 2023).
- Population data from 2021 and 2022 Population Bulletins (NBS, 2022 and 2023, respectively).

The full report of the **National Population and Housing Census (NPHC) 2022 report** was not released as at time of preparation of this report. However, data from the census was reflected in a Population and Vital Statistics Bulletin (PVSB) of the NBS, released in June 2023. As such, only select key indicators were recalculated using updated census information, and certain key points of relevance to health from the dissemination meeting (held by NBS, June 2023) were integrated into relevant sections of the report.

Wherever comparisons of year-on-year trends in rates are made, 2021 population estimates are used. **Once back casting is available, MOH will recalculate historic trends and include in future health reports.**

2 Leadership and Governance

LEAD AUTHOR(S): **SUSAN FOCK-TAVE**; TELMA ROSE; AGNES CHETTY

Leadership and governance are the key to functioning health systems. According to WHO, leadership and governance involves ensuring strategic policy frameworks exist and are combined with effective oversight, coalition-building, regulation, attention to system-design and accountability⁵. Governance is arguably the most important enabling function within the health system as governance actions provide both a foundation and a lever for the resource generation, financing and service delivery functions.

Viewed through the lens of health system performance, the governance function has four sub functions, linked to the WHO definition (Table 1)⁶. Governance is a crosscutting function, with each other function requiring its own governance mechanisms. This chapter looks at the sector's performance in each of these sub-functions during 2022.

Table 1 Governance sub-functions and definitions of governance

Governance sub-functions	Corresponding section of governance definition
Policy and vision	Strategic policy frameworks
Stakeholder voice	Coalition-building
Information and intelligence	Accountability
Legislation and regulation	Effective oversight, regulation

Source: Health system performance assessment: a framework for policy analysis (WHO, 2022).

NHSP 2022-2026 priorities for governance and leadership span multiple SDs:

- Clarify and strengthen MoH governance structures and processes (SD1).
- Improve communication, coordination and collaboration within and beyond MOH (SD1).
- Ensure accountability at all levels of the health sector (SD1).
- Develop needed synergistic health policies (SD1).
- Engage the community and build meaningful partnerships (SD1).
- Implement the RBM strategy (SD1).
- Drive changes required to improve quality in health (SD1, SD2).
- Revitalize the health in all policies (HiAP) approach (SD4) and mobilise high-level commitment for HiAP (SD1, SD4).
- Build one integrated Health Information System (HIS) for MOH (SD6).
- Implement eHIS (SD5, SD6).
- Improve data governance (SD6).
- Invest in health data (SD6).
- Strengthen research capacity and use research to improve practice (SD6).
- Enable data use (SD6).

2.1 Sector Stewardship and Management

The **Minister for Health leads the MOH**, with the Principal Secretary as the Chief Executive. The Ministry is responsible for policy development, planning, human resource development and international cooperation. The Ministry oversees the three statutory entities under its purview - the Health Care Agency, the Public health Authority and the National AIDS Council. **Sector level coordination is through weekly coordination meetings of leaders across the entities, and the extended management meeting** (every two weeks).

⁵ WHO https://www.who.int/health-topics/health-systems-governance#tab=tab_1

⁶ Health system performance assessment: a framework for policy analysis.

The **Health Care Agency (HCA)** was established under the Health Care Agency Act (2013) to:

- Plan, deliver and evaluate health services at primary, secondary and tertiary level.
- Ensure access to essential medicines and medical supplies that comply with established legal requirements.
- Educate patients on their rights and responsibilities relating to health care.
- Facilitate the participation of the community in health planning and evaluation.

The **Public Health Authority (PHA)** is a statutory entity under the Public Health Authority Act 2013. Its primary functions are to:

- Develop and implement legislation to protect health.
- Regulate, monitor and evaluate all health-related services in Seychelles, to ensure good practice in the interest of public health.

The National Aids Council Act 2013 guides the work of the **National Aids Council (NAC)** as the coordinating and monitoring body for HIV/AIDS response at national level, and provides strategic guidance on related matters.

The **National Institute of Health and Social Studies (NIHSS)** is a tertiary training institution within the MOH. The Tertiary Education Act (2011) guides its work and it is becoming more autonomous. The institute provides pre and in-service education and training for health and social work professionals.

The **Division for Substance Abuse Prevention, Treatment and Rehabilitation (DSAPTR)**, formerly Agency for the Prevention of Drug Abuse and Rehabilitation (APDAR) has been fully absorbed into the Ministry of Health (NHSP objective 1.1). The Director General leads the day-to-day operations and strategic oversight lies with the Principal Secretary for Health.

The **Public Health Emergency Operations Committee** continued to lead the national response to the COVID-19 pandemic in Q1-Q2 of 2022; the Integrated Disease Surveillance and Response (IDSR) committee reconvened from Q3 onwards.

The Minister for Health attended the 75th World Health Assembly in Geneva, Switzerland in May 2022.

Key positions in the Health Care Agency remain vacant at the end of 2022. These include the position of Deputy Chief Executive Officer, Director Seychelles Hospital and Principal Medical Officer for Seychelles Hospital. These vacancies have the potential to hinder the work of the Health Care Agency.

2.2 Policy and Vision

This section addresses progress made in providing strategic vision for the health sector.

2.2.1 National Health Strategic Plan 2022-2026

MOH launched its new National Health Strategic Plan (NHSP) for the years 2022 to 2026. This replaces the National Health Strategic Plan 2016-2020, the lifespan of which had been extended into 2021 due to the COVID-19 pandemic. NHSP 2022-2026 is expected to move the health sector beyond the pandemic, building on lessons learnt, increasing resilience and improving preparedness for future health emergencies.

The Policy Unit contributed to the development of the new **National Development Strategy** (NDS, led by Ministry Responsible for National Planning), ensuring coherence across the NHSP and health pillar of the NDS.

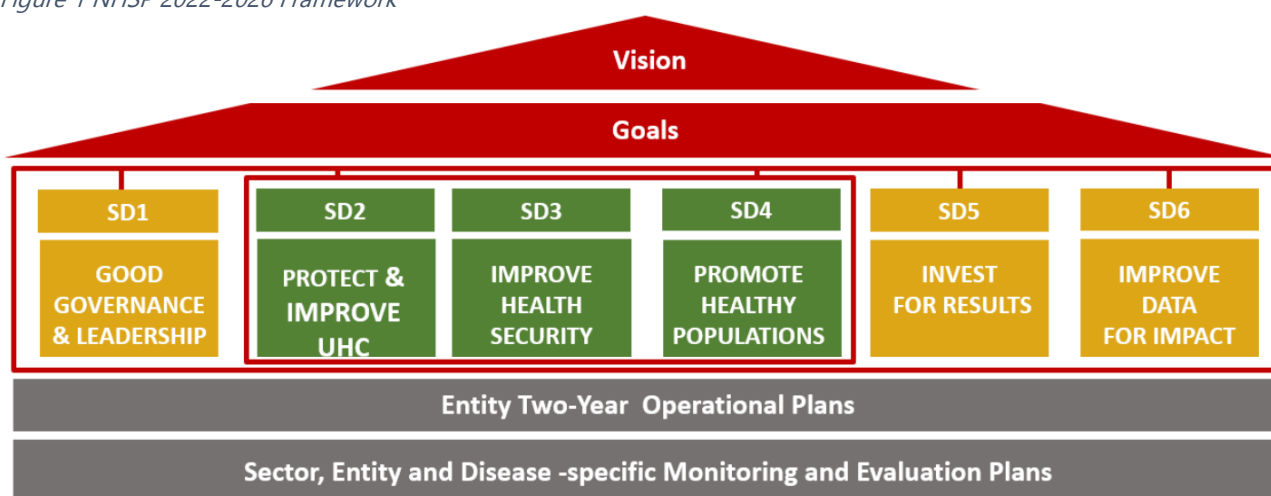
The **vision** of the health sector⁷ is “the attainment, by all people in Seychelles, of the highest level of physical, social, mental and spiritual health, and living in harmony with nature.”

The four **goals** set out in the NHSP 2022-2026 are:

1. Increase life expectancy and healthy life expectancy.
2. Achieve and sustain all dimensions of Universal Health Coverage (UHC).
3. Prevent, prepare for, detect and respond to health emergencies.
4. Promote healthy populations.

The MOH will work through the effective implementation of actions defined across six key **strategic directions (SDs)** to achieve the goals, these SDs are summarised in the Figure 1.

Figure 1 NHSP 2022-2026 Framework



Source National Health Strategic Plan 2022-2026.

2.2.2 Primary Health Care Package

MOH produced the first ever Primary Health Care (PHC) Package for Seychelles (NHSP objective 2.1). Although the Constitution mandates the State to take steps to provide for free primary health care in State institutions for all its citizens⁸, the country had not previously defined primary health care and the package of services to be provided.

The development of a package of services for primary health care was a recommendation of the 2013 Health Task Force Report and the Primary Health Care Conference held in Barbarons in 2019. The development started in 2020, with final cabinet approval in 2022, and implementation starting the same year.

The **PHC Package** clearly outlines the preventive, curative, rehabilitative and palliative care interventions across the life-course and different disease specific programmes (NHSP objective 4.1). The Package does not cut back on existing services, which include oral health and rehabilitation services. Instead, it now includes **packages of interventions under five new programmes – Adolescent and Youth Health, Home-based Care, Healthy Ageing, Men’s Health and Palliative Care.**

2.2.3 Policy Briefs

The Policy Unit in MOH developed policy briefs summarising the contents of three important pieces of legislation – the **Mental Health Care Act 2020, the Civil Status (Amendment) Act 2021 and the Inquest**

⁷ National Health Policy, 2016 (MOH, 2015).

⁸ Constitution of the Republic of Seychelles, Article 29(1)(a)

into Death Act (2021). The briefs outline the reasons and objectives of the Acts, and highlight the key provisions and implications for practice (NSHP objective 1.2).

2.2.4 Stakeholder Voice

The MOH can only achieve its strategic objectives with the collaboration of local and international partners (NHSP objective 1.6). MOH continued to rely on the strong collaboration forged with partner MDAs during the first pandemic year to continue its vaccination campaign. The good working relationship with the Ministry of Education was key to vaccinating children and adolescents against COVID-19, as well as for catching up on missed routine childhood immunisations.

Collaboration with Government Ministries, Departments and Agencies (MDAs) was strengthened by the creation of the **Policy Engagement Network**, which brings together policy analyst from all MDAs (NHSP objective 1.9), to address the social determinants of health, improve coordination for the development of policies in all sectors and build rapport between MOH and other MDAs.

Local partners continued to support pandemic response. Donations ranged from medical and personal protective equipment to other health consumables and IT equipment such as laptops and tablets. Other donations were to show appreciation of health care workers on the ground.

The **World Health Organization (WHO) remains the country's staunchest international partner.** The WHO and MOH signed a new Plan of Action for 2022-2023, estimated to cost USD 958 million⁹. WHO donated an ultra-cold freezer for COVID-19 vaccines as well as laptop computers, tablets and other IT equipment for surveillance and vaccination activities. The country also received a substantial donation of medical equipment from the Japanese Government (see section 5.1.2 Donations).

The MOH has long-standing cooperation agreements with several partner countries, especially for the provision of doctors to the local workforce. Towards the end of 2022, however, the **Cuban Embassy informed the MOH that Cuba would be reducing the size of its Medical Brigade.** There would be no replacements for physicians ending their contracts and no extensions. This decision was based on the difficulties the Embassy was facing in repatriating funds to Cuba. The decision to reduce the number of doctors affects both primary health care and specialized care.

In October 2022, the **MOH signed a cooperation agreement on research in health with the Le Centre Hospitalier Universitaire (CHU) de la Réunion.**

2.3 Information and Intelligence

This sub-function captures all actions that ensure accountability and transparent, data-driven and evidence-based decisions (NHSP objective 6.7).

2.3.1 Sector Accountability

The Minister for Health appeared before the National Assembly to answer 14 questions, including a Private Notice Question, an Urgent Question and a Question without Notice (NHSP objective 1.3). The questions covered a broad range of issues including:

- Availability and access to adolescent sexual reproductive health services for adolescents.
- Progress on sexual and reproductive health related SDGs.
- Strategies to address teenage pregnancy.
- Strategies to increase production and availability of blood and blood products.

⁹ WHO (2022) – Seychelles and the World Health Organisation signs joint two-year work plan to further support the country strengthen its health system and achieve national health goals; Link: <https://www.afro.who.int/countries/seychelles/news/seychelles-and-world-health-organisation-signs-joint-two-year-workplan-further-support-country>.

- Resumption of regular services and reduction of backlogs due to COVID-19 related service cutbacks.
- Further decentralization of haemodialysis services.
- The future of the NAC.
- Plans for revision of the Termination of Pregnancy Act.
- Strategies to increase the number of local pharmacists.

The Health Professionals Council (HPC) and the Seychelles Medical and Dental Council (SMDC) conducted elections to select peer representation on their respective governing boards. The three professional councils also published the list of registered practitioners as mandated by their respective acts. The HPC introduced minimum English language proficiency for new expatriate allied health professionals wishing to register to practice in Seychelles (NHSP objective 2.1.2).

In 2022, the Health Sector joined all MDAs to move forward with implementation of all four pillars of the **Results Based Management (RBM)** Strategy (NHSP objective 1.7). MOH leadership and management completed a series of training to successfully implement RBM across all its entities. The first sector RBM meeting was held in June 2022.

2.3.2 Information and strategic Intelligence

Continuing the practice of recent years, MOH used data collected and analysed from the public health sector to produce the **APR 2021**. This annual publication provides an overview of the breadth and depth of the achievements, strengths and challenges of the Health sector in each calendar year. It is an opportunity to monitor the performance of the sector against set local and global targets. The document is available in the public domain (www.health.gov.sc). MOH continued to create opportunities to disseminate the findings of the APR as widely as possible, through:

- Meeting on Mahé that brought together health leaders and service providers.
- Meeting on Praslin with service leaders and providers on Praslin and La Digue.
- Meeting with NGOs.

Other **routine reports** produced in 2022 include the (NHSP objectives 1.1, 6.3, 6.4 and 6.5):

- Vital Statistics Reports 2021 (annual and quarterly).
- Primary Health Care Services Statistical Report 2021 (annual and quarterly).
- Hospital Services Statistical Report 2021 (annual and quarterly).
- Public Health Statistical Report 2021 (annual and quarterly).
- Health Care Worker COVID-19 Infections Reports (monthly).
- National Immunisation Technical Advisory Group (NITAG) Report 2021.
- COVID-19 vaccination updates (weekly).

Although the MOH has completed five rounds of **National Health Accounts** and has an adequately trained team of professionals, it has not yet reached full institutionalization (NHSP objective 5.2).

Efforts to produce **National Health Workforce Accounts** started in early 2020 with training of a core group of health professionals (NHSP objective 5.1). The training ended a week before the first COVID-19 case was detected and no progress was made during the first two pandemic years.

2.3.3 Health Information Systems

A performance –monitoring and evaluation readiness assessment survey conducted in 2018¹⁰ illustrated the extensive data collected across the health system. The main data processing units are:

¹⁰ Performance Monitoring and Evaluation: Report of Readiness Assessment, MOH, 2018.

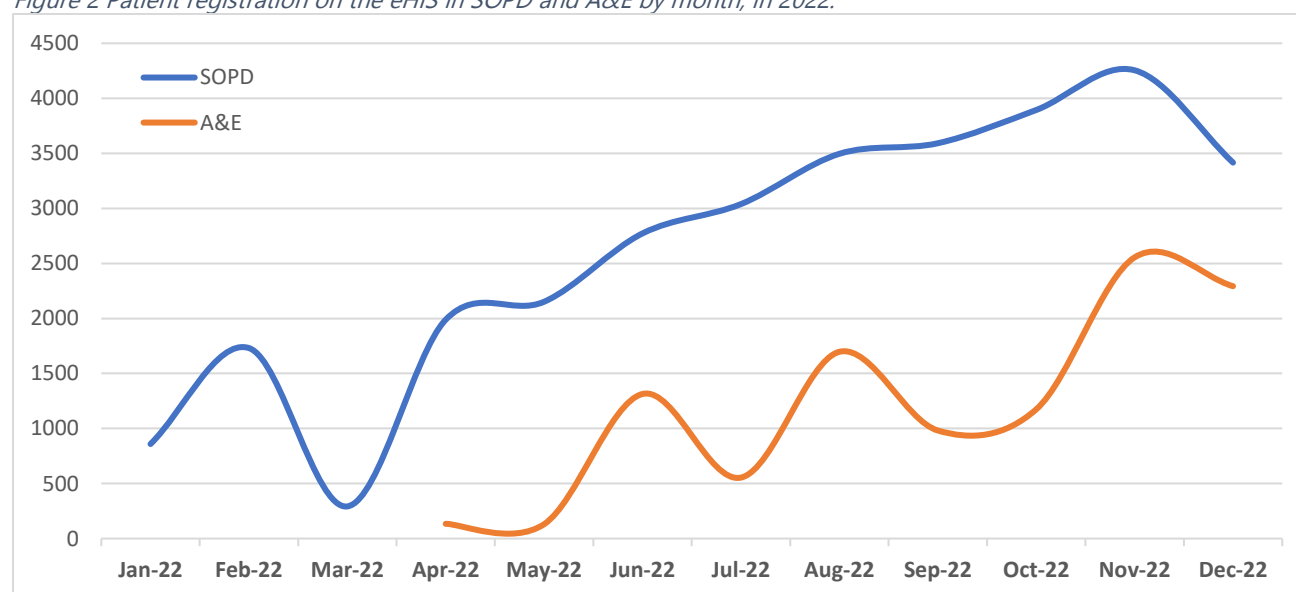
- **Statistics Unit**, which receives and compiles routine service utilisation data from community health services, hospital-based services and public health services. Information products derived from these data include quarterly and annual statistical reports. The Unit also contributes towards other reports as necessary. It is also a major contributor to the annual health sector performance reports. The Statistics Unit staff have had both in-house and external (with University of Oxford) training to build staff capacity, aimed at improving on the quality of information products.
- **Disease Surveillance and Response Unit (DSRU)**: compiles information on notifiable diseases and produces epidemiological reports of varying frequency. The unit has an online disease notification system, which captures data from both the public and private health sectors. DSRU is responsible for data management in disease outbreaks including the COVID-19 pandemic.
- **Cancer Registry** maintains the national cancer register, which contains data on cancer incidence, mortality, and survival. This electronic register was developed by the IARC¹¹, which also provides regular training for the registry staff.

e-HIS: The rollout of the electronic health information system, which began in 2021, made slower progress in 2022 than expected. After introduction of patient registration and appointment modules in outpatient facilities, the electronic medical records was introduced in 2022. The next steps will be to fully integrate other modules, e.g., the laboratory information system, the radiology information system and pharmacy modules to allow for paperless ordering and reporting of investigations as well as online prescribing. Key to the successful implementation will be the ongoing training of staff and effective change management strategies, to ensure all stakeholders are on board with the new system.

Figure 2 shows patient registration on eHIS at two sites where rollout of the registration module commenced since 2021, with gradual enrolment of various health service areas; a process that is still ongoing. Of the 39,068 Specialist Out-Patient Department (**SOPD**) attendances reported in 2022, **81% were registered on the eHIS**, whereas only **26%** of the 41,143 attendance at Accident and Emergency (**A&E**) were captured.

An improvement can be seen from January through to December 2022 in eHIS use in these two sites. Note that patient registration numbers do not imply that electronic medical records were completed for each visit.

Figure 2 Patient registration on the eHIS in SOPD and A&E by month, in 2022.



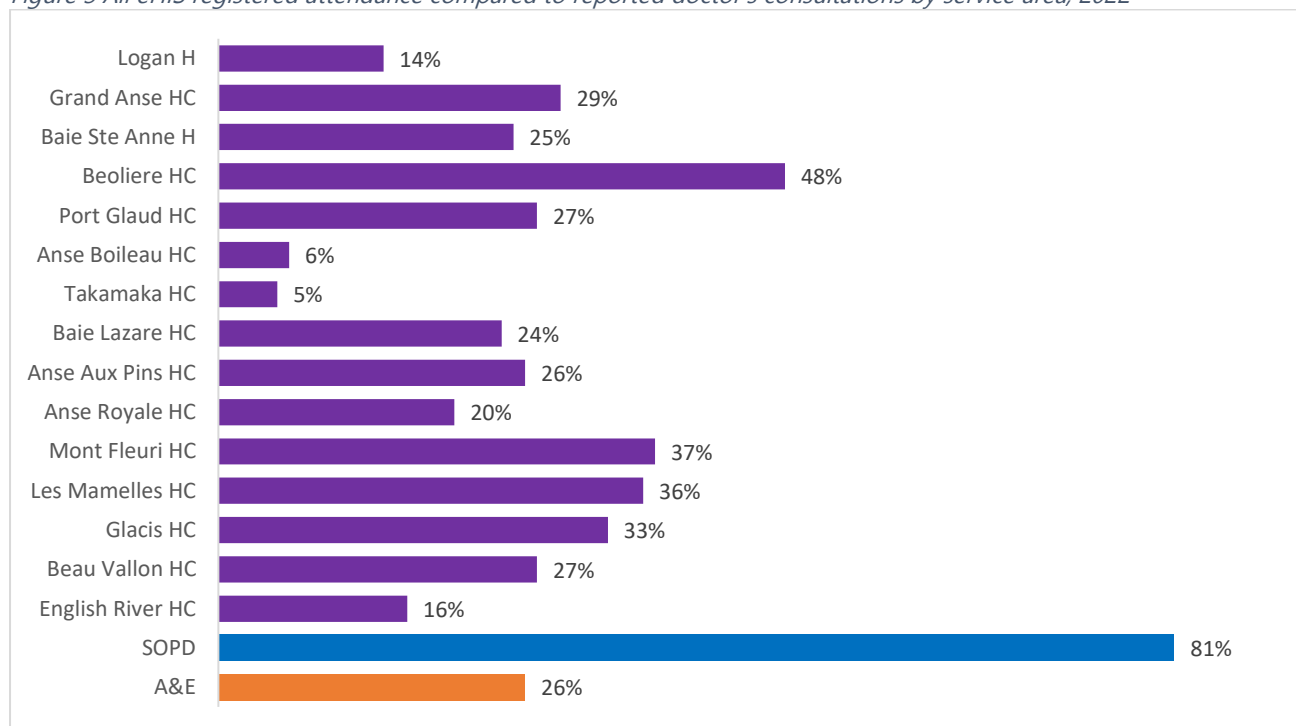
Data Source(s): Lifeline Enterprise Suite (Manorama Information Systems) Hospital Activity Dashboard (accessed 14 July 2023).

¹¹ International Agency for Research on Cancer (IARC) of the WHO; Link: <https://www.iarc.who.int/>.

Use in primary care were lower, 21% on average for all clinics registered on the eHIS (averages by clinic range from 5% to 48%), see Figure 3. Overall average utilisation in community health centres increased from 4% in January 2022 to 44% by December 2022. Rollout for inpatient service areas did not begin in 2022.

Overall, East region showed the highest average utilisation of eHIS for registration, followed by North, Inner Islands, South, Central, and lastly West regions. Many clinics show increasing use from January to December, but for some clinics, use peaks end-Q3 of 2022, then begins to drop in Q4.

Figure 3 All eHIS registered attendance compared to reported doctor's consultations by service area, 2022



H – Hospital; HC – Health Centre; **Note:** eHIS attendance includes all care services registered, beyond just doctors' consultations. Data Source(s): Lifeline Enterprise Suite (Manorama Information Systems) Hospital Activity Dashboard (accessed 14 July 2023); Annual Primary Health Care and Hospital Services Statistical Report 2022 (MOH, 2023).

The **partial use of eHIS implies the continued need for two parallel systems of reporting**: data capture through eHIS and non-eHIS ('traditional') reporting to statistics unit, DSRU and programmes, the latter being more complete. This presents the **risk of neither mechanism capturing the complete health data, and adds to reporting burden** for frontline staff.

Vital Statistics Information Systems: MOH has been strengthening vital statistics information systems since 2018; a key step occurred in 2022 – the transition from use of ICD-10 to ICD-11 coding. This implies classification of certain disease will change. Other improvements include:

- Capacity building for disease coding and certification of deaths amongst doctors and statisticians.
- Regular production and dissemination of vital statistics reports.

This has already led to improvements in quality of death certificates and causes of death datasets.

Research: The Research Unit is housed in the Public Health Authority and is understaffed. The Seychelles Child Development Study (summarised in 2.3.4 Seychelles Child Development Study) and the Seychelles Heart Study are two long-running research projects with key participation of local health actors.

Despite progress in data collection and improvement in the quality of derived information products, many of the challenges identified in previous annual reports remain¹².

The need for a unifying data management and governance system as well as a fully functional electronic health information system remains an imperative to improving data availability, access and use.

2.3.4 Seychelles Child Development Study

The Seychelles Child Development Study (SCDS) is a longitudinal observational study initiated in 1985. The original purpose of the international collaboration was to investigate the impact on child development from exposure to methylmercury from the consumption of fish during pregnancy. Early findings indicated that improved infant and child developmental outcomes were associated with fish consumption.

With the expansion of the study to include the potential impact of nutrition, further investigation demonstrated the beneficial effects of omega 3 fatty acids, from the maternal fish diet, on child development.

More recent work has focused on better understanding the interplay of nutritional factors, environmental exposures and genetics, and the use of laboratory measures such as inflammatory and epigenetic biomarkers that might be more sensitive indicators of effect compared to the traditionally used neurodevelopmental outcomes.

The overall public health message is that **consumption of fish in Seychelles has not been shown to adversely affect children's development**, and fish should remain an important component of the diet because of its nutritional benefits.

In 2022, the SCDS team completed the evaluation of the mothers of the Main Cohort who, together with their infants, were enrolled in 1989 and 1990 and are now aged 50 years and above.

Of the 678 mothers eligible for participation in this latest phase, 609 (89.8%) attended for a three-hour session involving interviews and examination at the Child Development Centre. Information gathered included general health status and fish consumption.

The test battery included standardised evaluation of cognitive function, physical strength and functioning and measures of frailty. Biological samples included hair and blood. The evaluation of this cohort will permit the investigation of lifetime fish consumption on ageing processes.

On completion of the evaluation of the mothers of the main cohort, the team embarked on the evaluation of the Second Nutrition Cohort, consisting of 1,500 children whose mothers were pregnant at enrollment in 2008 to 2011.

A number of Masters and Doctorate candidates at partner Universities participated in the SCDS, and one Seychellois was enrolled as a PhD candidate with the support of the SCDS. Publications in scientific journals during the year covered a variety of topics such as child development, statistical methods, experimental toxicology, genetics and inflammatory markers.

2.4 Legislation and Regulation

As the COVID -19 pandemic continued to evolve and indications for vaccines widened, MOH introduced **new regulations under the Public Health Act** to cater for new these changes. These regulations allowed for vaccination of children (5-11y) and adolescents (12-17y) against COVID-19.

¹² Annual Health-sector Performance Report 2021 (MOH, 2022).

Other legislation with major implications for the health sector were **Civil Status (Medical Certificate of Cause of Death) Regulations**¹³ and the **Civil Status (Notification of Birth and Death Forms) Regulations**¹⁴, which came into force on 1st June 2022. The regulations introduce new procedures for the registration of births and deaths as well as certification of cause of death aligned with international norms.

These regulations will ensure timely and accurate recording and reporting of vital events to the Civil Status. They also aim to improve the quality of cause of death data and information on and vital statistics in general.

Major pieces of legislation remain in various stages of the development process:

- **Amendment to the Tobacco Control Act (2009)** – Cabinet has approved amendment to include Electronic Nicotine Delivery System (ENDS) to be classified as tobacco products, thus extending all provisions applying to regular combustible cigarettes in the Seychelles Tobacco Control Act (2009) to the manufacture, distribution, sale and use of ENDS¹⁵ (NHSP objective 4.2).
- Others include the Medicines Bill, revisions of the Overseas Treatment Act 2018, the Medical Practitioners and Dentists Act 1994 and the Nurses and Midwives Act, 1985. The subsidiary legislation for the regulation of health and health related services is also outstanding.

The **regulatory and registration unit** of the PHA has a critical role to regulate health practices across the country (NHSP objectives 3.4 and 3.5), and oversee relevant regulatory boards of the PHA, including:

- Pesticide Control Board.
- Complementary Healthcare Services Board.
- Food Control Committee.
- National Tobacco Control Board.
- PHA Appeals Board.
- Health Research and Ethics Committee.

The **private health sector** constitutes a significant health service provider, but unlike the public health sector, is focused on medical and diagnostic services. All advanced care are provided only in the public sector. The number of practices have increased rapidly year-on-year (Table 2).

Table 2 Registered private health practices, 2021-2022.

Category of Private Practices	# Registered 2021	# Registered 2022
Private pharmacies	25	36
Optometrists	6	11
Opticians	3	2
Medical practices	20	27
Dental practices	8	8
Stand-alone Laboratories	1	1
Independent Service Providers		
Physiotherapist	1	1
Nurses	4	3
Dental Technicians	1	1
Dental Technologists	1	2

Data Source(s): Regulatory and Registration Unit Annual Report 2022 (PHA, 2023).

¹³ Link: <https://seylil.org/akn/sc/act/si/2022/56/eng@2022-05-05>

¹⁴ Link: <https://seylil.org/akn/sc/act/si/2022/57/eng@2022-05-05>

¹⁵ UPCCD Annual Report, 2022.

Small steps have been made towards improved engagement with private health sector (NHSP objective 2.3.1). As at 2022, the MOH did not receive information on service utilisation from private health service providers; the regulatory and registration unit of the PHA has an important role to play to improve reporting from private health service providers.

2.5 Progress towards Strengthening Leadership and Governance

The governance structure of the MOH is evolving, with the reintegration of substance use services under the DSAPTR, and discussions around the NAC. The health-sector leadership was trained on RBM, with sector-RBM committee set-up, and the first meeting in June 2022. Sector accountability was further supported by consistent production of annual sector performance report (2017-2021), including through the pandemic, and the production of a National Drug Observatory Report for the first time in 2022.

In addition to the launch of the NHSP 2022-2026, the PHC package of services was also rolled-out. New mechanisms were set-up through a policy network, to engage non-health sectors in understanding and working on HiAP: a policy network was established in 2022, which included policy analysts from across all Ministerial portfolios.

The HCA has set-up a QoC committee to lead efforts to improve QoC, including preparation of strategic guidance on quality improvement in the sector.

Recent years has seen **improved collaboration in regulatory functions** between the health professionals' councils and the PHA. There was ongoing work to revise laws governing the operations of the SMDC and SNMC in 2022, to allow councils to better fulfil their mandates.

New laws and processes were introduced for death certification, with training to support implementation. Research remains fragmented in MOH.

Key pending issues include: a **review of the implementation of the 2013 task force recommendations** (SD1); improving internal communications within the health sector (SD1); better engagement and regulation of NGOs and private health sector (SD1); and **the creation of one integrated health information system** for the MOH (SD6).

Reporting burden on nursing staff remained above pre-pandemic levels. Additional reporting mechanisms established during the COVID-19 pandemic response were only partially integrated into routine data management and reporting mechanisms.

The MOH undertook **inter-sectoral, national and global reporting**, including on progress towards SDGs, AU Agenda 2063, UNAIDS and IHR (eSPAR).

3 Health Status

WRITER(S): **SANJEEV PUGAZHENDHI**; **WILLIAMS ITUEN-UMANAH**;

The state of health of a population is a comprehensive reflection of its overall well-being, development, and the effectiveness of the healthcare systems. The **Social Determinants of Health (SDHs) and the quality of, and access to, health services, and effective public health, all contribute towards the health of a population**. This chapter aims to delve into core indicators used to measure health status of a population, and is organised as follows:

- Patterns of births and deaths (3.1.2 Births and 3.1.3 Deaths).
- Maternal and child mortality (3.1.4 Maternal and Child Deaths).
- Life expectancy (3.1.5 Life Expectancy).
- Key causes of death (3.2 Causes of Death).
- Excess mortality (3.1.6 Excess Mortality).
- Key patterns of morbidity (3.3 Morbidity).
- Fertility, including adolescent fertility (3.1.7 Fertility).
- Risk factors for Health (3.4 Risk Factors for Health).

It is important, when interpreting the significance of trends and changes in various rates, to take into account the small population of Seychelles, and the resulting large year-on-year variations that may occur in health data.

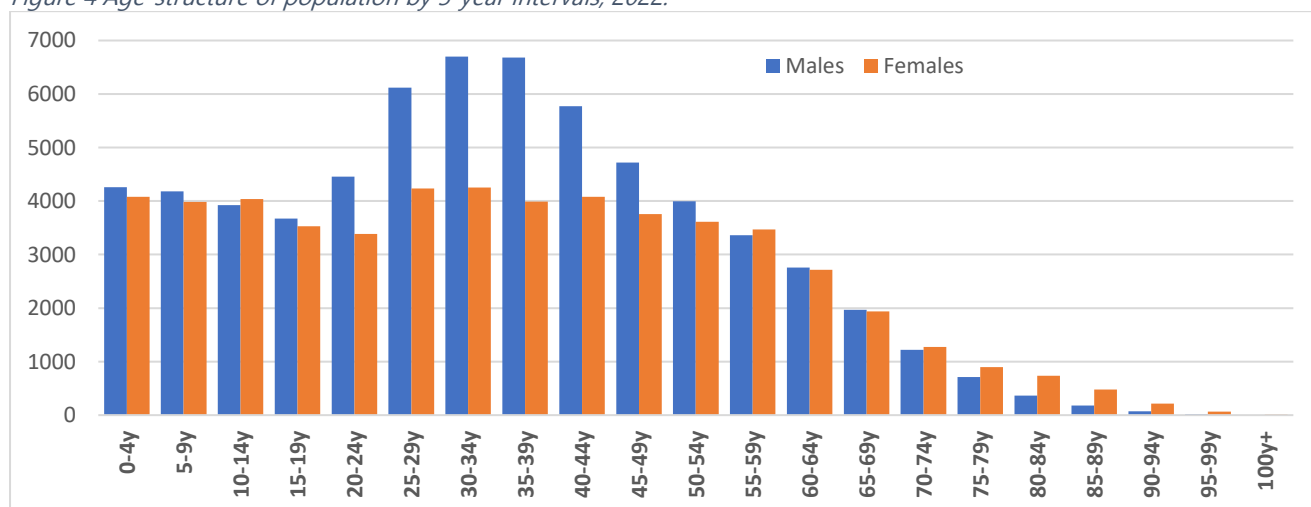
3.1 Vital Statistics

3.1.1 National Population and Housing Census 2022

The 2022 Population Bulletin (PVSB of NBS, June 2023), reported a 2022 mid-year estimated resident population was **119,878 (65,135 or 54% M, 54,743 or 46% F)**. **Seychellois constitute 84%** of resident population, and **non-Seychellois 16%**. This represented an increase of 20,620 (+21%) over 2021 population estimates, and a male-to-female ratio of 1.18:1.

Age structure is shown in Figure 4: 0-14y constitute 26.4% of the population, whereas 0-17y (children), 30.4%, 15-34y (youth), 30.3%, 35-64y, 40.8%, and 8.5% aged 65y+. The age-dependency ratio was 406; there were **significantly more males than females in the age-range 20 to 49y**, likely due to the fact that majority of expatriate workers are young men.

Figure 4 Age-structure of population by 5-year intervals, 2022.



Data Source(s): Population and Vital Statistics Bulletin (PVSB) 2022 (NBS, 2023).

The **increase in population has implications on all rates based on total or age-specific populations**, including total and adolescent fertility rates, birth and death rates, age- and cause-specific mortality rates, and life expectancy. It also affects coverage rate for vaccines, health spending per capita, health worker density per capita and hospital beds per capita.

Not all rates and coverage in this report were re-calculated using the new census estimates. **Where new census data was used, it is clearly stated as footnotes or as notes under respective figures and tables.**

3.1.2 Births

The **registered births numbered 1,569 in 2022 (52% M, 48% F)**, a slight reduction over 2021. The Crude Birth Rate (CBR) is 15.8 per 1,000 population (Table 3). Data from Maternity services show 1,531 births (1,522 live and 9 stillbirths¹⁶) in 2022, of which 93% were delivered at term¹⁷. Certain key characteristics of deliveries are summarised in Table 4.

Table 3 Registered live births, 2018-2022

Births (Registered at Civil Status)	2018	2019	2020	2021	2022
Total Live Births	1,650	1,605	1,554	1,665	1,569
Male	832	813	765	855	811
Female	818	792	789	810	758
Crude Birth Rate	17.1	16.4	15.8	16.8	15.8† (13.1*)

* Based NPHC 2022 (NBS, 2023, unpublished); † Based on 2021 Population Estimates; Data Source(s): DoCS and NBS.

Table 4 Characteristics of deliveries and new-borns, 2022

Parameters¹⁸	2022	Parameters	2022
Total deliveries in 2022	1,512	Substance use amongst mothers delivering	46 (3%)
Attended by skilled health personnel	1,501 (99.3%)	Total births recorded in service	1,531
Spontaneous Vaginal Deliveries	1,067 (70.6%)	Live births	1,522
Assisted Breech deliveries	9 (0.6%)	At ≥37wks ('term')	1,423 (93%)
Instrumental Deliveries	15 (1%)	<37wks (pre-term)	99 (6.5%)
Deliveries by Caesarean section	386 (25.5%)	Stillbirths	9
Births before arrival to health facility	11 (0.7%)	Characteristics of new-born	
Twin deliveries	18 (1.2%)	Macrosomia (>4kg birth weight)	68 (4.4%)
Teenagers delivering	157 (10.4%)	Low-birth weight (<2.5kg)	137 (9%)
Deliveries in mothers aged 35y+	222 (14.7%)	Congenital malformations	54 (3.5%)
Deliveries where mother was obese ¹⁹ at booking visit	197 (13%)	Reported cases of birth trauma to new-born	2
Caesarean rate amongst obese mothers	38.8%	Congenital Syphilis (identified at birth)	1
Tested positive for COVID at delivery	90 (6%)	Neonatal Abstinence Syndrome ²⁰	9

Data Source(s): Family Health Programmes Annual Report, 2022 (HCA, 2023) and Maternity Services (HCA, 2023).

3.1.3 Deaths

Civil status reported **929 deaths** (538, 58% M, 391, 42% F) in 2022, giving a Crude Death Rate of 9.4 per 1,000 population (Table 5); this was similar to 2021. The number of deaths remained above pre-pandemic levels since 2021 (see section 3.1.6 Excess Mortality).

¹⁶ Only 5 stillbirths were registered at Civil Status.

¹⁷ Numbers may differ from registered births (due to time lag with birth registration at Civil Status).

¹⁸ Numbers from Maternity services; may differ from NBS data (which is based on births registered in 2022).

¹⁹ Body Mass Index (BMI) 30 and above in adults.

²⁰ Neonatal abstinence syndrome is a condition that affects new-borns who receive opioids or addictive substances through the placenta during pregnancy. Babies have withdrawal symptoms during their first few days of life. Treatment is available to help the child overcome addiction (Cleveland Clinic, 2022: What is Neonatal Abstinence Syndrome? Link: <https://my.clevelandclinic.org/health/diseases/23226-neonatal-abstinence-syndrome>).

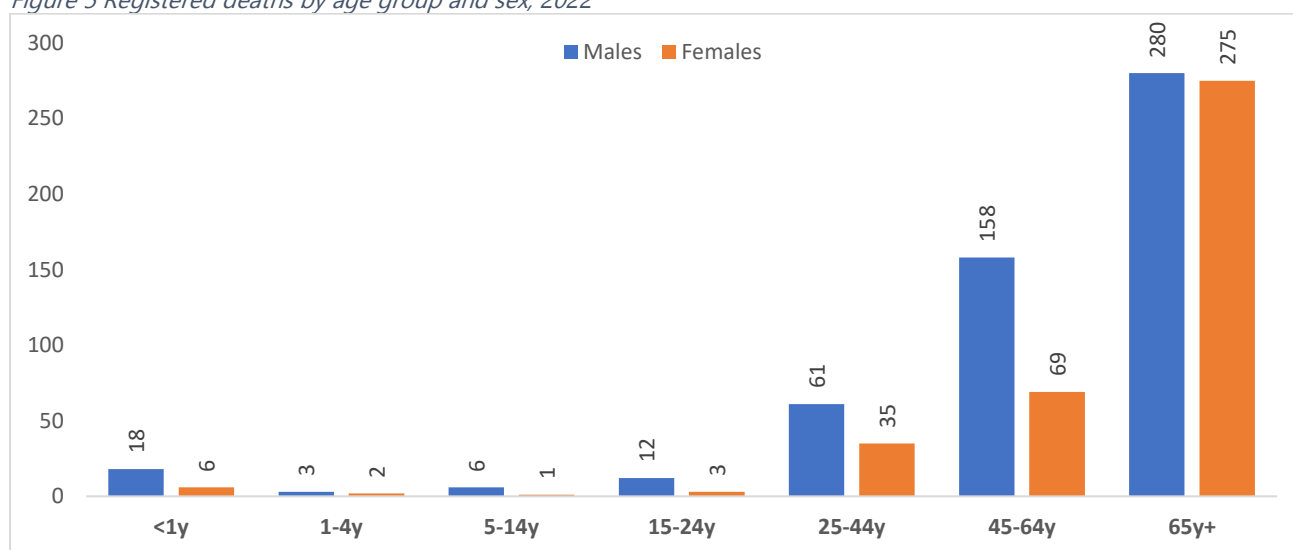
Table 5 Registered deaths, 2018-2022

Deaths	2018	2019	2020	2021	2022
Total	818	795	668	925	929
Male	470	438	397	480	538
Female	348	357	271	445	391
Crude Death Rate	8.5	8.1	6.8	9.3	9.4 [†] (7.7)*

[†] Based on 2021 population estimates; * Based on PVSb 2022 (NBS, 2023). Data Source(s): DoCS; NBS.

Sixty percent of all deaths in 2022 occurred in the 65y+ age group (Figure 5), there were more deaths amongst men, and men die disproportionately younger than their female counterparts do (also see section 3.2 Causes of Death).

Figure 5 Registered deaths by age group and sex, 2022



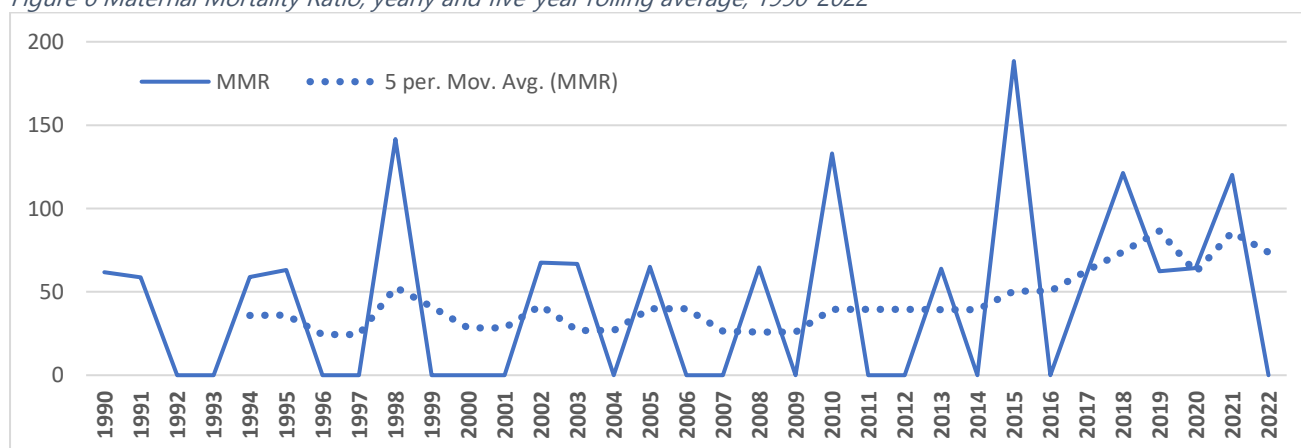
Source: Vital Statistics Report 2022, Statistics Unit, MOH.

Amongst deaths in 2022, 888 (95.6%) were amongst Seychellois, **41 (37M, 4F) were reported amongst Non-Seychellois** (include Gainful-Occupation-Permit or GOP-workers, non-Seychellois residents, and tourists).

3.1.4 Maternal and Child Deaths

No maternal deaths were reported in 2022 (compared to two deaths in 2021, both related to COVID-19 infection). Figure 6 shows the Maternal Mortality Ratio (MMR) trends, with five-year rolling averages from 1990 to 2022. NHSP and SDG targets for MMR is <70 deaths per 100,000 live births. Since

Figure 6 Maternal Mortality Ratio, yearly and five-year rolling average, 1990-2022



Data Source(s): Historic NBS reports; Maternity Services (HCA); Statistics Unit (MOH).

Seychelles has few births in absolute numbers, and the Maternal Mortality Ratio (MMR) reflects maternal deaths per 100,000 live births, annual MMR show high variability from year-to-year, between zero and 200. However, the **five-year rolling averages, which reflect overall trend better, shows an increase since 2018**.

Some **child mortality indicators show an increase over 2021, specifically infant mortality rate** (Table 6). Trends in child mortality indicators from 2018 to 2022 are illustrated in Figure 7.

Table 6 Maternal and child mortality indicators, 2022, compared to NHSP baseline and targets

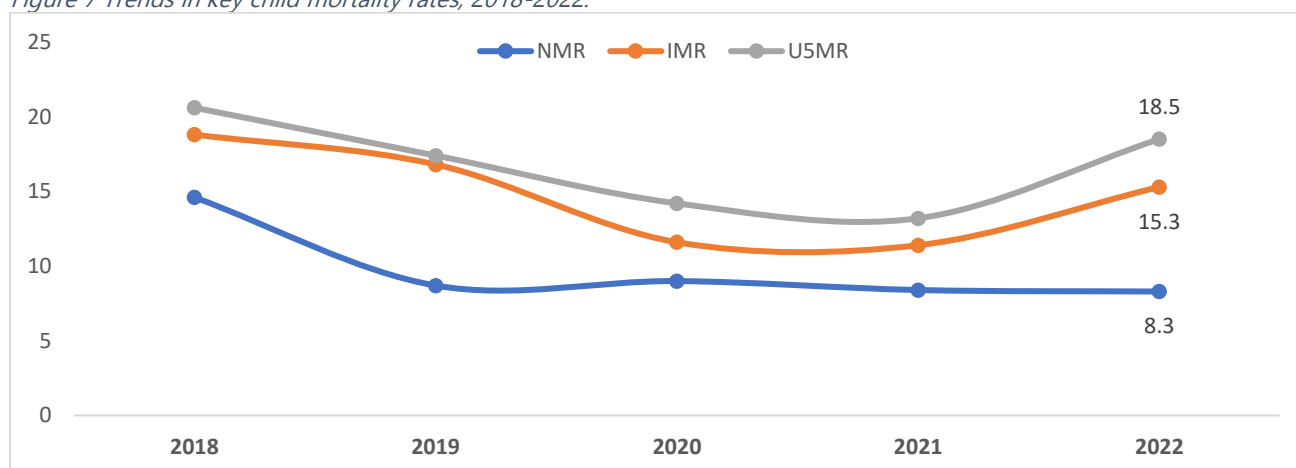
Indicator	2022	NHSP Baseline (2020)	End-Term Target
# Maternal Deaths	0	1	-
Maternal Mortality Ratio (MMR) (SDG 3.1.1)	0.0	64.4	<50
# Neonatal Deaths (<28 days of age)	13	14	-
Neonatal Mortality Rate (NMR) (SDG 3.2.2)	8.3	9	<5
# Infant Deaths (<1y of age)	24	18	-
Infant Mortality Rate (IMR)	15.3	11.6	≤10
# Under-five deaths	29	22	-
Under-five Mortality Rate (U5MR) (SDG 3.2.1)	18.5	14.2	≤12
# Stillbirths	9	8	-
Stillbirth rate	5.7	5.1	<5
# Perinatal deaths	12	13	-
Perinatal mortality rate	7.6	-	-

Notes:

- MMR is maternal deaths per 100,000 live births.
- NMR, IMR and U5MR are per 1,000 live births.
- Stillbirth and perinatal mortality rates are per 1,000 total births.

Data Source(s): Vital Statistics Report 2022 (MOH, 2023).

Figure 7 Trends in key child mortality rates, 2018-2022.



NMR – Neonatal Mortality Rate; IMR – Infant Mortality Rate; U5MR – Under-five Mortality Rate;
Data Source(s): Vital Statistics Report 2022 (MOH, 2023).

3.1.5 Life Expectancy

Life Expectancy (LE) at birth for both sexes decreased to 72.6y in 2022 based on 2021 population estimates. The changes were different for each sex – with male LE decreasing further from 71.3y to 68y, and female LE recovering from 75.5y to 77.7y.

Using End-2022 Population Bulletin (NBS, 2023) estimated resident population, the **LE at birth was 74.2y for both sexes, 69.8y for males, 77.8y for females; the census based estimate cannot be compared with past years until back-casting is available**. Both estimates remained below the NHSP end-term targets (Table 7). The wide gap between LE of males and females continued to persist.

Table 7 Life Expectancy 2022, compared to past years.

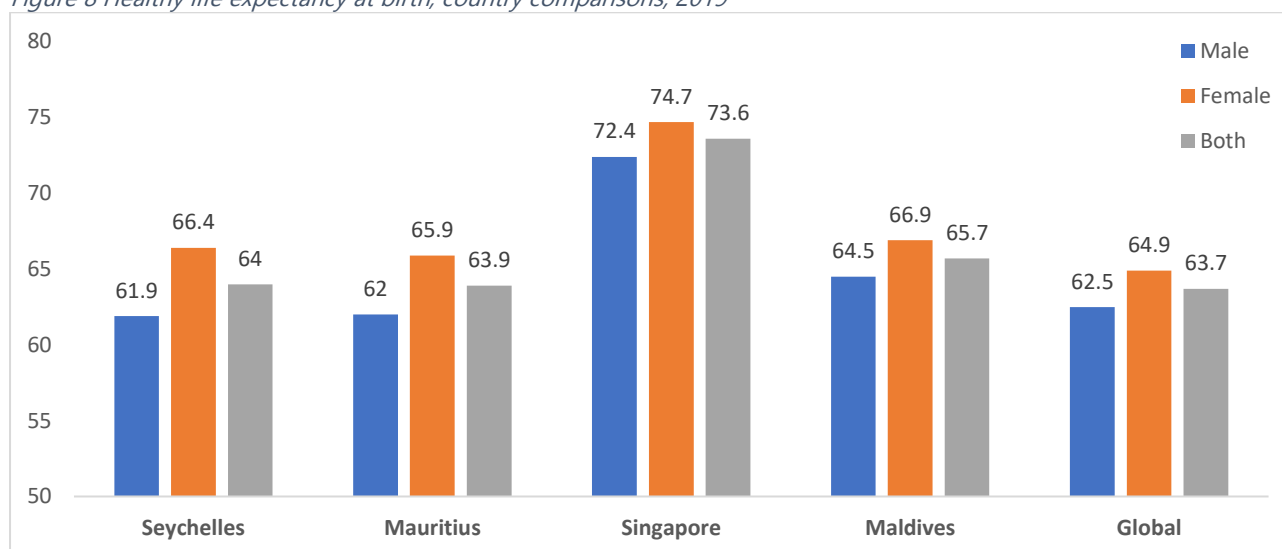
Sex	Average 2015-19	2020 (Baseline)	2021	2022 (using 2021 pop bulletin)†	2022 (using 2022 pop bulletin)*	NHSP End-Term Target
Male	69.6y	72.7y	71.3y	68.0y	69.8y	74y
Female	78.8y	82.0y	75.5y	77.7y	77.8y	83y
Both	74y	77.3y	73.5y	72.6y	74.2y	79y

Data Source(s): Statistics Unit, MOH; † 2021 Population Estimates (End-2021 Population bulletin of NBS, 2022); * PVSb 2022 (NBS, 2023).

Based on 2021 population estimates and standard life-table methods, **a male child born in 2022 had an estimated 73.7% chance of reaching age 65y, and a female, an 81.6% chance.** This has implications for some policies, e.g. for retirement age and elderly care.

The WHO estimates that the **healthy life expectancy at birth in Seychelles was 61.9y for males, 66.4y for females, and 64y overall** (2019 comparable estimates). This is comparable to peer countries in the region, such as Mauritius and Maldives, but is well-below most other high-income countries (Figure 8). The gap between LE and healthy LE crudely estimates the length of time people live in poor health; **both men and women in Seychelles live a significant portion of their lives, the equivalent of six and eleven years respectively, with illness and/or disability.**

Figure 8 Healthy life expectancy at birth, country comparisons, 2019



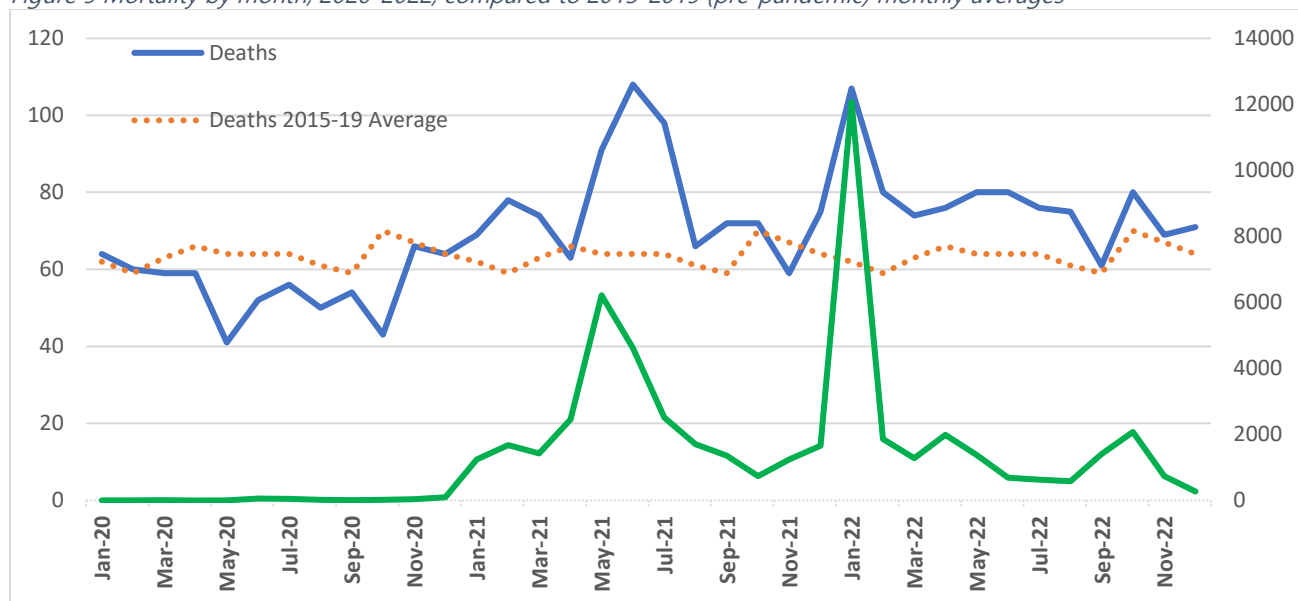
Data Source(s): World Health Statistics Report 2023 (WHO, 2023).

3.1.6 Excess Mortality

Despite fewer deaths attributable to COVID-19, an additional 167 deaths were registered in 2022, compared to pre-pandemic average (2015-2019) – i.e. **+21.8% excess deaths** (vs +21.2% in 2021).

Analysis by month (Figure 9) shows that **most excess deaths occurred in January 2022** (coinciding with the Omicron surge), where deaths were 73% above expected pre-pandemic levels, surpassing the previous peak in June 2021 (the Delta surge).

Figure 9 Mortality by month, 2020-2022, compared to 2015-2019 (pre-pandemic) monthly averages



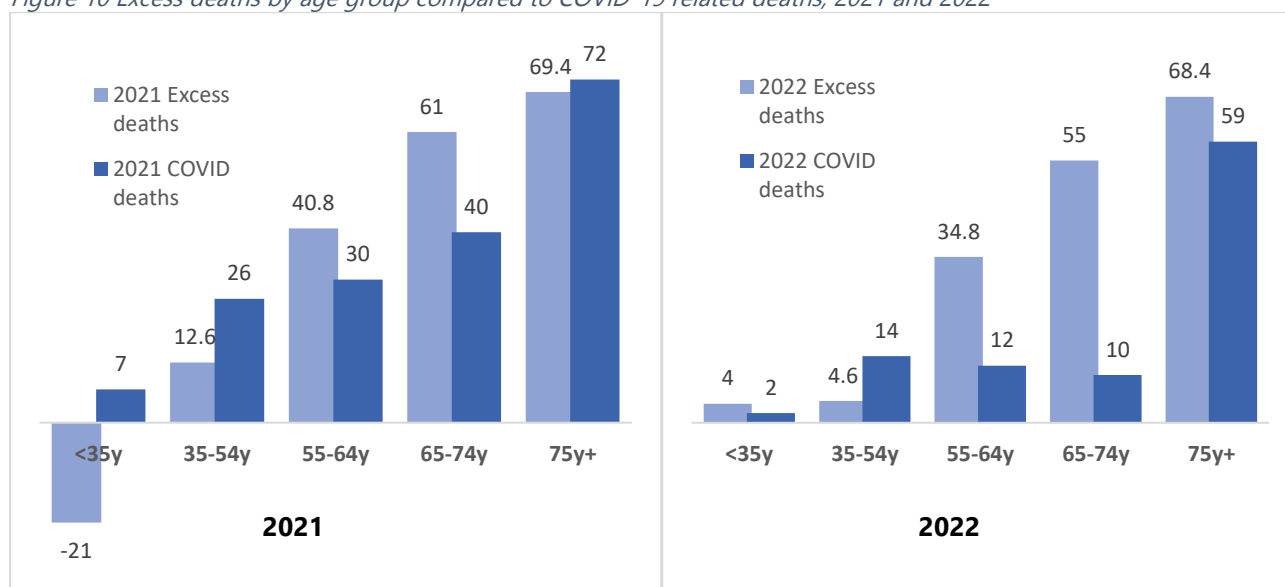
Data Source(s): Statistics Unit, MOH; SEIS Report (PHA, 2023).

Figure 10 shows a plot of excess mortality and COVID-19-related deaths by age group for 2021 and 2022.

Excess deaths in the older age groups (75y+) were equivalent to known COVID-19 related deaths, whereas in 55-74y age-range, excess deaths were well-above number of deaths associated COVID-19.

The difference between excess deaths and COVID-19 deaths in the 55-74y group is more evident in 2022 than in 2021. The reduction in mortality in 2021 in the under-35y group reflects the lower rates of mortality from external causes in 2021 (see 3.2.3 Deaths from External causes).

Figure 10 Excess deaths by age group compared to COVID-19 related deaths, 2021 and 2022



Data Source(s): Statistics Unit, MOH.

3.1.7 Fertility

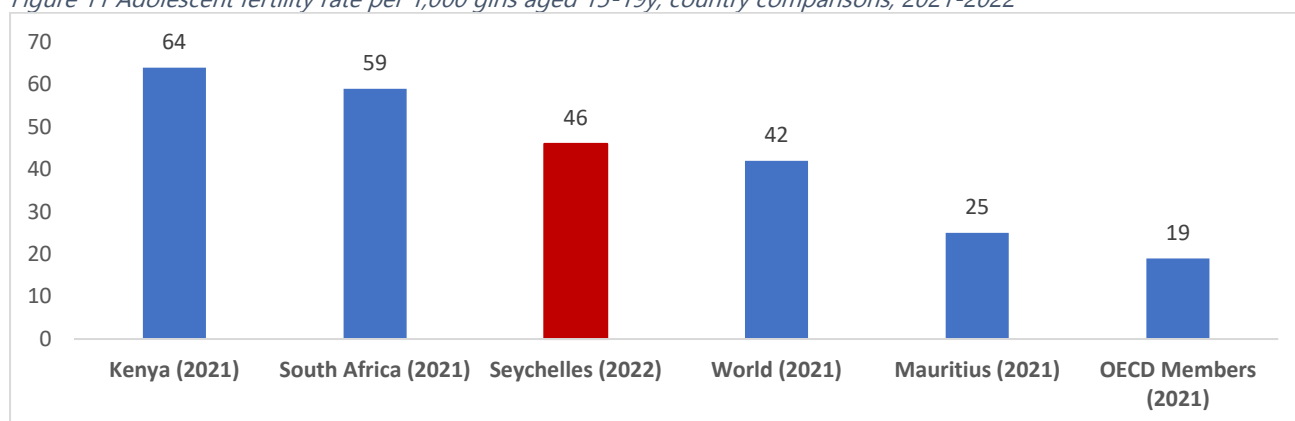
The NBS (using PVS2022 population estimates) reported a **Total Fertility Rate (TFR) of 2.01 (live births per woman aged 15-49y) in 2022²¹**, a drop from previous years (2.46 in 2021). The number of pregnancies

²¹ NBS Statistical Bulletin: Mid-year estimated population and demographic indicators 2022 (NBS 2023).

recorded amongst adolescent girls aged 15-19y was similar to previous years (164 births, and 61 abortions). The **adolescent fertility rate (SDG 3.7.2) was 46 per 1,000 girls in age-range 15-19y**, compared to 63 in 2021 (NHSP target of <30) – this change is due to use of population estimates from new census.

In the 10-14y age group, three births and five abortions were recorded. There were **422 reported cases of abortions**, similar to previous years²². This included **69 Medical Termination of Pregnancy**, of which 15 (21.7%) were amongst girls 10-19y. Figure 11 compares adolescent fertility rates of Seychelles to other countries.

Figure 11 Adolescent fertility rate per 1,000 girls aged 15-19y, country comparisons, 2021-2022

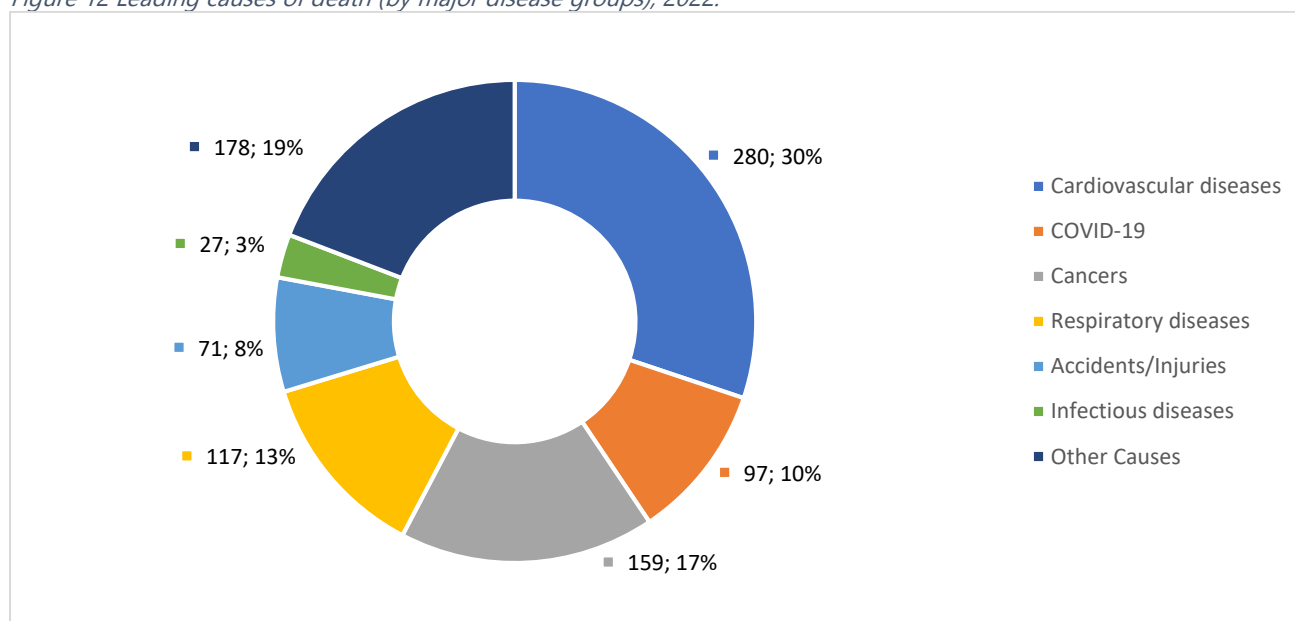


Data Source(s): World Bank; Seychelles data from NBS.

3.2 Causes of Death

Cardiovascular diseases (CVDs) remained the underlying cause for most deaths, before and during COVID-19 pandemic. Cancers were the second leading cause of deaths in 2022 (replacing COVID-19), followed by respiratory diseases (predominantly lung infections), and COVID-19 (Figure 12). **Deaths from accidents and injuries, although only the fifth leading cause, increased significantly in 2022** compared to 2021.

Figure 12 Leading causes of death (by major disease groups), 2022.

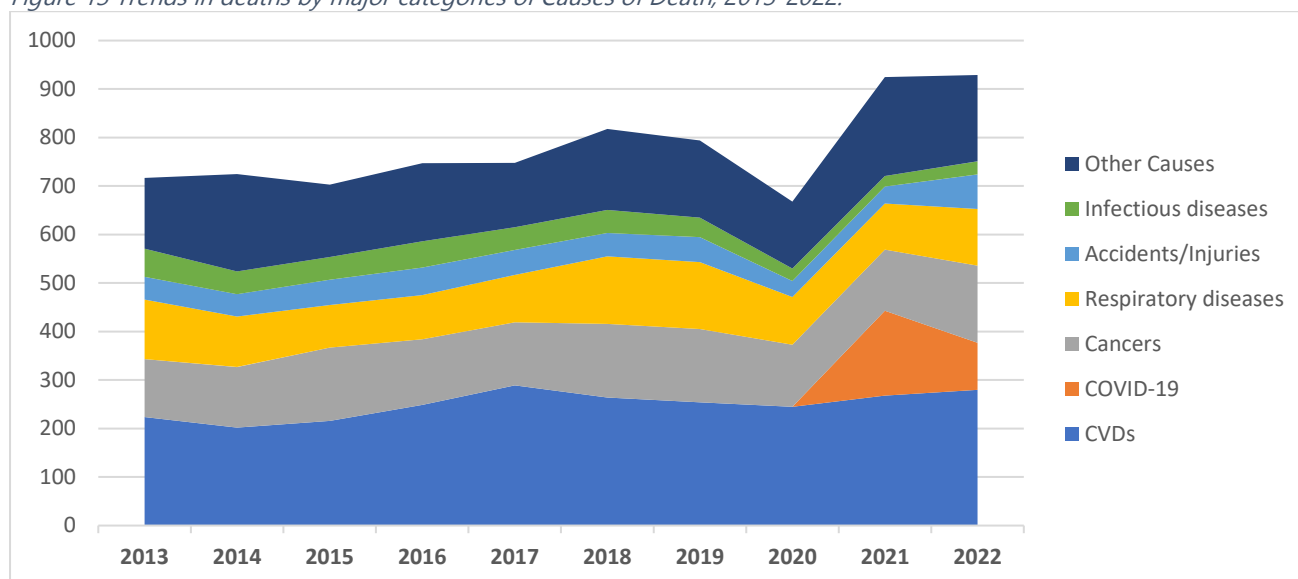


Data Source(s): DoCS; Statistics Unit, MOH.

²² Annual Health-sector Performance Report 2021 (MOH, 2022).

The ten-year trends in these key causes of death (Figure 13) show the significant entry of COVID-19 as from 2021, the continued increase in CVD deaths (in terms absolute numbers) and total deaths, and the increase in number and proportion of deaths from external causes.

Figure 13 Trends in deaths by major categories of Causes of Death, 2013-2022.



Note: Deaths from respiratory diseases includes both Chronic Respiratory Diseases (CRDs, numbering 12 deaths) and pneumonias (77 of remaining 105 deaths).

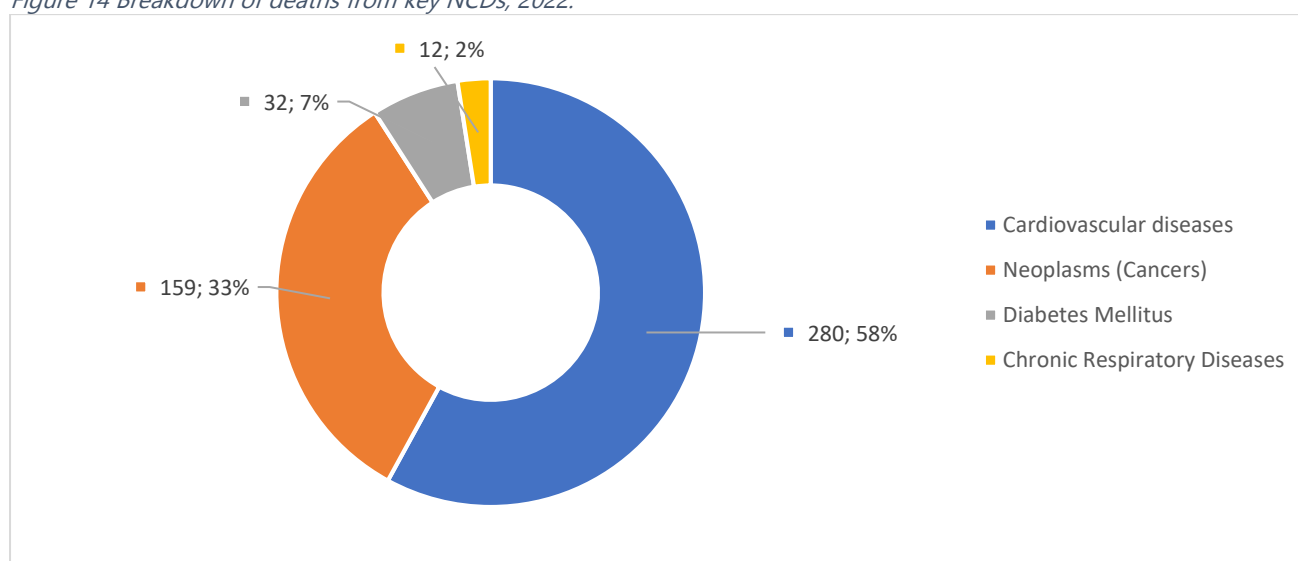
Data Source(s): Adapted from Vital Statistics Report 2022 (MOH).

The MOH is transitioning from ICD-10 to ICD-11 for coding of deaths and hospitalisations. The quality of cause of death reporting has improved over recent years (through capacity building of medical practitioners and statistics officers in the MOH).

3.2.1 Mortality from Key NCDs

NCDs were the underlying cause of death for 52% of all deaths in 2022. The four key NCD categories (CVD, cancers, diabetes and chronic respiratory diseases) accounted for 483 deaths (273 M, 210 F). CVDs and cancers predominate amongst NCD deaths (Figure 14).

Figure 14 Breakdown of deaths from key NCDs, 2022.



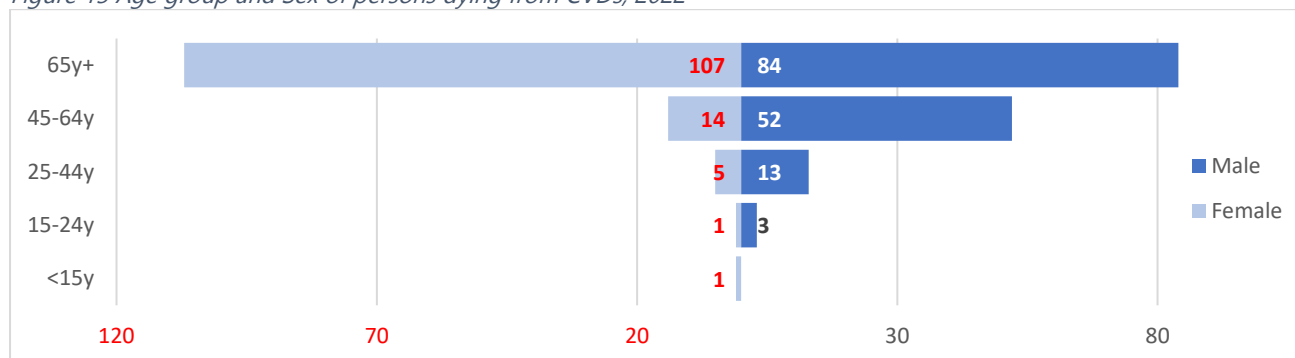
Chronic Respiratory Diseases refers to primarily to Chronic Obstructive Pulmonary Disease and Asthma.

Data Source(s): Vital Statistics Report 2022, Statistics Unit, MOH.

Cardiovascular Diseases

Deaths from CVDs as a percentage of deaths show an apparent reduction in 2021 and 2022 compared to pre-pandemic years; but absolute numbers of deaths actually increased serially from 2020, through 2021, to 2022 (see Figure 13). At younger ages, deaths from CVDs amongst men are more than in women (Figure 15).

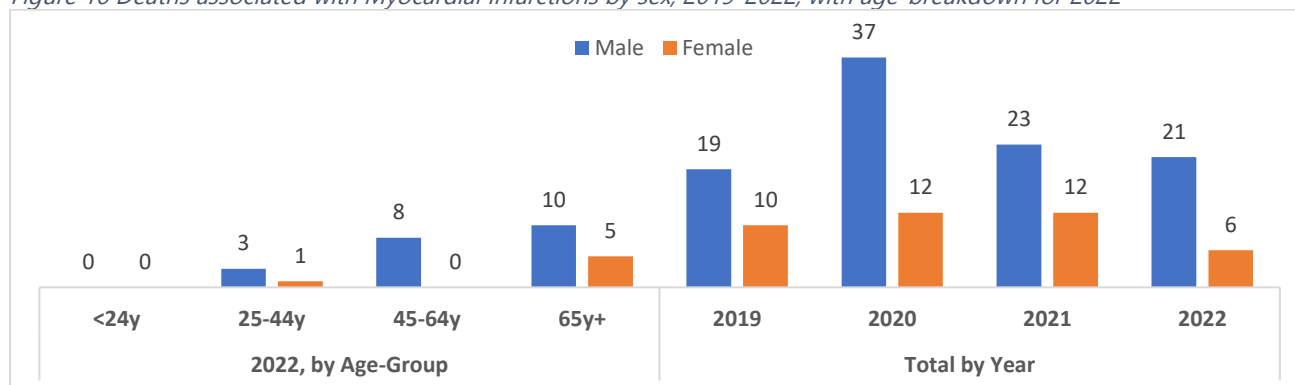
Figure 15 Age group and Sex of persons dying from CVDs, 2022



Data Source(s): Statistics Unit, MOH.

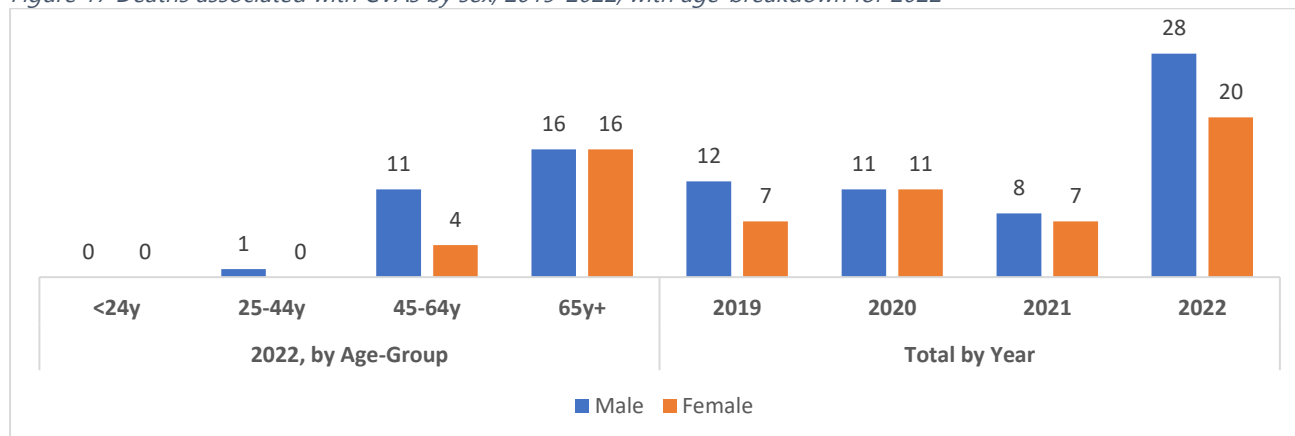
In 2022, **27 deaths (21M, 6F) were associated with Myocardial Infarctions (MIs, or Heart Attacks)**. More MI-related deaths occur in men in all age-groups (Figure 16). In comparisons, **46 deaths in 2022 (28M, 20F) were associated with Cerebrovascular Accidents (CVAs, or strokes)**; a significant increase over 2019-2021 (Figure 17). More stroke-related deaths occur in younger men. See section 3.3.1 Heart Attacks and Strokes for admissions for these conditions.

Figure 16 Deaths associated with Myocardial Infarctions by sex, 2019-2022, with age-breakdown for 2022



Data Source(s): DoCS; NBS; Statistics Unit, MOH.

Figure 17 Deaths associated with CVAs by sex, 2019-2022, with age-breakdown for 2022



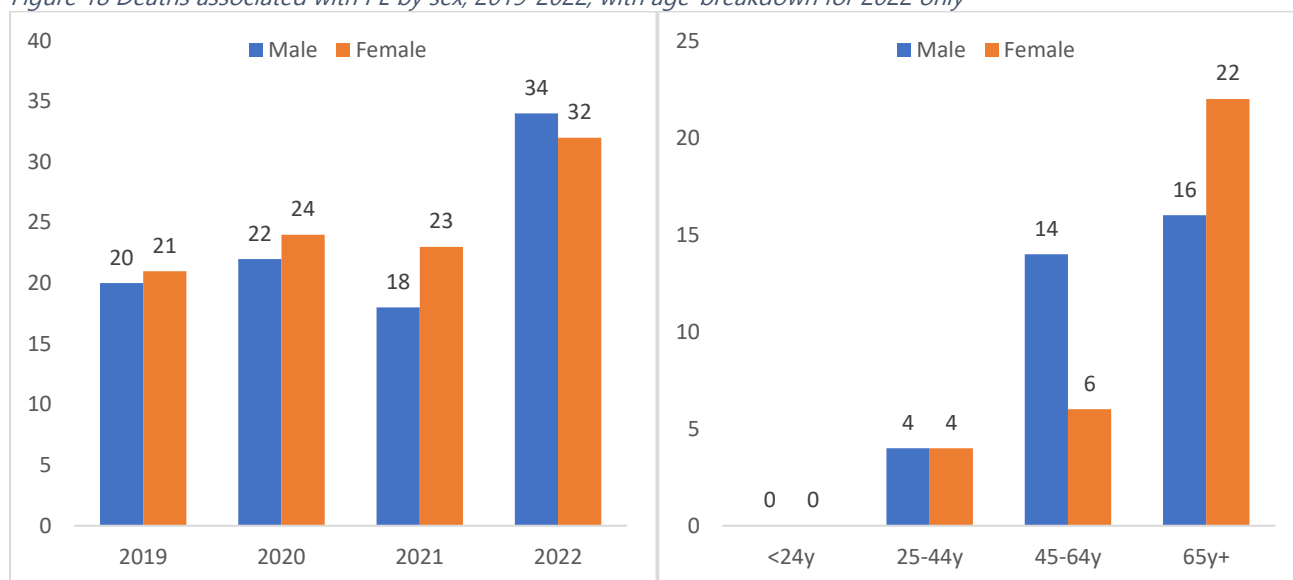
Data Source(s): DoCS; NBS; Statistics Unit, MOH.

Pulmonary Embolism

Pulmonary Embolism (PE) is typically a complication of a deep vein thrombosis (DVT), often leading to death; both these conditions together are often referred to as Venous Thrombo-Embolism (VTE). VTE is a leading morbidity amongst the Seychellois population and **66 deaths (7.1% of total) were associated with PE, often with PE as the immediate cause of death**. In comparison, only 43 deaths were associated with PE in 2021. PE deaths appear to be more common amongst **men at younger age groups, and women at 75y+** (Figure 18).

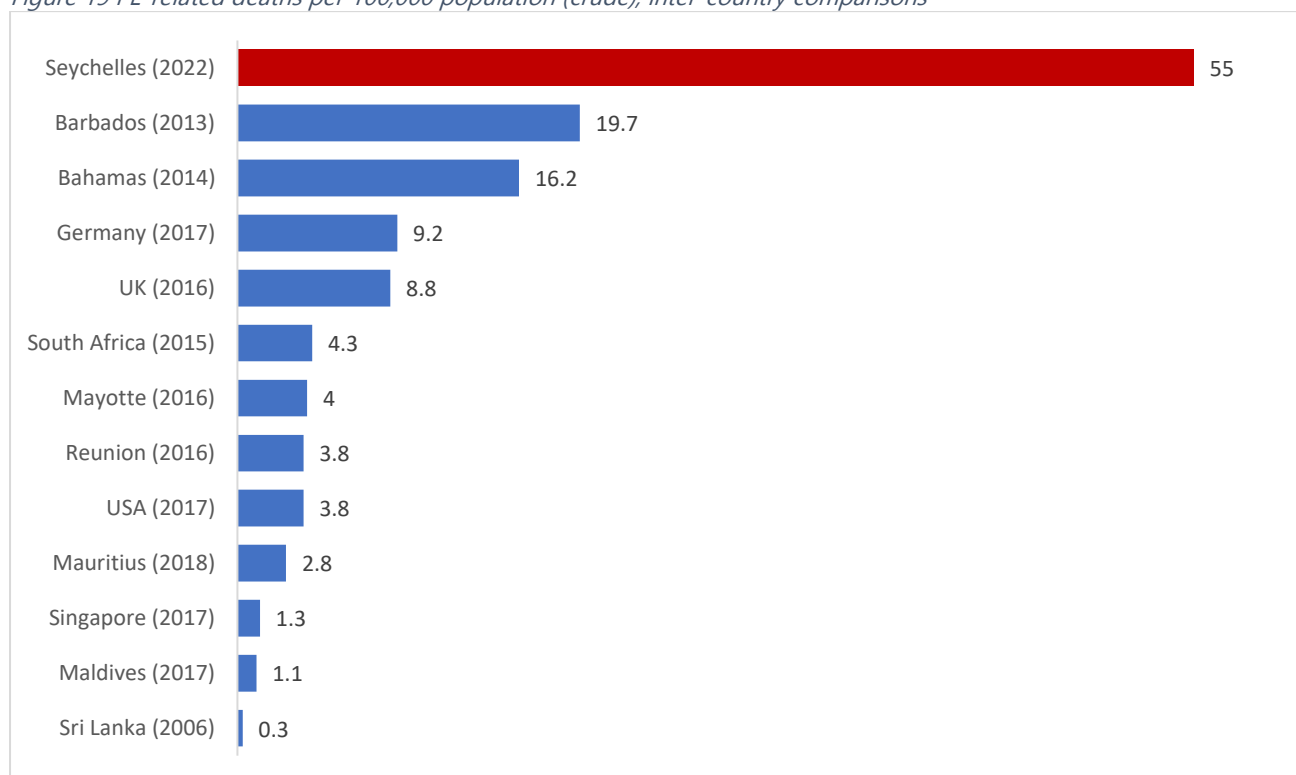
PE-related deaths in Seychelles in 2022 appear to be amongst the **highest in the world** (Figure 19).

Figure 18 Deaths associated with PE by sex, 2019-2022, with age-breakdown for 2022 only



Data Source(s): DoCS; NBS; Statistics Unit, MOH.

Figure 19 PE-related deaths per 100,000 population (crude), inter-country comparisons



Data Source(s): Barco et al. (2021); Statistics Unit, MOH.

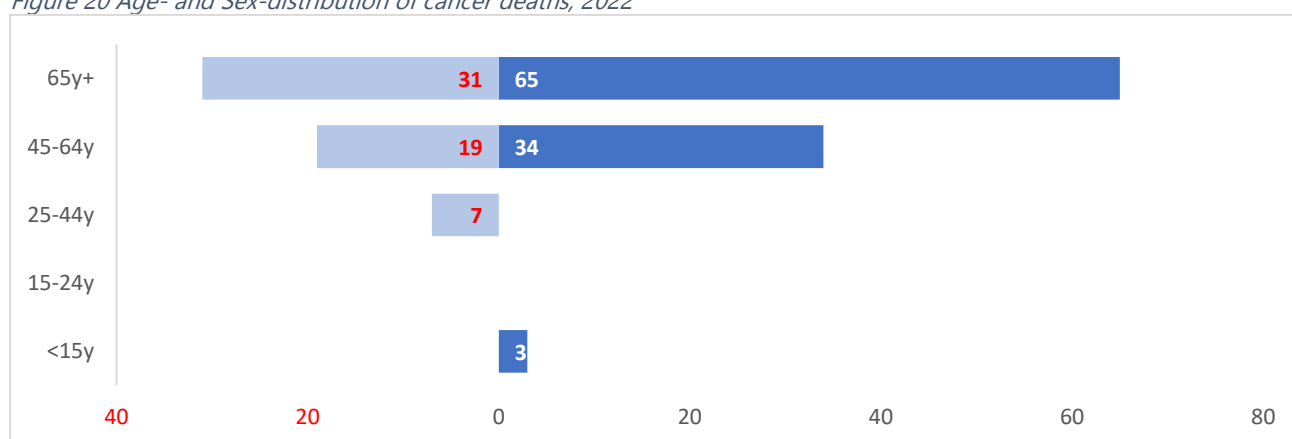
The age groups most affected in Seychelles also appear to be older (60y+), compared to international patterns, where the 20-49y olds are most affected²³. Globally, small-island states in the Caribbean appear to have disproportionately high mortality from PE.

The high level of population exposure to COVID-19 in 2021 and 2022, and existing research pointing to increased risk for thromboembolic complications for up to a year post-infection²⁴, could explain the increase in PE-related and stroke-related deaths.

Cancer deaths

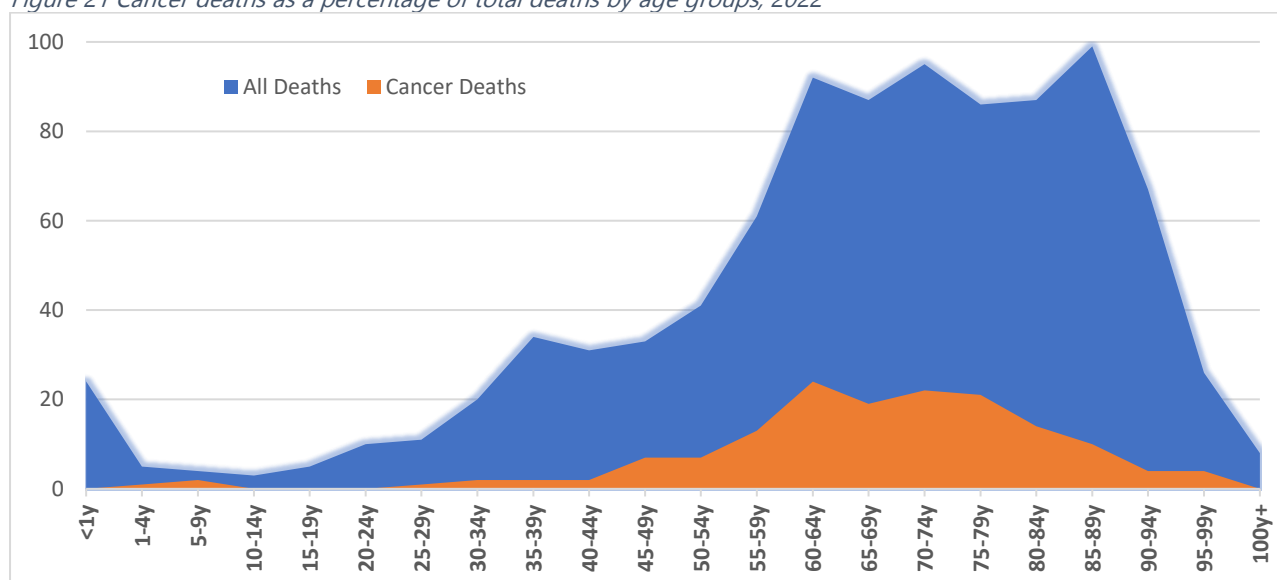
Both cancer incidence (see section 3.3.2 Cancer) and deaths were **higher amongst men** in 2022, with large variations between individual cancer types (Figure 20). **Cancer deaths recorded were 159 (102 M, 57F)** for the year 2022, with significantly more cancer deaths amongst men. Analysis of cancer deaths (only as underlying cause), showed that cancer caused a large fraction of deaths in the 45-80y age-range (Figure 21).

Figure 20 Age- and Sex-distribution of cancer deaths, 2022



Data Source(s): Statistics Unit, MOH; based on civil registration data.

Figure 21 Cancer deaths as a percentage of total deaths by age groups, 2022



Data Source(s): DoCS; NBS; Statistics Unit, MOH.

²³ Barco, S., Valerio, L., Gallo, A., Turatti, G., Mahmoudpour, S. H., Ageno, W., ... & Konstantinides, S. V. (2021). Global reporting of pulmonary embolism-related deaths in the World Health Organization mortality database: Vital registration data from 123 countries. *Research and practice in thrombosis and haemostasis*, 5(5), e12520.

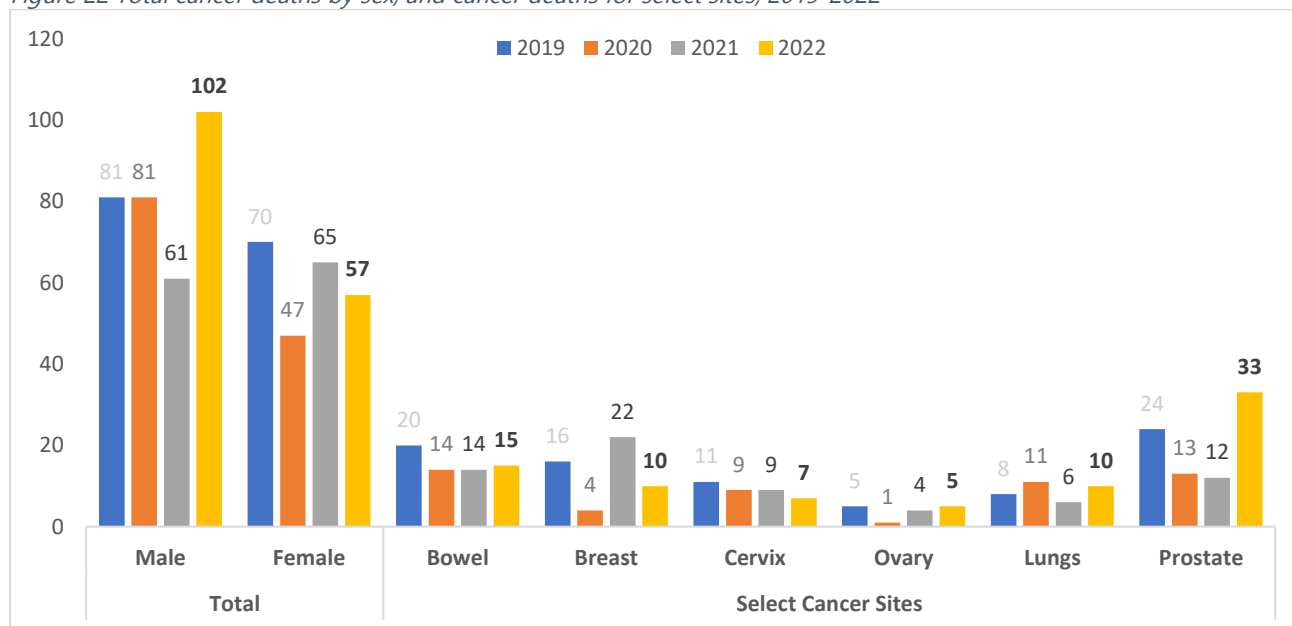
²⁴ Scharf, R. E., & Anaya, J. M. (2023). Post-COVID Syndrome in Adults—An Overview. *Viruses*, 15(3), 675.

Prostate cancer was the most common cause of cancer death in men; **all 33 deaths from prostate cancer in 2022 were in men age 60y+**, with majority in age group 75y+ (Figure 22 and Figure 23). This implies that prostate cancer is an important cause of death in older men.

All 20 deaths from head and neck cancers were amongst men, as were the majority of deaths from cancers of the lungs, airways, oesophagus, stomach, liver, pancreas and biliary system.

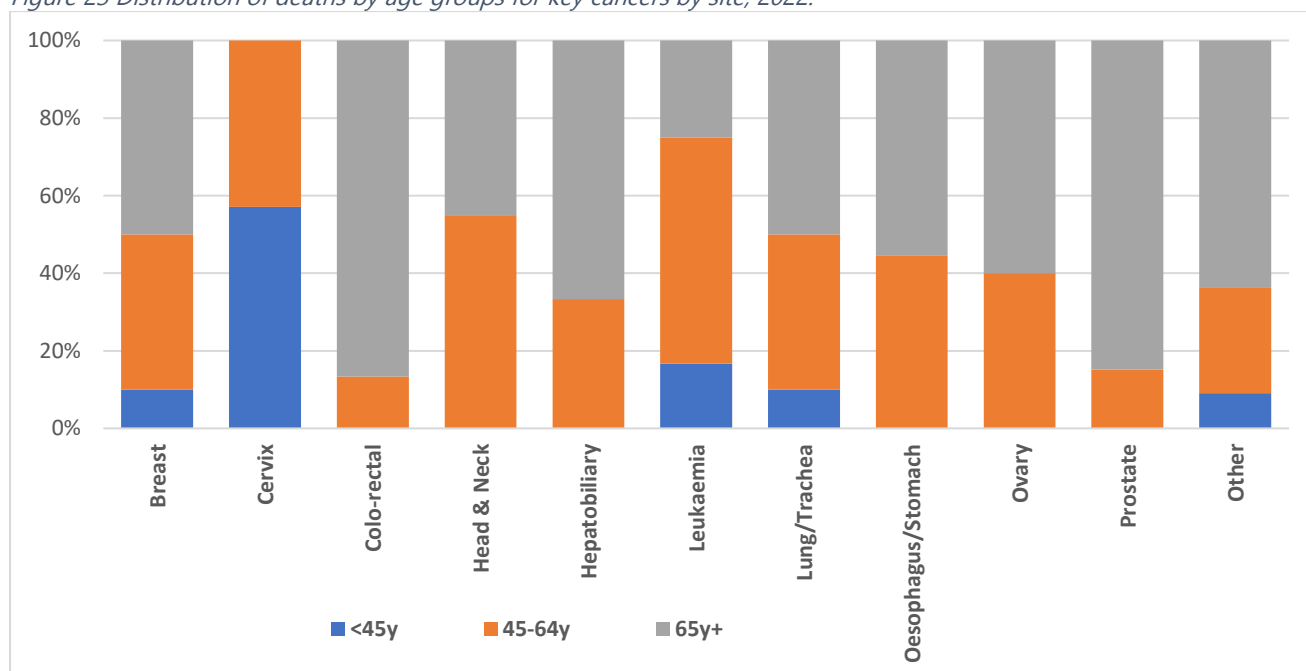
Seven deaths due to cervical cancer were amongst relatively young women aged between 35 and 59y; whereas deaths from breast cancers were amongst women in a broad age-range (30-100y). Bowel cancers (15 deaths) affected both sexes equally.

Figure 22 Total cancer deaths by sex, and cancer deaths for select sites, 2019-2022



Data Source(s): Statistics Unit, MOH; based on civil registration data.

Figure 23 Distribution of deaths by age groups for key cancers by site, 2022.



Data Source(s): Statistics Unit, MOH; based on civil registration data.

Diabetes, Hypertension and Kidney Disease

Analysis of the complete sequence of cause of death in death certificates show that **Diabetes and Hypertension were noted in 128 (14%) and 248 (27%), respectively, of all deaths registered in 2022.**

These could be in either part 1 (immediate, intermediary and underlying causes) or part 2 (contributory causes) of the certificate. Diabetes and Hypertension are often under-recorded, and the reported data remain under-estimates.

One in 20 people (4.7%) who died in 2022 also suffered from chronic kidney disease (CKD). Eighteen people on long-term haemodialysis died in 2022 (see Haemodialysis services).

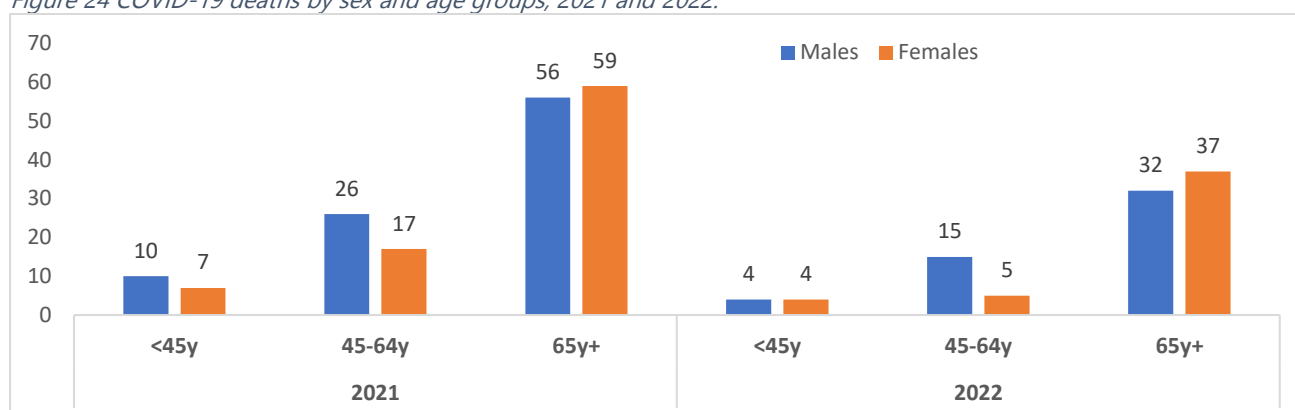
3.2.2 Deaths from Key Communicable Diseases

COVID-19 deaths

The **absolute number of deaths with COVID-19 as the underlying cause have reduced from 175 (2021) to 97 (2022).** COVID-19 moved from the second to the fourth leading cause of death in 2022. COVID-19 deaths counts are from analysis of Civil Status records of deaths, and are *not* cross verified with mortality surveillance data.

The **majority of COVID-19 deaths occurred in Q1 and Q2** of 2022, and were **concentrated amongst older individuals** (71% in 65y+, Figure 24). In the over 65 group, both sexes were affected equally, but deaths in younger age groups occurred more amongst men.

Figure 24 COVID-19 deaths by sex and age groups, 2021 and 2022.



Data Source(s): Vital Statistics Report, 2022, Statistics Unit, MOH.

Pneumonia deaths

A total **77 deaths (42 M, 35F) were attributed to pneumonia as the underlying cause**; 56 (73%) of these deaths occurred in the 65y+ age group, 15 (20%) in the 45-64y group. The Statistics Unit did not have data on causative agents of pneumonia.

HIV/AIDS-related deaths

The Communicable Disease Control Unit (CDCU) reported **27 (19M, 8F) deaths among people living with HIV/AIDS (PLHIV)²⁵** in 2022, all aged 15y+. Amongst them, the main underlying causes of death were COVID-19 (7, 26%) cerebral toxoplasmosis (6, 22%) and cancers (6, 22%)²⁶.

²⁵ CDCU Annual Report, 2022 (PHA, 2023).

²⁶ Global AIDS Monitoring (GAM) Report – Seychelles 2022 (NAC, 2023).

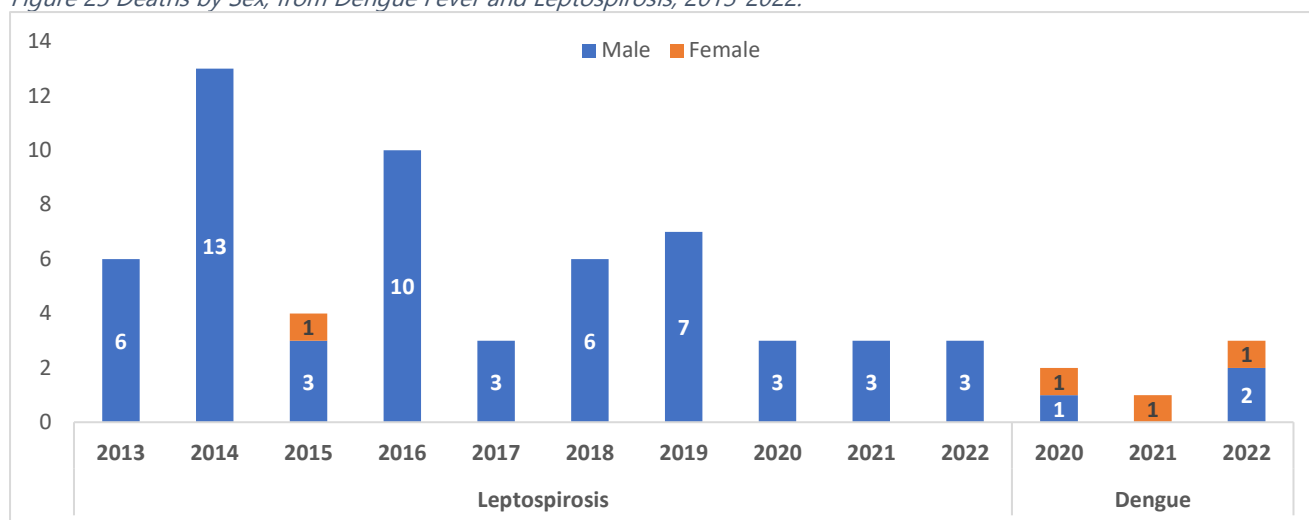
Deaths from Tuberculosis

One (male) death of a Tuberculosis (TB) patient was reported in 2022. For longer-term trends in TB incidence and mortality, please refer to the 2021 report²⁷.

Deaths from Leptospirosis and Dengue Fever

In 2022, **Leptospirosis caused three deaths (3M) and Dengue fever three deaths (2M, 1F)**. Figure 25 shows deaths from Leptospirosis from 2013 to 2022 (almost exclusively males), and from Dengue 2020 to 2022. Leptospirosis deaths were fewer in recent years (2020-2023 vs 2013-2019).

Figure 25 Deaths by Sex, from Dengue Fever and Leptospirosis, 2013-2022.



Data Source(s): Statistics Unit, MOH.

3.2.3 Deaths from External causes

Deaths from external causes refer to those deaths following road traffic accidents (RTAs), other accidents and injuries (including workplace injuries), homicides and self-harm (suicides).

A total of 71 deaths (7.6% of total) were due to external causes were reported in 2022 (Figure 26). This represents **an important increase compared to 2020 and 2021**, and significantly more than pre-pandemic levels (average of 6.9% of total deaths, or 52 deaths annually, from 2015-2019).

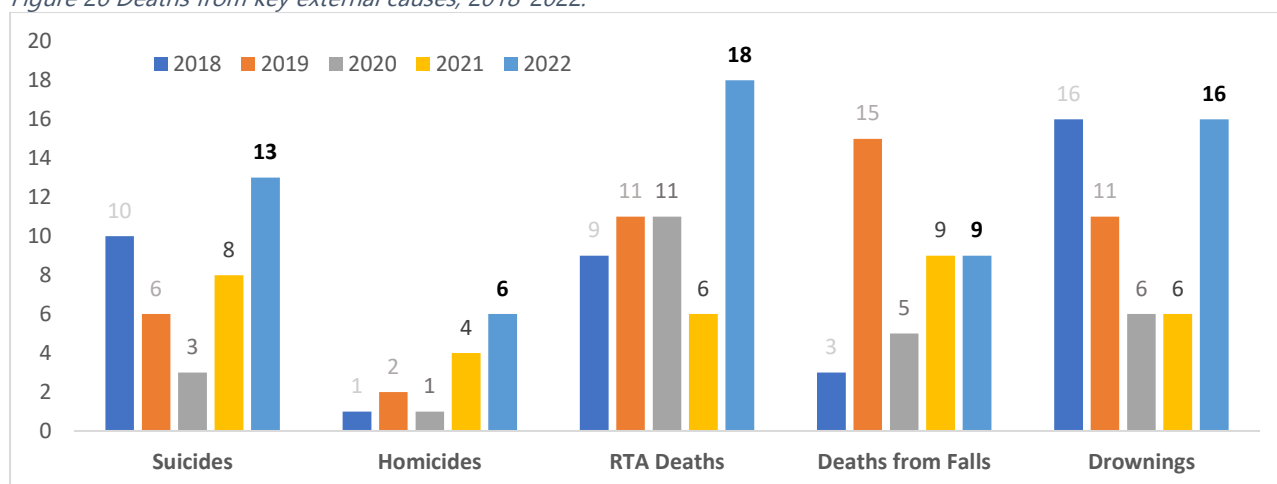
RTAs related mortality rate (SDG 3.6) was 15 per 100,000 population, and **Suicide Mortality Rate (SDG 3.4.2) was 10.8** per 100,000 population, using 2022 population estimates (PVSB, NBS, 2023).

Figure 27 shows five-year trend in mortality rates from external causes, using 2021 population estimates (NBS, 2022). The trends show mortality rates from all external causes increased from 2021 to 2022. The biggest increases were deaths from RTAs, and suicides and homicides.

Comparisons to other countries are shown in Figure 28, using 2022 population estimates. Seychelles RTAs mortality rates are comparable to Mauritius, below most mainland African countries, but far above high-income countries. Suicide mortality rates in Seychelles were comparable to high-income countries.

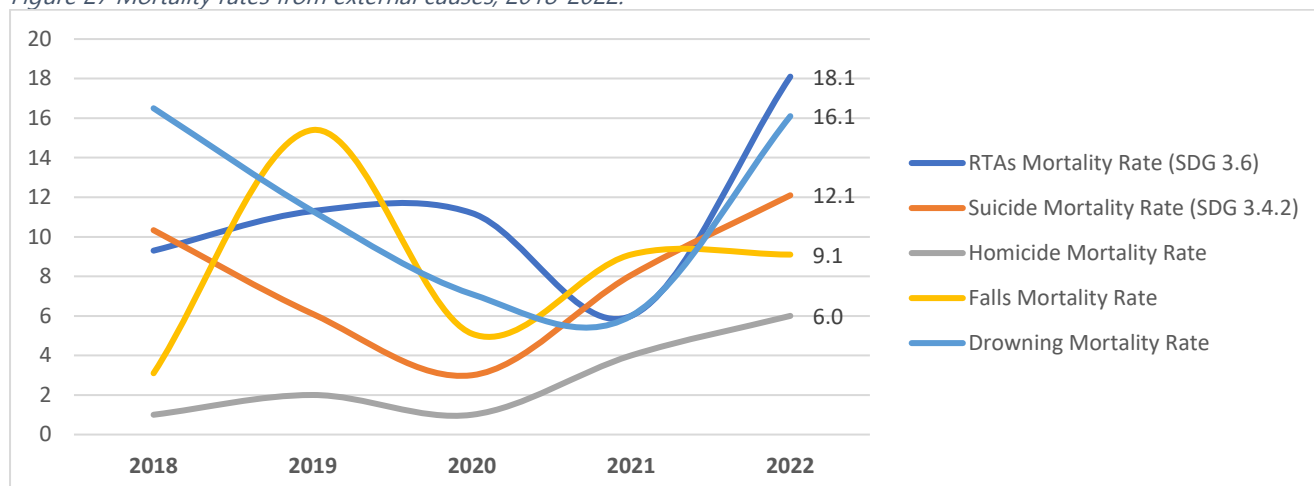
²⁷ Annual Health-sector Performance Report (MOH, 2022).

Figure 26 Deaths from key external causes, 2018-2022.



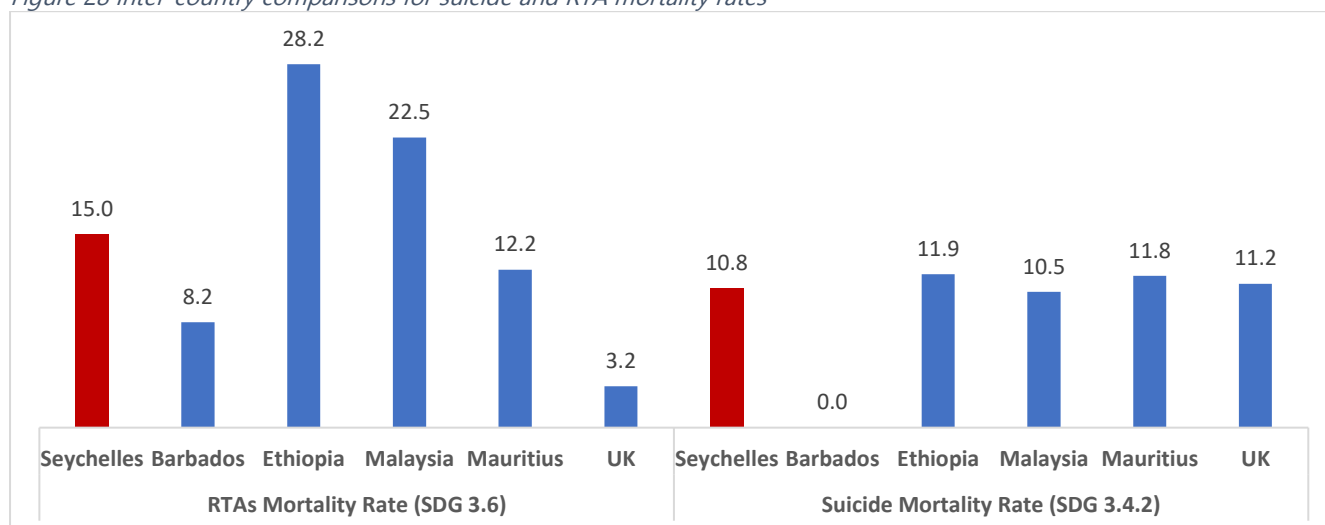
Note: Age-standardised rates will be calculated once back casting data is available. Data Source(s): Statistics Unit, MOH.

Figure 27 Mortality rates from external causes, 2018-2022.



Note: Using NBS 2021 population estimates for 2022 (End-2021 Population Bulletin); New population data can only be used for trend analysis once back casting is available.
Data Source(s): Statistics Unit, MOH.

Figure 28 Inter-country comparisons for suicide and RTA mortality rates



Data Source(s): Seychelles data from 2022, using 2022 estimated residential population (NBS, 2023); all other country data are estimates for 2019 from WHO Global Health Observatory.

3.2.4 Understanding premature deaths

Premature mortality refers to "death that occurs before the average age of death in a certain population"²⁸. Considering small numbers of deaths amongst children, and significant differences in patterns of causes of death across age sub-groups, an analysis of all deaths amongst 0-19y olds from 2015 to 2022 was undertaken.

Infant Deaths (<1y)

Twenty-four deaths (18M, 6F) were reported under age of one, all with underlying cause of prematurity or congenital malformations. Analysing longer-term data (2015 to 2022), **prematurity is the main underlying cause of infant deaths** (42.5% of all deaths in <1y), followed by congenital syndromes and malformations (26.8%), especially of the heart.

Early Childhood Deaths (1-4y)

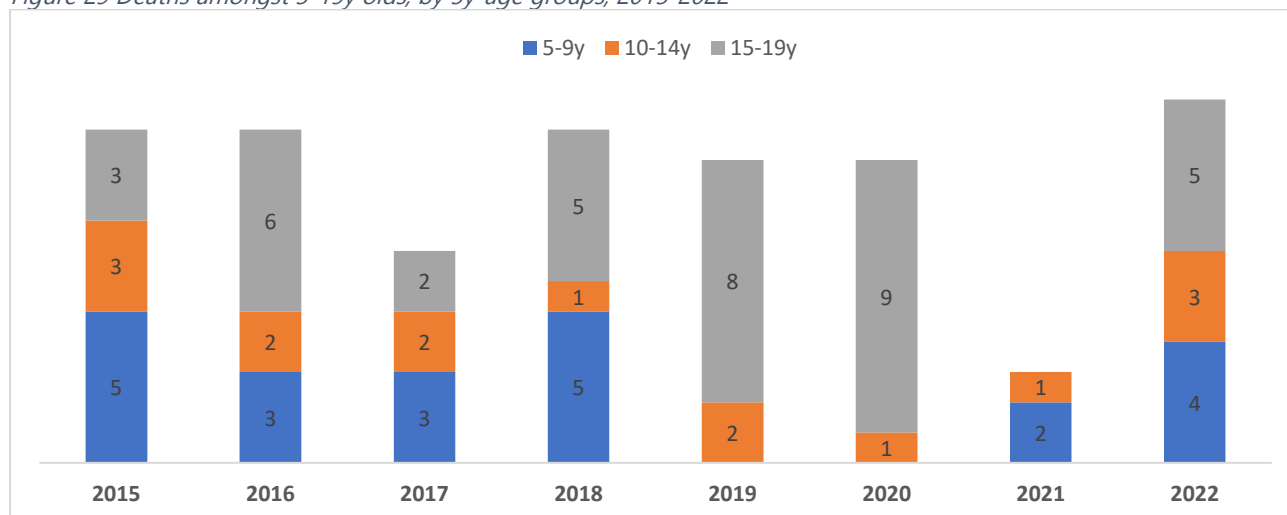
Five deaths (3M, 2F) occurred in the 1-4y age group in 2022. Analysis of 2015-2022 data showed, on average, three deaths occur per year in this age group, with boys and girls affected equally. The main causes were: (1) Cerebral Palsy; (2) Pneumonia; (3) Congenital syndromes and malformations; and (4) External causes (drowning, burns, head injury, road traffic accidents).

School-aged children (5-19y)

An average of 9 to 10 children aged 5-19y died each year during 2015-2022, accounting for about 1.2% of total deaths each year (Figure 29). Most of these deaths are amongst males (72%) and their distribution by age groups shows that there were, on average, 2.8 deaths annually amongst 5-9y olds, 1.9 amongst 10-14y olds, and 4.8 amongst 15-19y olds.

Twelve deaths (10M, 2F) occurred in age group 5-19y in 2022. Within this age group, **majority of deaths occurred amongst boys – 72%** of total deaths, and about half occur in the 15-19y age sub-group.

Figure 29 Deaths amongst 5-19y olds, by 5y-age groups, 2015-2022



Data Source(s): Mortality Database of Statistics Unit, MOH, 2023.

The common causes of death by five-year age groups, from cumulative data from 2015-2022, are summarised in Table 8 next. **Deaths from external causes were the major cause of death amongst schoolchildren – particularly road traffic accidents in 15-19y olds, and drowning in 5-9y olds.**

²⁸ National Institutes of Health, National Cancer Institute, Dictionary of Cancer Terms, link: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/premature-death>

Table 8 Top causes of death in children and adolescents aged 5-19y, 2015-2022

Age-group	Deaths	Top causes of death (2015-2022)
5-9y	4M	• Drowning • Cerebral palsy
10-14y	2M, 1F	• External causes (drowning, fall, domestic accident) • Congenital heart disease
15-19y	4M, 1F	• Road traffic accidents
Overall (age-range 5-19y)	-	• External causes (accidents, injuries, violence) – 40% of all deaths. • Cancers (14%) • Congenital and inherited abnormalities (11%) • Cerebral palsy (11%)

Data Source(s): Mortality Database of Statistics Unit, MOH, 2023.

Premature deaths in Adults

External causes (accidents, drowning and suicide) feature strongly as cause of premature deaths in adults aged up to 44y, followed by COVID-19, cancers and HIV/AIDS (Table 9, 2022 data only). In the 45-64y age group, NCDs, particularly cancers, predominate.

Table 9 Top causes of pre-mature deaths (in 15-64y range), 2022

Age group	Deaths	Key underlying causes of death in 2022	# deaths	% in age group
15-24y	15 (12M, 3F)	• RTAs	6	40%
		• Accidental Drowning	2	13%
25-44y	96 (61M, 35F)	• All external causes (Suicide, RTAs, falls, homicide)	21	22%
		• Suicide	9	9%
		• COVID-19	8	8%
		• Cancers	6	7%
		• HIV/AIDS	5	5%
45-64y	227 (158M, 69F)	• Cancers	53	23%
		• Ischaemic Heart Diseases	21	9%
		• COVID-19	20	9%
		• Pneumonia	15	7%
		• Strokes	12	5%

Note: HIV deaths are from analysis of causes of death from DoCS data. Data Source(s): Statistics Unit, MOH.

3.3 Morbidity

Morbidity (illness) is an essential component of estimating the burden of various disease conditions.

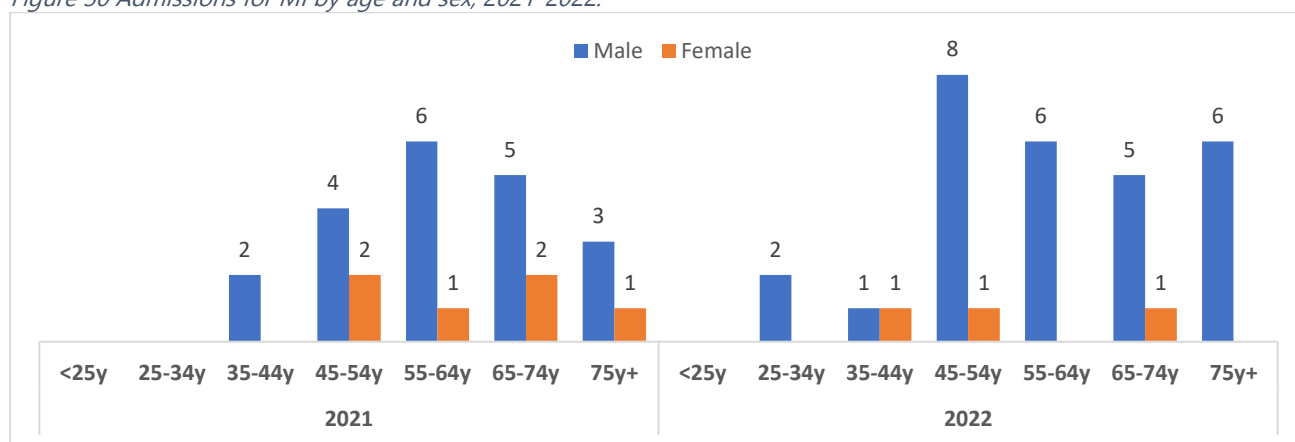
Understanding morbidity patterns is **particularly important for chronic medical conditions**, disabilities, and mental health conditions, which typically do not lead to death in the short-term, but still cause much suffering.

Morbidity data reported is mostly from hospitalisation records and attendance to certain specialist outpatient services. Community health services morbidity data have not been analysed. MOH does not have access to relevant data from the private health sector.

3.3.1 Heart Attacks and Strokes

There were **31 admissions for heart attacks** (Myocardial Infarction or MI), 28 M (90%), 3 F (10%), with the highest incidence in the 45-54y age group; a pattern similar to 2021 (Figure 30). Not all heart attacks are captured using data on admissions, as some die before hospitalisation.

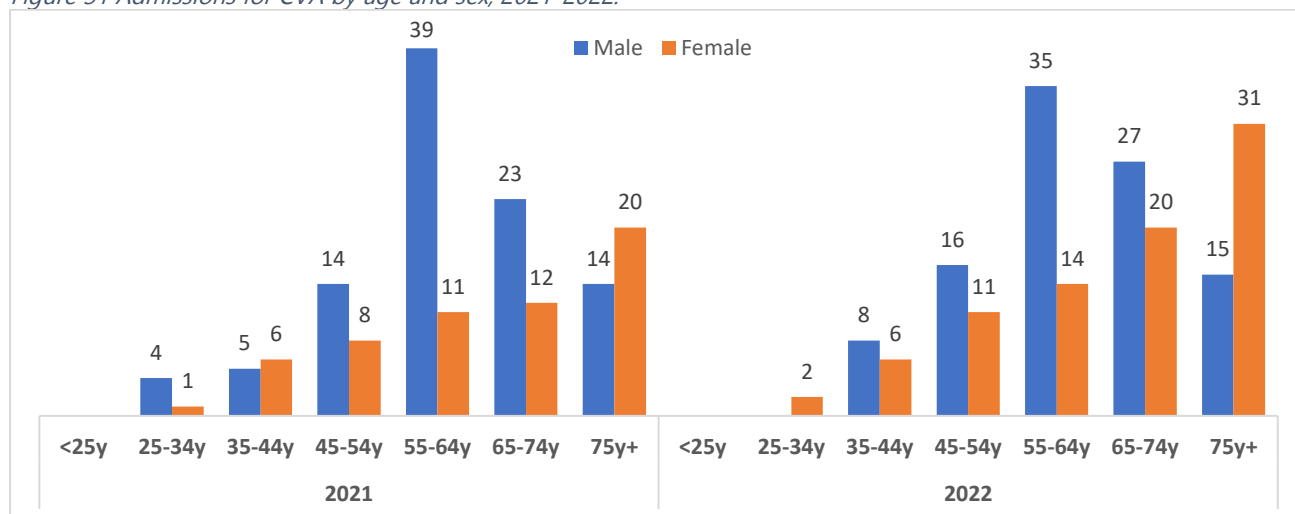
Figure 30 Admissions for MI by age and sex, 2021-2022.



Data Source(s): Statistics Unit, MOH.

In comparison, there were **185 admissions for new strokes** (Cerebrovascular Accident or CVA) in 2022, 101 M (56%), and 84 F (45%). The peak incidence were in men aged 55-64y, a pattern similar to 2021, but more admissions for stroke were noted amongst women in older age group (65y+) in 2022 compared to 2021, see Figure 31. Many more people survive the immediate phase of stroke than die of it (see APR 2021), contributing to more people living with varying degrees of disability (see section 3.3.6 Functional Limitations).

Figure 31 Admissions for CVA by age and sex, 2021-2022.

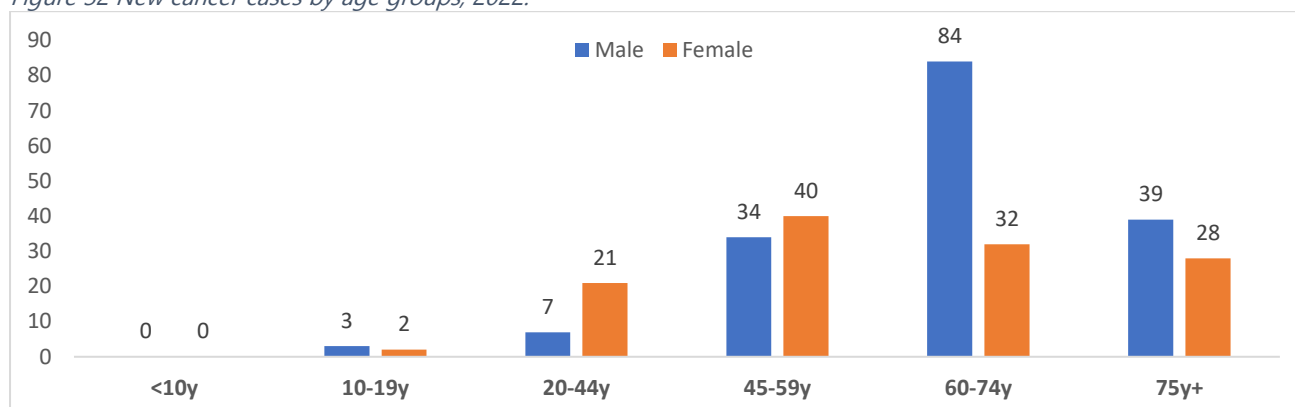


Data Source(s): Statistics Unit, MOH.

3.3.2 Cancers

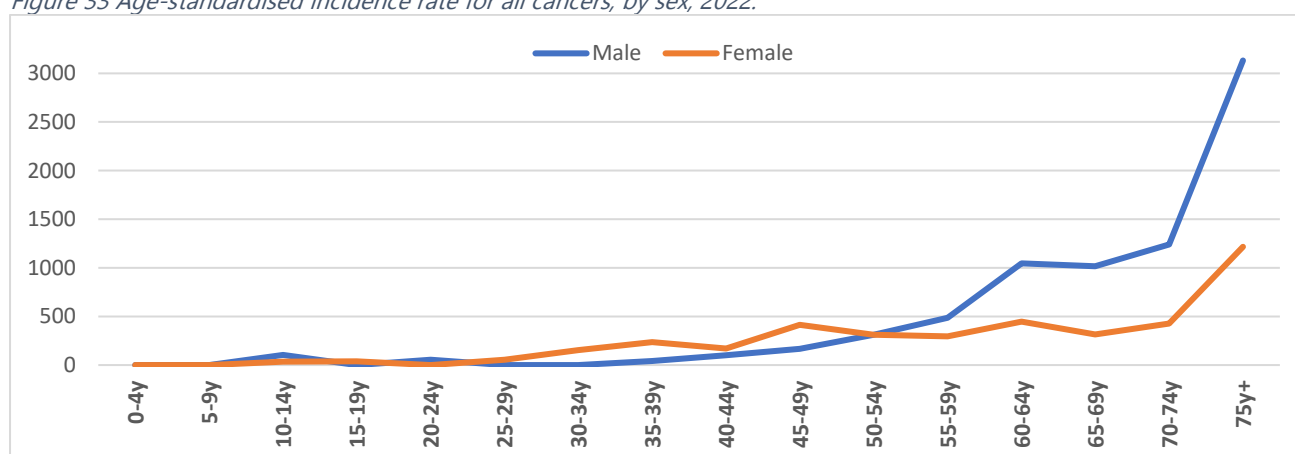
There were **290 new cases (167M, 123F) of cancer diagnosed in 2022**. Incidence of cancer is lower in women, and total new cases peak at younger ages in women (Figure 32). Overall age-standardised rate²⁹ of new cancers is 217.8 in males and 155.2 in females (Figure 33).

Figure 32 New cancer cases by age groups, 2022.



Data Source(s): Cancer Registry, PHA.

Figure 33 Age-standardised incidence rate for all cancers, by sex, 2022.



Note: 2021 Population Estimates (NBS 2022) used. Data Source(s): Cancer Registry, PHA.

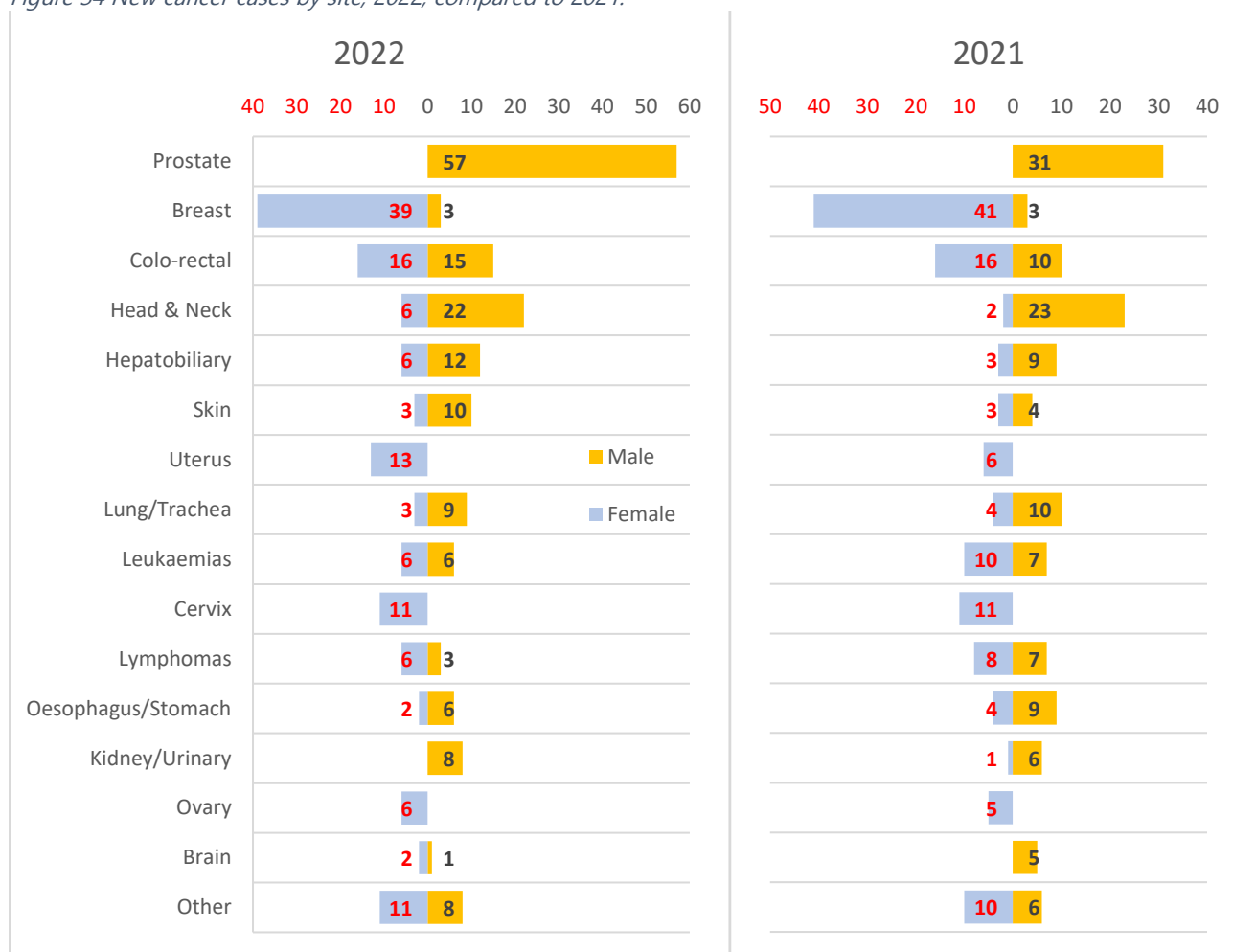
With reference to Figure 34, the following key patterns were notable in 2022:

- **Prostate cancers (57M) remained the most common cancer detected in men**, with a sharp increase (of 84%) from 2021 to 2022, with 90% of the cases detected after age 60.
- **Breast cancers were the most common cancer diagnosed in women (39F, 3 M)**, with cases being detected from age 35 onwards, and peaking in the 45-49y age group.
- **Colorectal cancers were reported in 15 men and 16 women.**
- **Head and neck cancers (includes cancers of the mouth, nose, throat, sinuses and voice box), were predominantly detected in men (22M, 6F)**, consistent with higher smoking rates in men).
- **Cervical cancer (11F, same as 2021)** remains a key source of morbidity and mortality amongst women in 2022. The MOH offers free HPV vaccination to all school girls to protect against cervical cancer, and free cervical cancer screening programmes are available at all clinics to detect changes even before cervical cancer develops.

²⁹ An age-specific rate is the incidence or mortality rate for a specified age group, in which the numerator and denominator refer to the same age group; it is expressed as the number of new cases or deaths per 100,000 population at risk (European Cancer Information System, link: <https://ecis.jrc.ec.europa.eu/info/glossary.html>).

- Cancers of the kidney, urinary system, liver, pancreas, biliary system, skin, lungs, trachea, oesophagus and stomach, predominantly affect men (associated with differential exposure to risk factors).

Figure 34 New cancer cases by site, 2022, compared to 2021.



Data Source(s): Cancer Registry, PHA.

3.3.3 Key Communicable Diseases

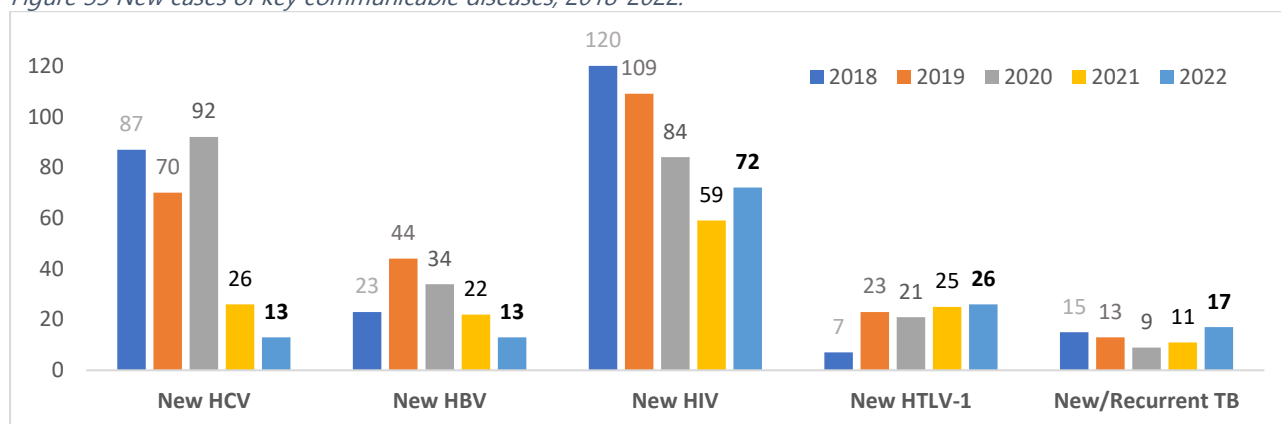
CDCU reported **72 new cases of HIV** (46M, 26F) in 2022, an increase over 59 detected in 2021 (43M, 16F). Of the 72, five had Hepatitis C co-infection. Amongst newly diagnosed cases, HIV transmission mode was reported to be **predominantly heterosexual**, 56% of new cases, followed by Men-who-have-sex-with men (MSMs) at 24%, then injection drug use (14%). HIV testing rates have improved – 19,418 HIV tests conducted in 2022, a 22% increase over 2021. Refer to Chapter 6 Progress towards NHSP and SDG3, for advancement towards UNAIDS targets.

In 2022, **13 new cases (8M, 5F) of Hepatitis C (HCV)** and **13 new cases (8M, 5F) of Hepatitis B (HBV)** were reported. Most new HBV infections were amongst non-citizens (5M, 3F). **Twenty six new cases (7M, 19F) of HTLV-1** (Human T-cell Lymphotropic Virus-1) were detected, of which 11 were amongst pregnant women.

CDCU reported **17 new cases (13M, 4F) of Tuberculosis (TB)** in 2022; 11 amongst Seychellois, and 5 amongst non-citizens. Among the TB cases, eight were smear-positive, five extra-pulmonary, one case was co-infected with HIV.

One TB death (male) was reported in 2022; and estimated **treatment coverage** for was reported as **87%**³⁰ (NHSP Target 100%). Figure 35 describes the trends in new cases of HCV, HBV, HIV, HTLV-1 and TB from 2018-2022. No cases of **Leprosy** were detected in 2022.

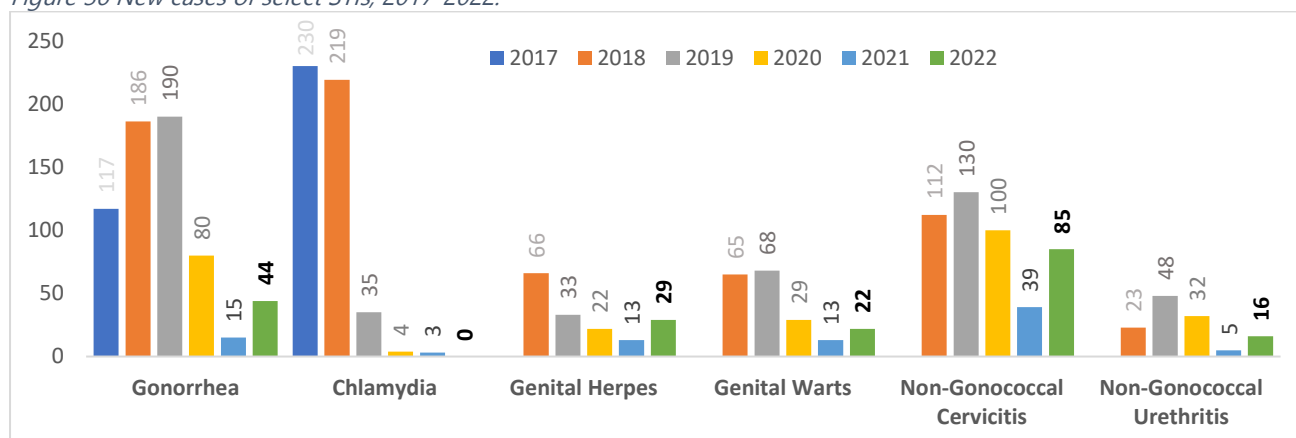
Figure 35 New cases of key communicable diseases, 2018-2022.



Note: Attendance to CDCU were lower since 2020; Data Source(s): CDCU Annual Report 2022 (PHA, 2023).

A total of **354 cases (115M, 239F) of other Sexually Transmitted Infections (STIs)** were reported in 2022. Forty-four cases of Gonorrhoea were detected at CDCU in 2022 (Figure 36). Over the past five years, 88% of Gonorrhoea and 73% of Chlamydia infections were detected amongst men. An **outbreak of Syphilis** also occurred in 2022 – this is covered under Chapter 4 Health Security.

Figure 36 New cases of select STIs, 2017-2022.



Note: Chlamydia test kit shortage since 2019. Data Source(s): CDCU Annual Report 2022 (PHA, 2023).

3.3.4 Intentional Self-Harm

In 2022, **83 cases (44M, 39F) of intentional self-harm (ISH)** were reported – a significant increase over 2021 and 2020 (Table 10). Age- and sex-distribution of reported ISH is depicted in Figure 37; it shows that reported cases span both sexes and both young and old.

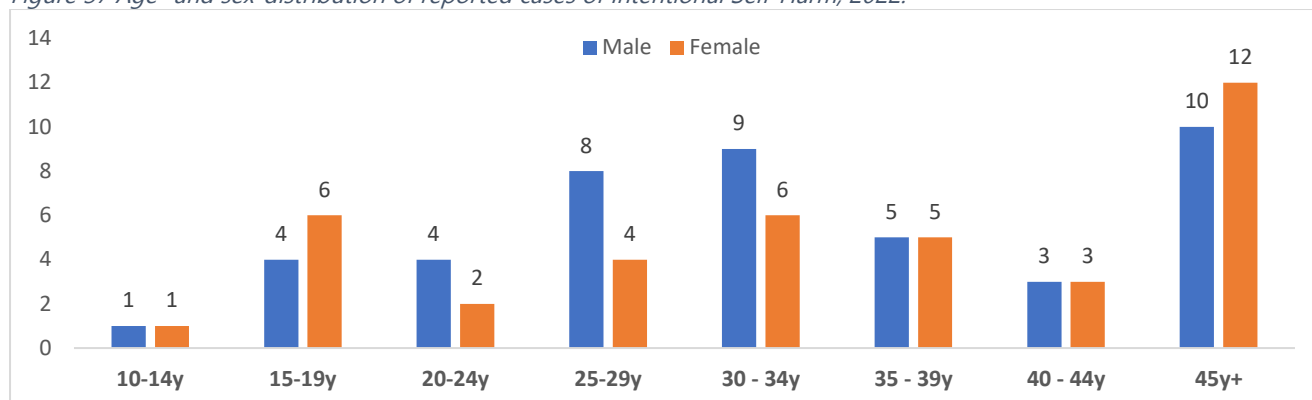
Table 10 Reported cases of ISH 2020 to 2022.

	2020	2021	2022
Total Reported Intentional Self-Harm	58 (21M, 37F)	55 (31M, 24F)	83 (44M, 39F)
Of which from self-poisoning	53 (16M, 37F)	41 (18M, 23F)	55 (25M, 30F)
Of which from hanging and/or strangulation	4 (all M)	12 (all M)	18 (13M, 5F)

Data Source(s): Statistics Unit, MOH.

³⁰ CDCU Communication to Global TB Programme of the WHO (12 Jul 2023).

Figure 37 Age- and sex-distribution of reported cases of Intentional Self-Harm, 2022.



Data Source(s): Statistics Unit, MOH.

The **majority of ISH reported are from self-poisoning with medications, drugs or other biological substances** – 55 in 2022 (25M, 30F). **Hangings and strangulations accounted for 18** (13M, 5F).

3.3.5 Unintentional Poisoning

Twenty-three cases of unintentional or accidental poisoning were reported in 2022; a little less than 2020 and 2021 (Table 11). The **majority (87%) were amongst children aged <15y**.

Table 11 Cases of unintentional poisoning reported, 2020-2022.

	2020	2021	2022
Unintentional Poisoning (Total)	28 (11 M, 17 F)	29 (16 M, 13 F)	23 (11 M, 12 F)
Of which are children aged <15y	26 (93%)	20 (69%)	20 (87%)

Data Source(s): Statistics Unit, MOH.

3.3.6 Functional Limitations

NPHC 2022 questions (NBS, 2023, unpublished) assessed on functional limitations (amongst people aged 5y+) across six domains (sight, hearing, mobility, concentration, care and communication) showed that **4.2% of population in age-range had at least one disability** (about 5,000 people).

Amongst these, 8.5% were aged 5-24y, 37.2% aged 25-64y, and 54.3%, 65y+. This means **almost 1,900 people were living with significant disability in their prime working-age** (25-64y). This has implications for adaptations across society for people with functional limitations.

The **most common domain of limitation was mobility** (walking or climbing steps), with 2,043 people living with mobility limitations. Of these, 2% are aged 5-14y, 35% 15-64y (working-age), and 64% aged 65y+.

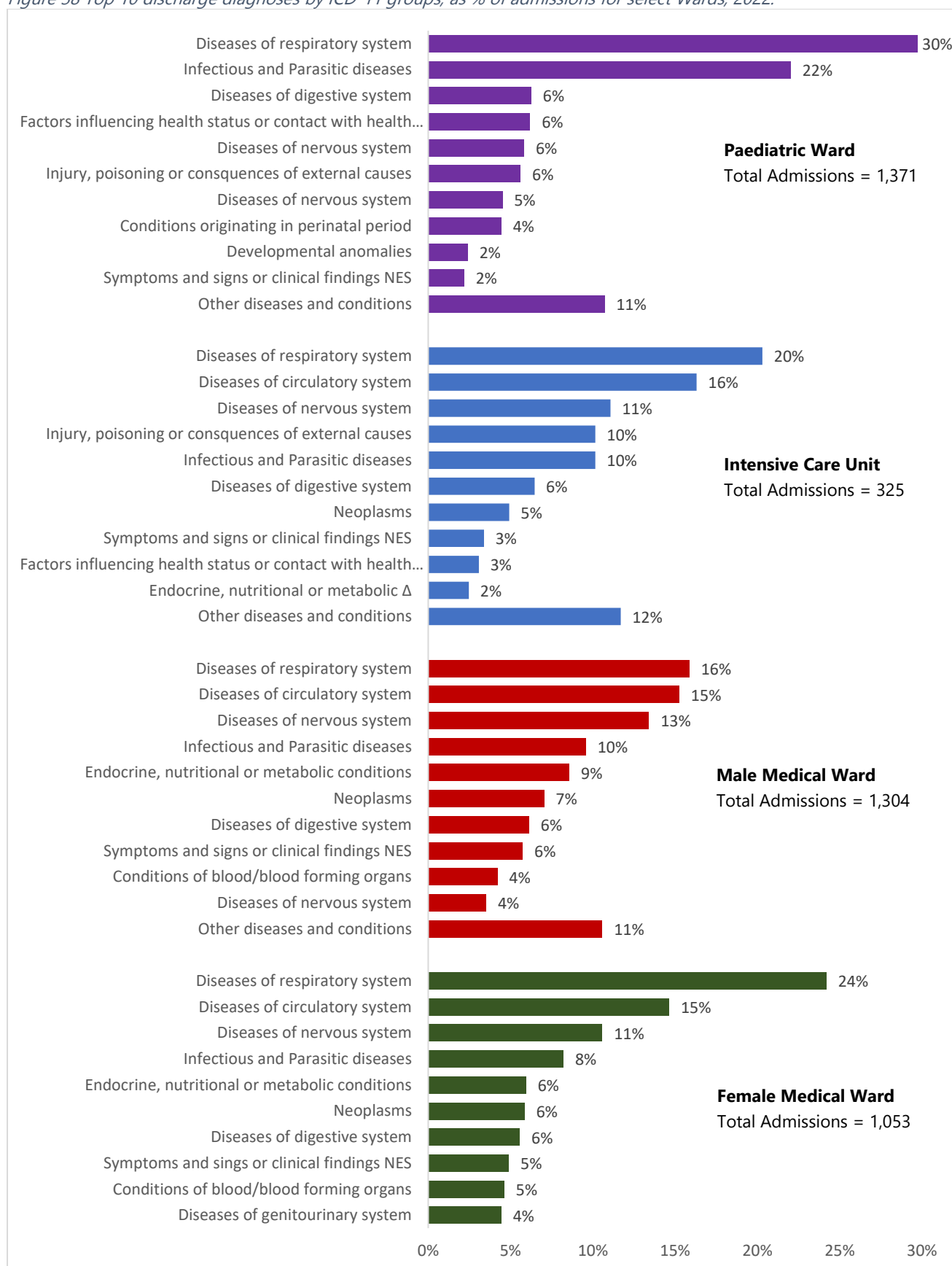
The NPHC 2022 did not include questions on cognitive function or dementia. In an ageing population, these are likely to become increasingly important, and represent an area of underestimated and unmet needs.

3.3.7 Inpatient Morbidity

The discharge diagnosis of patients admitted to male medical (MMW), female medical (FMW) and paediatric (PW) wards, and Intensive Care Unit (ICU) were analysed, and top 10 reasons for admission (as per ICD-11 broad diagnostic categories) for each ward are illustrated in Figure 38.

The patterns clearly show that **diseases of the respiratory system are, by far, the most common reason for admissions (and a major cause of death) across all four wards analysed**. This pattern is consistent across years and indicates the need for new interventions to prevent pneumonias and transmission of respiratory pathogens, reduce modifiable risk factors for severe pneumonia (e.g. obesity), and improve care of patients admitted with the condition.

Figure 38 Top 10 discharge diagnoses by ICD-11 groups, as % of admissions for select Wards, 2022.



NES – not elsewhere specified. **Note:** Strokes (Cerebrovascular Accidents) are classified under diseases of the neurological system in above categorisation (per ICD-11 convention). Data Source(s): Inpatient admissions database, Statistics Unit, MOH.

The second and third most common discharge diagnoses on adult wards (MMW, FMW and ICU) were cardiovascular diseases and diseases of the nervous system, respectively. In the PW, infectious and parasitic diseases and diseases of the digestive system (which includes gastroenteritis) ranked second and third.

An analysis of admissions to Psychiatric Ward in 2022 show a **pre-dominance of substance-use disorders** (49% of total) as the underlying reason for admission (Table 12), a pattern similar to 2021.

Table 12 Reasons for admission, Psychiatric Ward, 2022

Reason for Admission	Total	%
Mental & behavioural disorders associated with Alcohol-use	37	16%
Mental & behavioural disorders associated with Opioid-use	9	4%
Mental & behavioural disorders associated with Cannabis-use	42	18%
Mental & behavioural disorders associated with Multiple Drug-use	26	11%
Schizophrenia	32	14%
Delusions/Psychosis	35	15%
Mood disorders	28	12%
Others	19	8%
Total	228	-

Data Source(s): Mental Health Services Annual Report 2022 (HCA, 2023) and Annual Hospital Services Report 2022 (MOH, 2023).

Male surgical ward (D'Offay) admissions in 2022 show that majority of diagnoses were coded as symptoms, signs, or for conduct of certain procedures. The underlying reasons for admissions were not well captured.

Admissions data for female surgical (Hermitte) ward, Maternity, and Family Hospital Isolation, and morbidity data from primary care and specialist clinics, were not analysed at time of preparation of this report. As such, patterns of morbidity in these inpatient and outpatient settings are not described in this report. For occupancy-related information for all wards, please see section 5.4 Service Delivery.

3.4 Risk Factors for Health

Year 2022 marked seven years of implementation of the **Seychelles Strategy for Prevention and Control of NCDs 2016-2025** (MOH, 2016), which sets out targets for obesity, tobacco, alcohol and physical activity as follows (levels in 2025 compared to 2010):

- 10% reduction in prevalence of alcohol use.
- 10% reduction in prevalence of insufficient physical activity.
- 30% reduction in prevalence of tobacco use.
- No increase in prevalence of obesity.

The School Screening Programme Report³¹ for 2022 showed updated trends in overweight and obesity in 9-15y olds, and estimates of risk-behaviours (self-reported): tobacco use, alcohol use and substance use. This survey was co-led by the Unit for Prevention and Control of CVDs (UPCCD) and School Health Programme, with support from Unisanté, University Centre for Primary Care and Public Health (Lausanne, Switzerland).

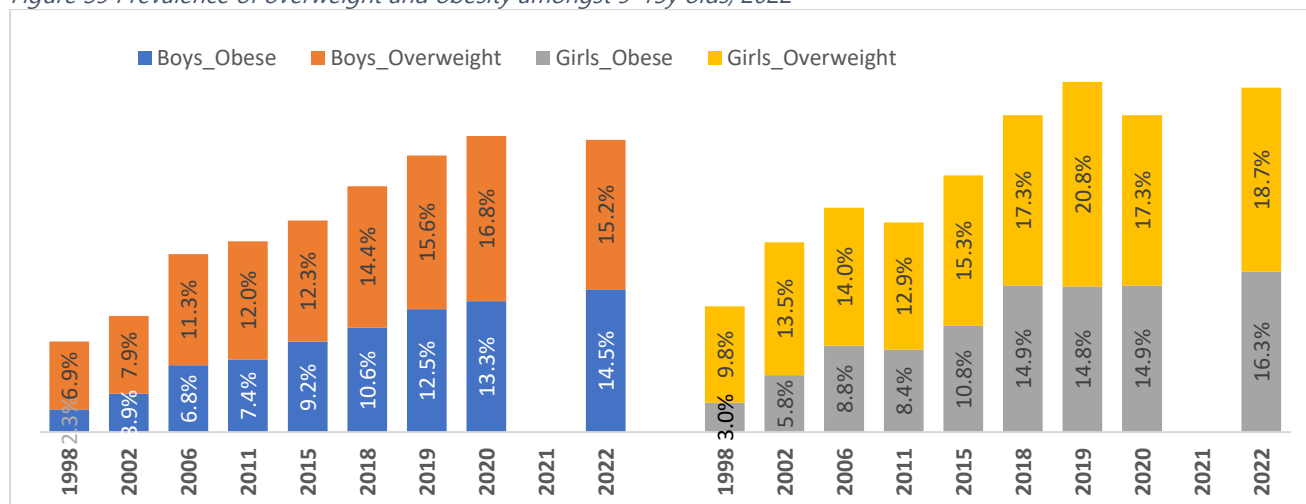
The survey could not be conducted in 2021 in view of COVID-19 school service interruptions. Overall participation rate in 2022 was higher (79% of targeted schoolchildren), compared to 2020 (73.5%).

³¹ P Bovet, G Mangroo, B Viswanathan, S Charlette (2023): School screening programme: prevalence of overweight and obesity and health behaviours in children and adolescents aged 9 to 15 years in Seychelles in 2022.

Overweight and Obesity in School children

The report shows that **just over one in three girls (i.e. 35%) and just under one in three boys (i.e. 30%) are overweight or obese**; similar to previous available data from 2020. However, as illustrated in Figure 39, there has been a steady increasing trend from 1998, when the first survey was carried out.

Figure 39 Prevalence of overweight and obesity amongst 9-15y olds, 2022



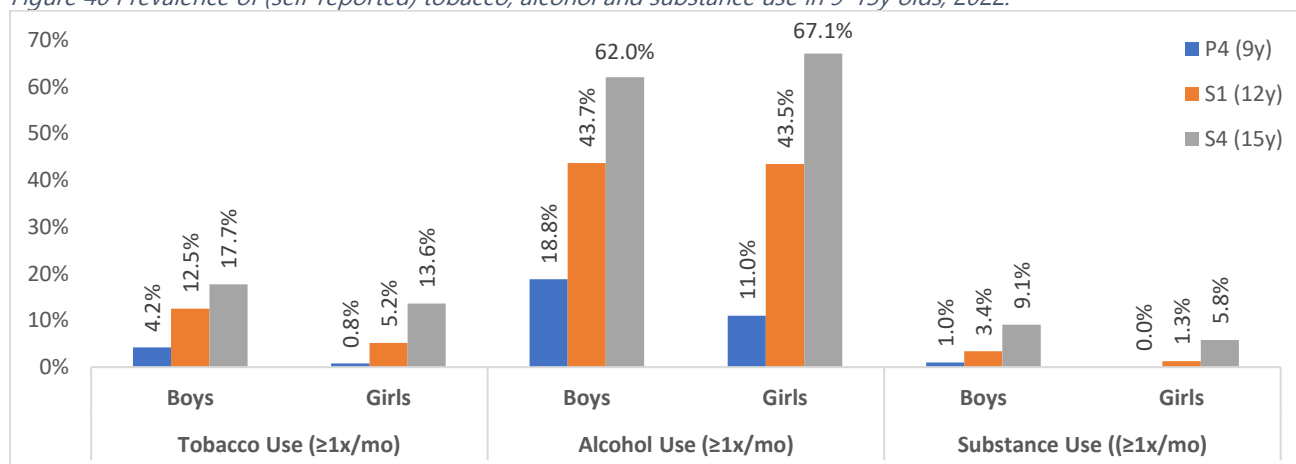
Note: Screening was not conducted in 2021. Data Source(s): School Health Survey (UPCCD, School Health Programme, 2023).

Substance use amongst school children

Alcohol use was high in the school health screening, even amongst 9y olds. Self-reported alcohol use (at least once in preceding month) was 18.8% in boys at P4, increasing to 62% at S4, and 11% in girls at P4, increasing to 67.1% in S4 (Figure 40).

Prevalence of self-reported tobacco use (at least once in preceding month) rose from 4.2% in P4 to 17.7% in S4 amongst boys, and from 0.8% to 13.6% in girls. **Substance use (self-reported)** amongst boys increased from 1% in P4 to 9.1% in S4 amongst boys, and from 0% to 5.8% amongst girls (Figure 40).

Figure 40 Prevalence of (self-reported) tobacco, alcohol and substance use in 9-15y olds, 2022.



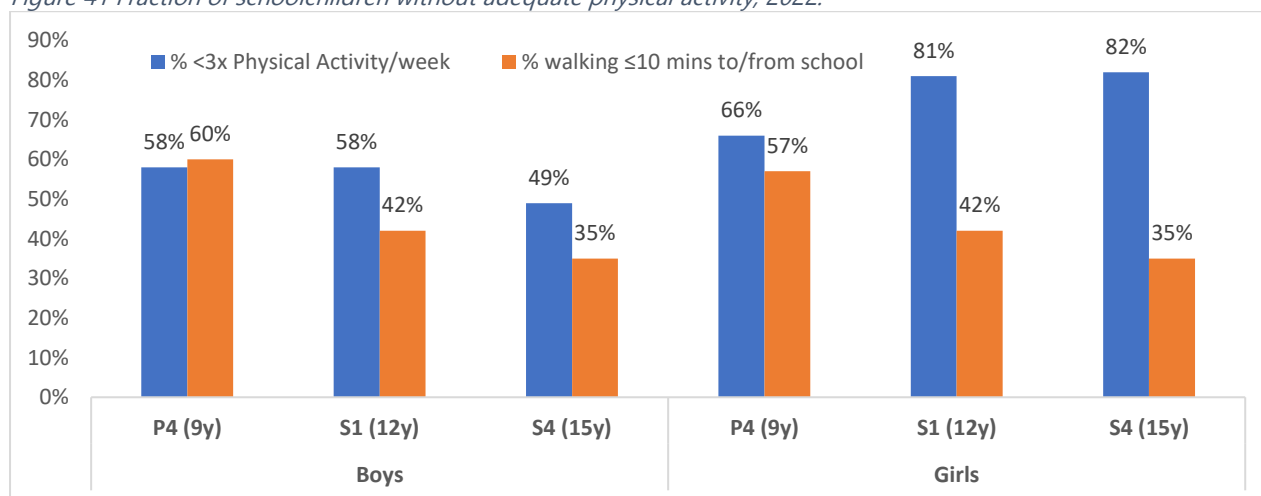
Data Source(s): School Health Survey (UPCCD, School Health Programme, 2023).

Physical activity amongst school children

Two questions were asked to assess physical activity – walking time to and from schools and number of times the child undertakes physical activity during leisure time per week/month. The fraction of children **not** walking at least 10 minutes to get to school and return home was similar in both boys and girls, and reduced with age.

The fraction of children **not** undertaking physical activity at least thrice a week during leisure time was higher amongst girls, increased with age in girls, and reduced with boys. An estimated four in five girls do not undertake physical activity at least thrice a week at S4, compared to only half of boys (Figure 41).

Figure 41 Fraction of schoolchildren without adequate physical activity, 2022.



Data Source(s): School Health Survey (UPCCD, School Health Programme, 2023).

The report also outlines key recommendations for action. The annual school screening provides a key mechanism to track obesity and overweight amongst children. The Institute for Early Childhood Development (IECD), Child Health Programme and Nutrition Unit conduct relevant surveys and/or analysis of routine programmatic data for younger children. Also see section 2.3.4 Seychelles Child Development Study.

Breastfeeding of infants

The NPHC 2022 (NBS, 2023, unpublished) results show the age at which infants were weaned off breastfeeding: **30% stop breastfeeding within 2 months of life, another 33% between 3 and 5 months**, 20% between 6-8 months, and 17% after 9 or more months of life. Adequate breastfeeding is protective against a wide range of illnesses, including NCDs; the global health recommendation is for exclusive breastfeeding for first six months of life, with mixed feeding to age 2 years³².

3.5 Key points on Health Status

Annual registered births were unchanged, but deaths remained high in 2021 and 2022, leading to both **excess deaths** and a significant fraction of premature deaths. The **year-on-year reduction in life expectancy for men from 2020 to 2021, and from there to 2022, is cause for concern**, and indicates need for more efforts to protect the health of men.

The available data and trends on the state of health of the population show that **NCDs remain the most important burden of disease**, with cardiovascular disease, diabetes, stroke and venous thromboembolism being important disease conditions. **COVID-19 was a key infectious disease burden in Q1-Q2 of 2022. HIV deaths were high** in 2021 and 2022. **Pneumonia** deaths remained a key disease burden in 2022. An **increase in deaths from external causes**, in particular due to RTAs and suicides, was a new trend in 2022, as the country recovers from the impact of the COVID-19 pandemic.

Main **risk factors for NCDs and many communicable diseases start since childhood**, as shown by the school health screening results. The complexities of causal chains of obesity and substance abuse require that the MOH reaches beyond the health sector and drives coordinated actions to tackle the root causes.

³² WHO Breastfeeding: Overview; link: https://www.who.int/health-topics/breastfeeding#tab=tab_1.

4 Health Security

WRITER(S): **MEGGY LOUANGE**; **SANJEEV PUGAZHENDHI**;

The COVID-19 pandemic has been ongoing in the world for the past three years and Seychelles has not been spared. The first case of COVID-19 was reported in Seychelles on 14th March 2020, since then, we have been affected in varying degrees over the course of the pandemic with distinct waves of varying amplitudes.

The NHSP 2022-2026 was developed during the pandemic and covered action for health emergencies primarily under SD3 Improve Health Security. The **key priorities** outlined are as follows:

- Strengthen the One Health approach³³.
- Implement the IHR³⁴ to prevent and swiftly detect outbreaks.
- Implement Integrated Disease Surveillance and Response (IDSR).
- Develop and strengthen regulatory function of the PHA.
- Improve public health standards and legislative documents.
- Strengthen public health law enforcement.
- Strengthen public health programmes.

Year 2022 also saw outbreaks of Syphilis, Ciguatera fish poisoning, and the global threat of spread of Mpox. The health sector, under the leadership of the PHA, strived to prepare, prevent and respond effectively to these new and re-emerging challenges.

4.1 Key Disease Outbreaks

4.1.1 COVID-19 Response

The **PHEOC (Public Health Emergency Operations Committee)** response coordination mechanism for the health sector response remained active until May 2022, where after the committee was inactive and roles were devolved to the **IDSR committee**, and relevant teams under PHA and HCA (contributing towards COVID-19 response institutionalisation, NHSP objective 2.4.2). The response continued to be supported by the WHO, country and regional offices.

Throughout the pandemic the **Surveillance and Epidemic Intelligence Section (SEIS)** of the PHA has been heavily involved with all activities pertaining to COVID-19 such as data management, case investigation, contact listing, contact tracing, contact follow up along with other duties pertaining to mitigation of the pandemic. The devolution of functions and reduced societal emphasis on controlling COVID-19 in 2022, reduced availability of resources for the response, with implications on effectiveness.

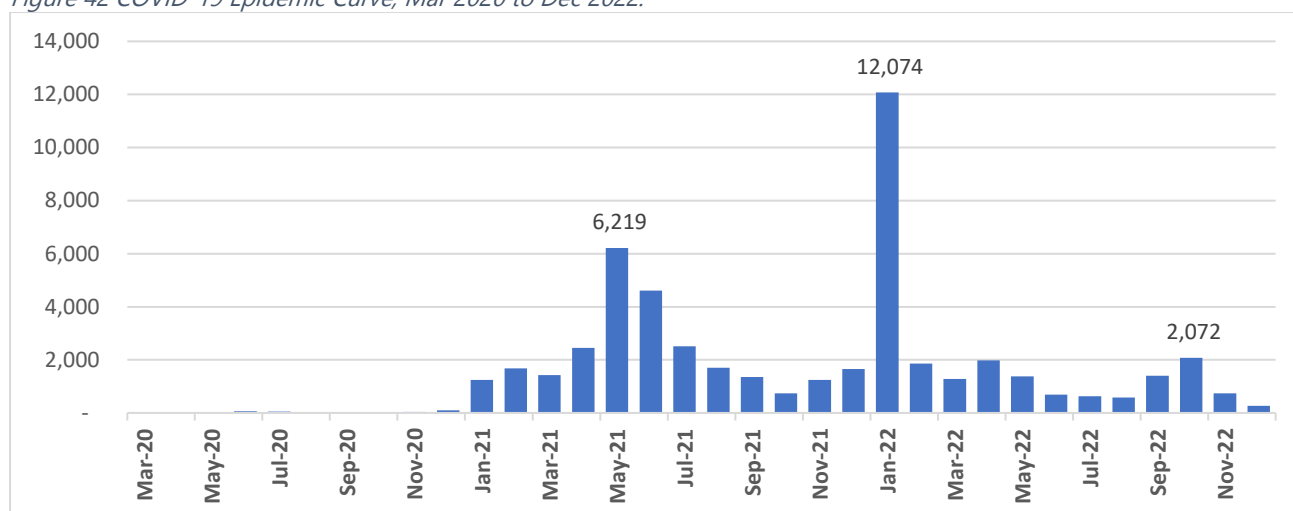
According to latest consolidated figures, a **cumulative of 52,359 cases of COVID-19 by the end of 2022**. The third year of the COVID-19 pandemic started with a **new surge of cases (Omicron variant), with over 12,000 infections confirmed in January 2022** alone (Figure 42). New cases over the rest of the year remained well below this peak. After an interlude, a much smaller surge was noted around September-October 2022.

Public health and social measures continued to be in-effect in 2022, although significantly less stringent than 2021; and the **COVID-19 vaccination campaign** continued through 2022, but with lower uptake (see COVID-19 Vaccination). Coverage of effective **contact tracing** amongst known cases reduced to 21% in 2022, from 80% in 2021, and almost 100% in 2020.

³³ WHO One Health; link: <https://www.who.int/news-room/questions-and-answers/item/one-health>.

³⁴ WHO International Health Regulations; link: https://www.who.int/health-topics/international-health-regulations#tab=tab_1.

Figure 42 COVID-19 Epidemic Curve, Mar 2020 to Dec 2022.



Data Source(s): Surveillance and Epidemic Intelligence Section Report (PHA, 2023).

Testing for COVID-19 was primarily through rapid antigen tests, but not all tests carried out were captured through routine tracking systems. The number of COVID-19 PCR tests conducted in 2022 were 75,447, a 27% reduction over 2021 (102,955)³⁵. **Genomic surveillance** of 2,295 samples were conducted in 2022, both at the SPHL (mutation panel screening and sequencing) and overseas (genomic surveillance at KEMRI³⁶, with support of WHO and Africa CDC); of these 1,962 (85.5%) were **Omicron** variants, followed by 202 (8.8%) Delta variant.

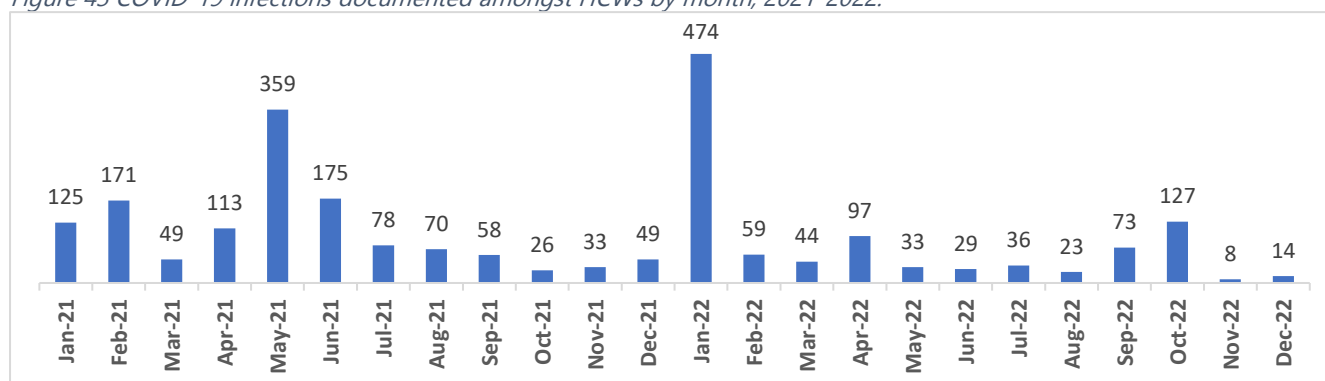
For **COVID-19-related deaths**, see section on COVID-19 deaths.

Training of HCWs in Infection Prevention and Control (IPC) and public sensitisation through mass and social media continued in 2022, although at a smaller scale. There were no stock outs of **essential COVID-19 response supplies**, and interruptions to essential health services were minimal. Majority of **staff involved in the pandemic response were re-deployed back to their pre-pandemic service areas**, with gradual resumption of routine healthcare services and public health functions.

Surveillance of COVID-19 infection amongst Healthcare Workers

The IPC Unit, in collaboration with DSRU and the office of the CMO, coordinated **screening of HCWs for COVID-19**. Figure 43 shows documented COVID-19 infections amongst HCWs by month, 2021-2022.

Figure 43 COVID-19 infections documented amongst HCWs by month, 2021-2022.



Data Source(s): IPC Unit Annual Report 2022 (HCA, 2023).

³⁵ SPHL Annual Report 2022 (PHA, 2023).

³⁶ Kenya Medical Research Institute (Link: <https://www.kemri.go.ke/>)

A cumulative **2,331 infections** recorded, with peaks in HCW infections coinciding with COVID-19 surges in the general population (May 2021 and January 2022), and affecting health workforce availability. It must be noted that adherence to COVID-19 surveillance guidelines amongst HCWs was incomplete throughout the COVID-19 pandemic, as such the numbers are likely underestimates, but can still allow certain inferences to be made.

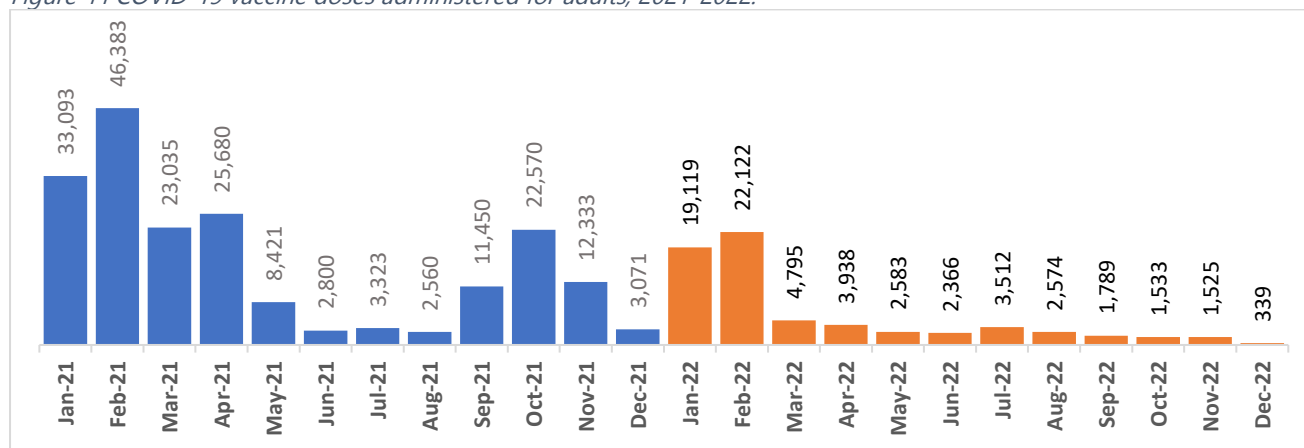
COVID-19 Vaccination

COVID-19 Vaccination Campaign continued throughout 2022, with **administration of 66,195 doses: 64,595 doses to adults** (32,274 Sinopharm, 32,321 Pfizer) and **1,600 doses to adolescents** (Pfizer only). The cumulative total of doses administered under the campaign from its start on 10th January 2021 was 260,914 doses (248,507 in adults and 12,407 in adolescents) by end 2022.

Doses of COVID-19 vaccine administered by month are shown in Figure 44, and cumulative doses administered by dose-sequence in Figure 45. Most doses in 2022 were administered in January and February, coincident with the Omicron surge.

In 2022, **vaccine uptake was primarily amongst adults seeking Dose 3 and 4** of their vaccine. Uptake other doses amongst adults and uptake of any doses by adolescents was low. This is well-below rates seen in 2021, but understandable in view of the high vaccine coverage and exposure to natural infection by end-2021.

Figure 44 COVID-19 vaccine doses administered for adults, 2021-2022.



Data Source(s): EPI Programme (HCA) and Statistics Unit (MOH).

COVID-19 vaccine coverage amongst adults for dose 1 and 2 has surpassed NHSP target of >80% of the target population. Amongst **adolescents, dose 1 coverage is 78.5%, dose 2, 71.9%, still below NHSP target of >80%.**

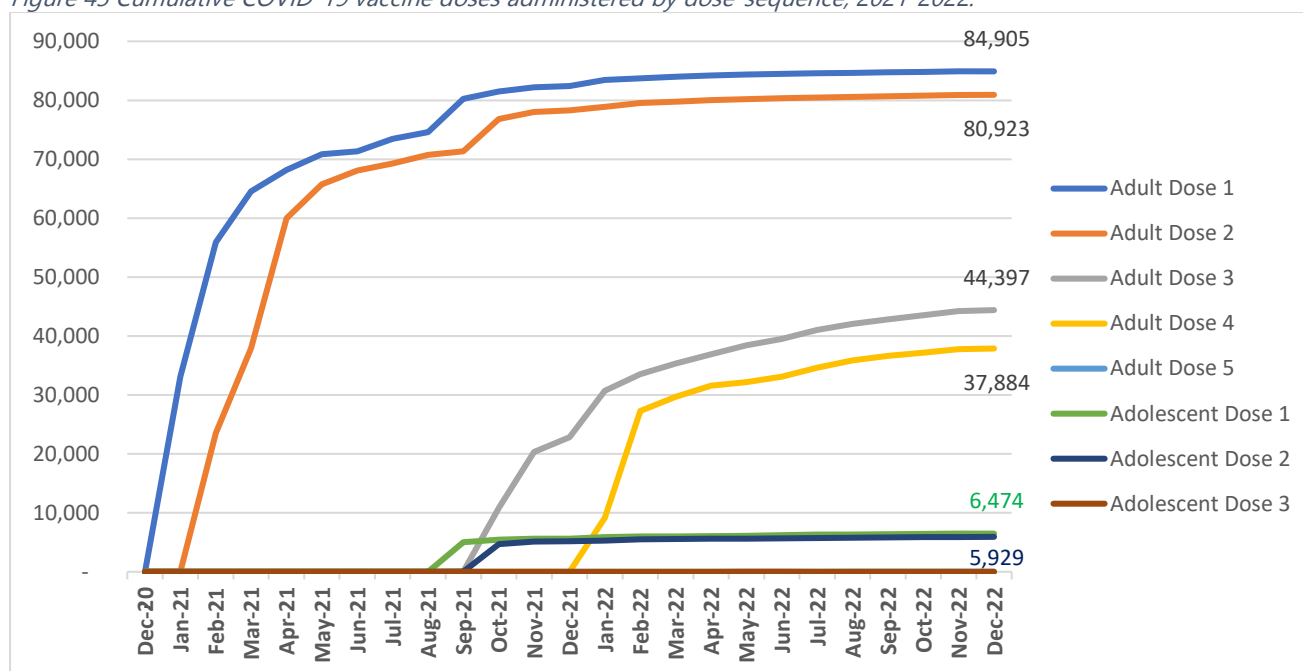
Table 13 summarises vaccine uptake by educational settings (target group for vaccination of adolescents aged 11-17y) as at end 2022. Uptake in private schools were higher than in public schools, and lowest amongst post-secondary institutions.

The 2022 census³⁷ **estimated 12,932 residents aged 12y+ were unvaccinated (16.7% of people in age-range)** as at June 2022, with highest fraction unvaccinated in the Central (28.7%) and East-South (25.6%) regions of Mahé, and lowest on Praslin (9.5%) and La Digue (4.2%).

The EPI programme issued 17,706 **electronic COVID-19 vaccination certificates** in 2022.

³⁷ Presentation of findings of National Population and Housing Census 2022 (NBS, May 2023).

Figure 45 Cumulative COVID-19 vaccine doses administered by dose-sequence, 2021-2022.



Data Source(s): EPI Programme (HCA) and Statistics Unit (MOH).

Table 13 COVID-19 Vaccine doses administered to adolescents aged 12-17, 2022.

Educational Setting	Eligible	Dose 1	%	Dose 2	%
State Secondary Schools, including special schools ³⁸	6,382	4,899	77%	4,453	70%
Private Secondary Schools	920	603	66%	763	83%
Tertiary Education Institutions	945	840	89%	566	60%
Administered at Regional Health Centres	-	132	-	147	-
Total	8,247	6,474	78.5%	5,929	71.9%

Data Source(s): EPI Programme (HCA) and Statistics Unit (MOH).

4.1.2 Syphilis

Seventy-one cases (45 M or 63%, 26 F or 37%) of Syphilis were reported in 2022, with majority of cases being detected in the 15-39y age group, where both males and females were affected. In the 40y+ age-range, almost all reported cases were in males. This count includes **one case of Congenital Syphilis** (infected in-utero or at birth) that was reported by Maternity Services and CDCU.

By district, the highest number of cases reported, both absolute numbers and relative to population, was in Grand Anse Mahé (8 cases), followed by Cascade (6 cases) and Saint Louis (5 cases)³⁹.

This represents a **twelve-fold increase in reported cases of Syphilis in 2022, compared to 2021**. Between 2009 and 2014, an average of 11 cases of Syphilis were reported annually, from 2015 to 2021, this reduced to three cases, followed by a spike in 2022. Similar increases were noted in other countries across the World⁴⁰, including in congenital Syphilis⁴¹.

³⁸ School for the Exceptional Child, School for the Deaf;

³⁹ Surveillance Report: Syphilis surge of cases in Seychelles (SEIS, Feb 2023).

⁴⁰ UK Health Security Agency Press release (6 Jun 2023): Gonorrhoea and syphilis at record levels in 2022; Link: <https://www.gov.uk/government/news/gonorrhoea-and-syphilis-at-record-levels-in-2022>.

⁴¹ CDC (11 Apr 2023): U.S. STI Epidemic Showed No Signs of Slowing in 2021 – Cases Continued to Escalate; Link: <https://www.cdc.gov/media/releases/2023/s0411-sti.html>.

4.1.3 Mpox (Monkeypox)

WHO declared a multi-country outbreak of Mpox a Public Health Emergency of International Concern⁴². Surveillance for **Mpox** (previously called Monkeypox) was strengthened in Seychelles following the announcement. Mpox preparedness and response plan and case management guidelines were developed (included case definitions, sample collection and transport, and case management); and public education materials developed and disseminated. Eight samples from suspect cases were sent overseas for testing through SPHL, **all returned negative**⁴³.

4.1.4 Ciguatera Poisoning

An outbreak of Ciguatera food poisoning was recorded in July 2022⁴⁴, with 13 individuals affected from across six households on Mahé. Key symptoms were gastrointestinal and neurologic, and started after consumption of salted fish of the species locally known as Varvara (two-spotted red snapper). Samples collected from one household and two vendors were sent overseas for testing, which confirmed Ciguatoxin-like toxicity in all the samples, with levels well-above limits set by the European Food and Safety Authority and Food and Drug Administration of the USA.

The source of the fish, and the location of the catch, were not confirmed, but was believed to be possibly outside Seychelles' territorial waters. The PHA issued a national alert, advising consumers of the outbreak and requesting retailers to withdraw the product from the market. The alert was withdrawn in late November 2022, with resumption in sale of the fish species.

4.2 Key Public Health Functions

4.2.1 Public Health Services

A team of public health officers under the PHA ensure that relevant standards, laws and regulations are followed across a range of areas, to reduce health and safety risk (NHSP objectives 3.4 and 3.5).

Port health inspected all sea vessels and airplanes for public health risks, including sick passengers/crew and invasive species. As the travel and tourism sector recovered, there was a significant increase in airplanes screened in 2022 (3,332), compared to 2021 (1,955).

Of the **354,928 travellers reported to have been screened** by public health at airports in 2022, only one passenger was identified as ill. This implies a very low case-detection rate through airport screening, and suggests the need for a review of the efficiency and effectiveness of the current approach to screening. In contrast, 17 sick passengers or crew were reported through sea port screening. Quarantine was imposed on a total of six sea vessels in 2022, a reduction from 82 in 2021, when COVID-19 was more prevalent.

Vector control activities in 2022 focused on containing the ongoing dengue epidemic through outbreak investigations and fogging, but activities in 2022 remained below 2021 levels. Inspection of housing conditions, and water and sanitation systems have resumed in 2022, with 10,452 visits conducted (compared to 254 in 2021).

The Public Health Engineering Unit oversees waste management and ensures adequate sanitation standards. Of **8,729 inspections** of sites, 138 (1.6%) were deemed unsatisfactory.

⁴² WHO (2022) Second meeting of the International Health Regulations (2005) (IHR) Emergency Committee regarding the multi-country outbreak of monkeypox; Link: [https://www.who.int/news/item/23-07-2022-second-meeting-of-the-international-health-regulations-\(2005\)-\(ihr\)-emergency-committee-regarding-the-multi-country-outbreak-of-monkeypox](https://www.who.int/news/item/23-07-2022-second-meeting-of-the-international-health-regulations-(2005)-(ihr)-emergency-committee-regarding-the-multi-country-outbreak-of-monkeypox).

⁴³ SPHL Annual Report 2022 (PHA, 2023).

⁴⁴ Direct communication from Public Health Services and SPHL (PHA, 2023).

The country completed the **2022 GLAAS (UN-Water Global Analysis and Assessment of Sanitation and Drinking-Water) survey** and contributed towards the biannual UN report⁴⁵. Seychelles was on-track to meet its SDG targets for WASH, and estimated an expenditure per-capita on WASH of 0.57% of GDP or USD 65.

Chemical and bacteriological analysis of drinking water are conducted through the SPHL based on referrals, tests conducted and proportion unsatisfactory are summarised for bottled and drinking water in Table 14 (also see 4.2.3 Public Health Diagnostic Services).

Table 14 Drinking and bottled water analyses conducted in 2022.

	Total	Unsatisfactory	% Unsatisfactory
Bacteriological Analysis			
Bottled Water	428	110	25.7%
Treated Drinking Water	285	70	24.6%
Chemical Analysis			
Bottled Water	8	0	0%
Treated Drinking Water	10	1	10%

Data Source(s): Public Health Services (PHA, 2023).

Over 2,000 visits to inspect food establishments were conducted in 2022, for 1,586 sites. The most visited were food retail outlets. Of these, in 139 visits conditions were deemed unsatisfactory (6.9% of total visits), with issuance of warning letters (116), notices (49) and/or closures (48).

Six-hundred-sixty complaints were received by Public Health Services in 2022; after investigation 109 (16.5%) were reported as unjustified or malicious. The most common reason for complaint was defective wastewater disposal (158, 24%).

4.2.2 Surveillance and Epidemic Intelligence

The Surveillance and Epidemic Intelligence Section (SEIS) worked towards gradual restoration of routine functions and **sentinel surveillance** systems in the latter half of 2022 (NHSP objective 3.3). In 2022, the overall timely notifiable disease reporting was only 29% (580 out of 2,028 reports due from government facilities); therefore, the available information is not a true representation of the overall pattern for the country.

Thirty-two cases of Leptospirosis and 213 cases of Dengue were reported in 2022, far lower than pre-pandemic patterns⁴⁶, but conclusions on patterns of local transmission cannot be made from this.

Surveillance of **anti-microbial resistance patterns** (NHSP objectives 2.2.1 and 3.1) amongst cultured organisms from 2017-2022 (Table 15) show that *Klebsiella pneumoniae* and *Acinetobacter baumannii* are the most commonly isolated multi-drug resistant organisms in Seychelles (MDROs). Extended spectrum beta-lactam resistance is the most common type of resistance identified, followed by vancomycin resistance (Figure 46). Resistance patterns of the top six most commonly isolated MDROs are shown in Figure 47.

Table 15 Commonly isolated multi-drug resistant organisms, 2017-2022.

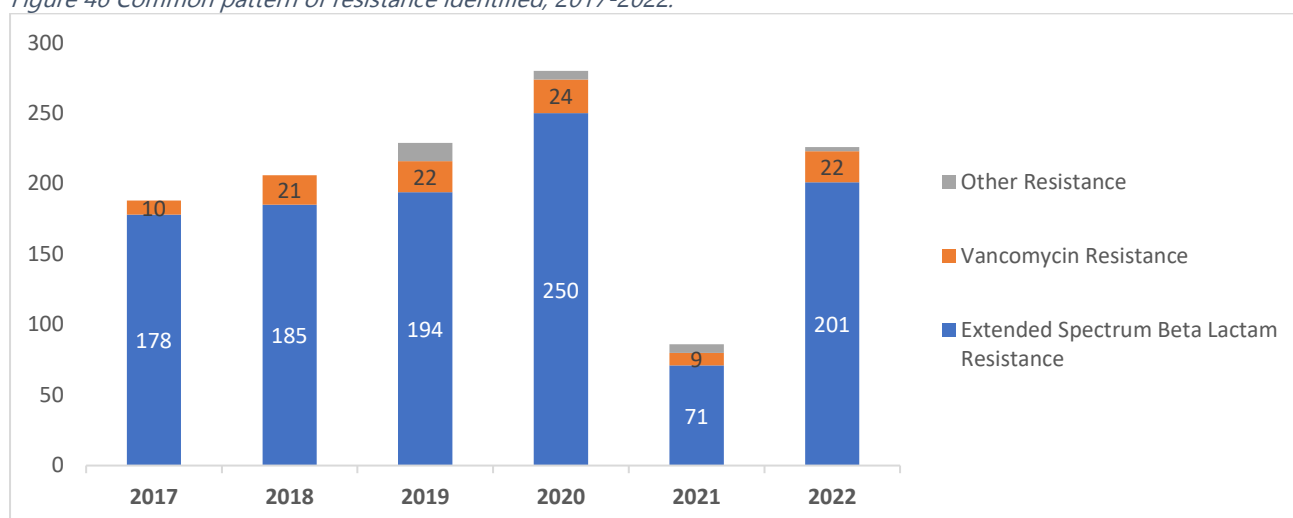
Rank	Species	# Isolated	Rank	Species	# Isolated
1	<i>Klebsiella pneumoniae</i>	524	6	<i>Burkholderia cepacia</i>	28
2	<i>Acinetobacter baumannii</i>	411	7	<i>Proteus mirabilis</i>	14
3	<i>Streptococcus Group D</i>	110	8	<i>Enterobacter cloacae</i>	10
4	<i>Escherichia coli</i>	66	-	Other	26
5	<i>Pseudomonas species</i>	46	-	Total	1,235

Data Source(s): Surveillance and Epidemic Intelligence Section Report (PHA, 2023).

⁴⁵ Strong systems and sound investments: evidence on and key insights into accelerating progress on sanitation, drinking-water and hygiene. UN-Water global analysis and assessment of sanitation and drinking-water (GLAAS) 2022 report. Geneva: World Health Organization; 2022. Licence: CC BY-NC-SA 3.0 IGO.

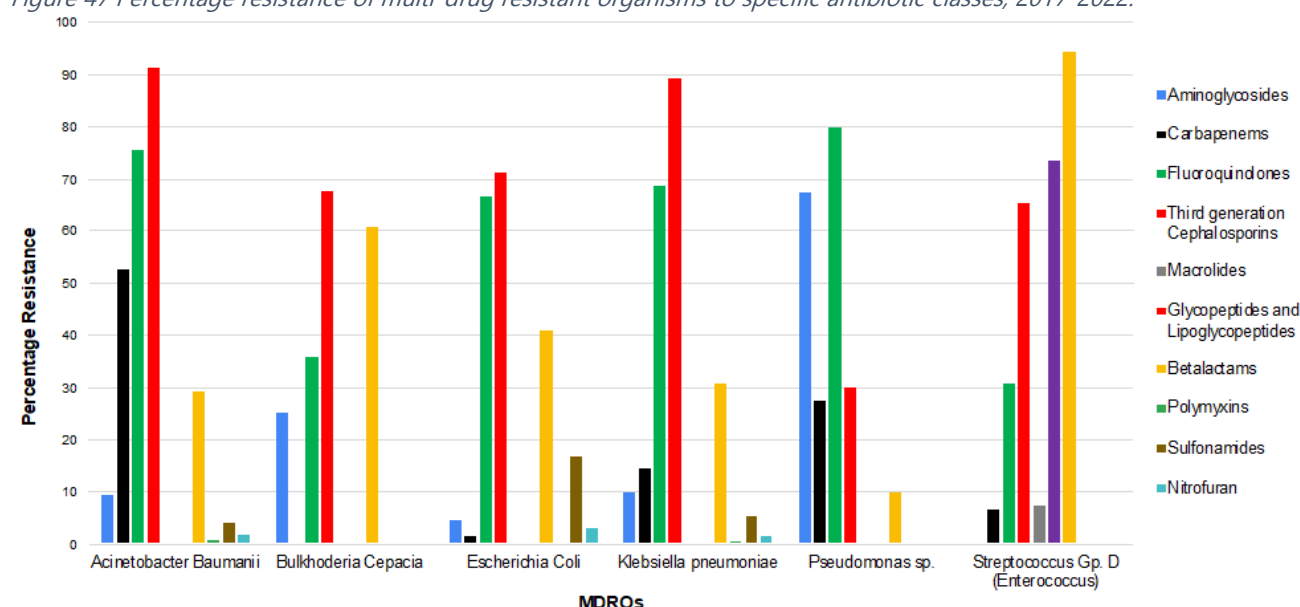
⁴⁶ See Annual Health Sector Performance Report 2017 and 2018 (MOH, 2018-2019).

Figure 46 Common pattern of resistance identified, 2017-2022.



Data Source(s): Surveillance and Epidemic Intelligence Section Report (PHA, 2023).

Figure 47 Percentage resistance of multi-drug resistant organisms to specific antibiotic classes, 2017-2022.



Source: Surveillance and Epidemic Intelligence Section Report (PHA, 2023).

Antimicrobial sensitivity testing of Gonococcal isolates are conducted at SPHL, although fewer tests were conducted from 2020 onwards (Table 16). The 2019-2022 cumulative data shows 59% of samples were resistant to Ciprofloxacin, 20% to Ceftriaxone and 25% to Cefixime; it also hints at increasing resistance to Ceftriaxone amongst Gonococcal isolates (NHSP objective 2.2.1).

Table 16 Antibiotic sensitivity tests conducted on Gonococcal isolates, with resistance patterns, 2019-2022.

	# Tests	Ciprofloxacin Resistant		Ceftriaxone Resistant		Cefixime Resistant	
		#	%	#	%	#	%
2019	167	106	63%	27	16%	42	25%
2020	66	41	62%	11	17%	14	21%
2021	7	0	0%	3	43%	0	0%
2022	35	14	40%	13	37%	14	40%
Total	275	161	59%	54	20%	70	25%

Data Source(s): SPHL Annual Report, 2022 (PHA, 2023).

4.2.3 Public Health Diagnostic Services

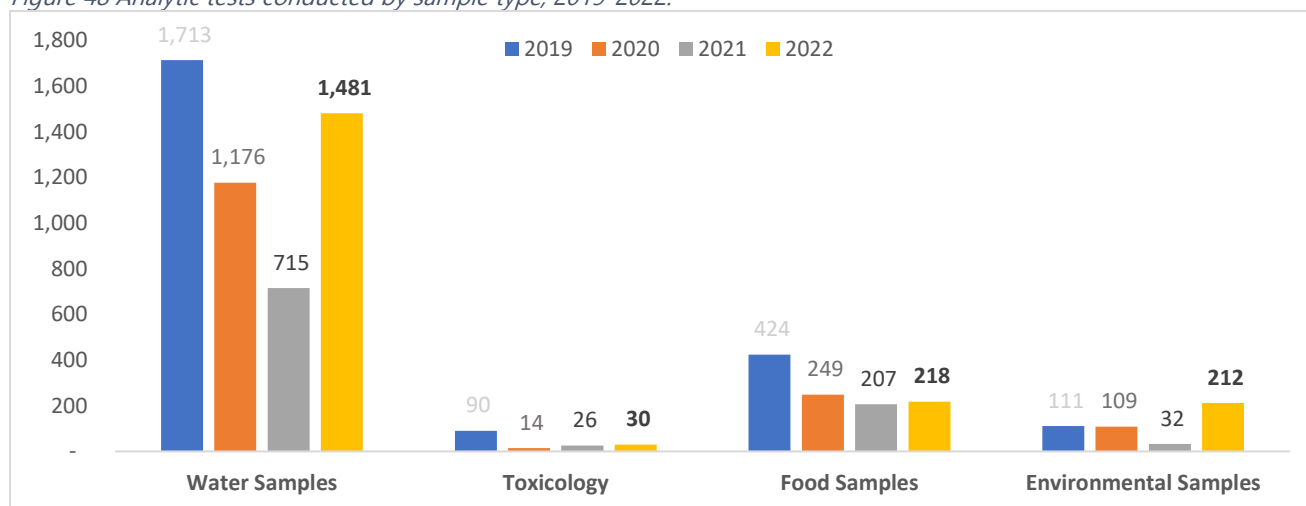
The Seychelles Public Health Lab (SPHL, ISO 9001:2008 certified) provides reference testing facilities for the fulfilment of public health and epidemiological needs of the country, including the One-Health approach, in partnership with other sectors (NHSP objectives 3.1 and 3.3).

The SPHL has served a regulatory role since 2021, for quality assurance of private laboratories, conduct of inter-laboratory checks, and enrolment of all labs undertaking COVID-19 PCR testing in WHO and other external quality assurance schemes (NHSP objective 3.4).

As in 2021, the majority of SPHL resources were dedicated to **COVID-19 PCR testing** (see 4.1.1 COVID-19 Response); but a degree of recovery of routine testing is noted in 2022.

The SPHL conducted **chemical, physical and microbiologic tests for analysis of water, food and environmental samples**, trends in number of samples tested are shown in Figure 48.

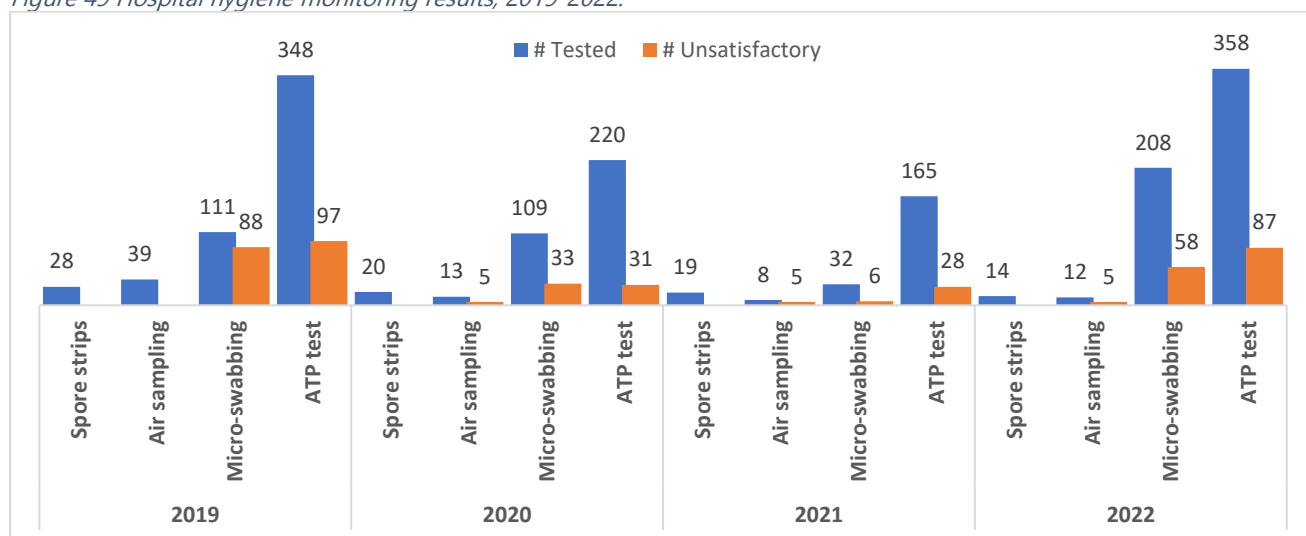
Figure 48 Analytic tests conducted by sample type, 2019-2022.



Data Source(s): SPHL Annual Report, 2022 (PHA, 2023).

The SPHL, in collaboration with IPC unit, conducted **hospital hygiene monitoring** annually, results are depicted in Figure 49. Number of samples taken for tests had increased in 2022, above pre-pandemic levels.

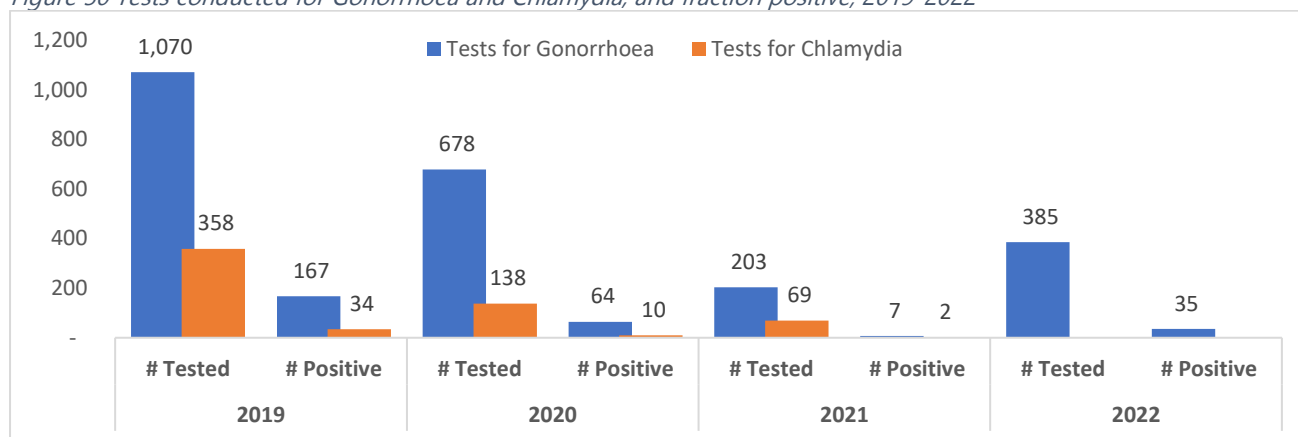
Figure 49 Hospital hygiene monitoring results, 2019-2022.



ATP – Adenosine Triphosphate; Note: All tests mentioned are different methods for assessing hygiene in health settings; Data Source(s): SPHL Annual Report, 2022 (PHA, 2023).

Confirmatory testing for HIV and **testing for common STIs** such as Gonorrhoea and Chlamydia are conducted at the SPHL. Figure 50 shows patterns in tests conducted for Gonorrhoea and Chlamydia in recent years; note that there had been a Chlamydia test-kit shortage since 2019. Also see section 3.3.3 Key Communicable Diseases, and section on Communicable Disease Control Unit.

Figure 50 Tests conducted for Gonorrhoea and Chlamydia, and fraction positive, 2019-2022



Data Source(s): SPHL Annual Report, 2022 (PHA, 2023).

Disease surveillance testing for 2022 is shown in Table 17. A significant number of tests were for screening for Salmonella and Staphylococcus aureus, usually as part of monitoring food handlers.

Routine sentinel surveillance was halted since 2021. With the setting up of the SARS-CoV-2/Flu Multiplex Assay, the SPHL did screen some samples as trial run and three samples tested positive for **Influenza**, indicating circulation of other respiratory viral pathogens apart from COVID-19 in 2022.

Table 17 Tests conducted for disease surveillance by organism type, 2022

Pathogen	# Tests	# +ve	Pathogen	# Tests	# +ve
Salmonella	1,859	8	MRSA	54	11
Staphylococcus aureus	2,487	380	Measles & Rubella	1	0
Rotavirus	45	4	Influenza	282	3
Tuberculosis	66	6	Mpox	8	0

MRSA – Methicillin-Resistant Staphylococcus Aureus; Data Source(s): SPHL Annual Report, 2022 (PHA, 2023).

Confirmatory testing for Leptospirosis and Dengue Fever were not conducted in 2021 or 2022.

4.2.4 Surveillance of Vaccine-Preventable Diseases

EPI published a report in 2022, on **surveillance of Rotavirus amongst diarrhoeal admissions in children from 2013 to 2021**⁴⁷. The report describes trends before and after introduction of Rotavirus vaccination into the routine childhood immunisation schedule in Seychelles.

The surveillance system was set-up in Q2-Q3 2013, with support of CDC and WHO-AFRO, and conducted screening of all admissions to Paediatric Ward, with support from the ward team and SPHL, under supervision of the EPI manager.

With reference to Table 18, prior to introduction of Rotavirus vaccination into the routine childhood immunisation, the surveillance system reveals that the estimated fraction of Rotavirus diarrhoeal admissions to

⁴⁷ Report on Rotavirus Surveillance in Seychelles, 2013-2021 (HCA, 2022).

paediatric ward was 29% of all admissions for diarrhea that were screened for Rotavirus. The primary genotype was G9 P8.

Post-introduction, the positivity rates reduced to 11% (using 2017-2019 data only – pre-pandemic). Data for 2020 and 2021 were excluded in this estimate, as many routine care and laboratory functions were interrupted; if this data were considered, the Rotavirus positivity rate will reduce to about 6%.

Table 18 Rotavirus surveillance test positivity and genotypes, 2013-2021

Stage	Year	Primary Genotype	Rotavirus test Positivity
Pre-Rotavirus Vaccine Introduction	2013 (Jul-Dec)	Included with 2014	19%
	2014	G9 P8	33%
	2015	G9 P8	29%
	2016	G4 P8	35%
Rotavirus Vaccine Introduced	2017	-	14%
Post-Rotavirus Vaccine Introduction	2018	G2 P4	4%
	2019	G3 P8	16%
	2020*	-	0%
	2021*	-	3%

Data Source(s): Report on surveillance of Rotavirus amongst diarrhoeal admissions in children, from 2013 to 2021 (EPI, 2023).

Coverage of Rotavirus vaccinations has remained high since introduction (99%). Interestingly, **admissions from non-Rotavirus diarrhoea also reduced** post-introduction of the vaccine across all paediatric age groups.

Acute Flaccid Paralysis surveillance (for Poliomyelitis) continued in 2022, and report submitted in July 2022. No cases of Polio were reported in Seychelles in 2022 (WHO GPW13 indicator). Refer to routine surveillance reports for additional information on vaccine-preventable diseases.

4.2.5 Climate and Health

No tangible progress was made towards key NHSP objectives (2.4.1) towards climate resilience in the health sector in 2022. Engagement with Ministry responsible for Climate Change was re-established towards end of 2022, as the sector recovered routine functions.

The Seychelles was represented at the Conference of Parties (COP) 27 United National Climate Change Conference (held at Sharm-el-Sheikh, Egypt, November 2022), but did not include representation from MOH⁴⁸.

4.3 Progress towards improving Health Security

The country responded to **multiple infectious and communicable disease threats** in 2022. These threats included existing (e.g. COVID-19, HIV), new (Mpox) and re-emerging (Syphilis) diseases. **Routine disease surveillance mechanisms were partially restored** in 2022.

AMR surveillance showed highly resistant organisms in circulation in the Seychelles; identified need for strengthening anti-microbial stewardship in MOH. A report on surveillance of Rotavirus showed significant reduction in prevalence of this infection amongst admissions to Paediatric ward.

An amendment was made in 2022 to **strengthen the Tobacco Control Act (2009)**. However, **challenges remained for development of required standards and regulations, and conduct of monitoring and enforcement of health-related practices**.

⁴⁸ https://unfccc.int/process-and-meetings/conferences/sharm-el-sheikh-climate-change-conference-november-2022/five-key-takeaways-from-cop27?gclid=Cj0KCQjwO-kBhDIARIsAL6Lorcsamv1QpZkrOBMUAtcCE80h6mssMZCK1yQ2wMdujOHv2oDW6Q93DUaAguzEALw_wcB

Strengthening of public health programmes is another key issue; with the HIV/AIDS programme manager post being vacant, and one person staffing two key disease-specific programmes – cancer and mental health.

5 Health System

WRITER(S): **EMELYN SHROFF**; **DOREEN ZELIA**; **SANJEEV PUGAZHENDHI**; **AGNES CHETTY**;

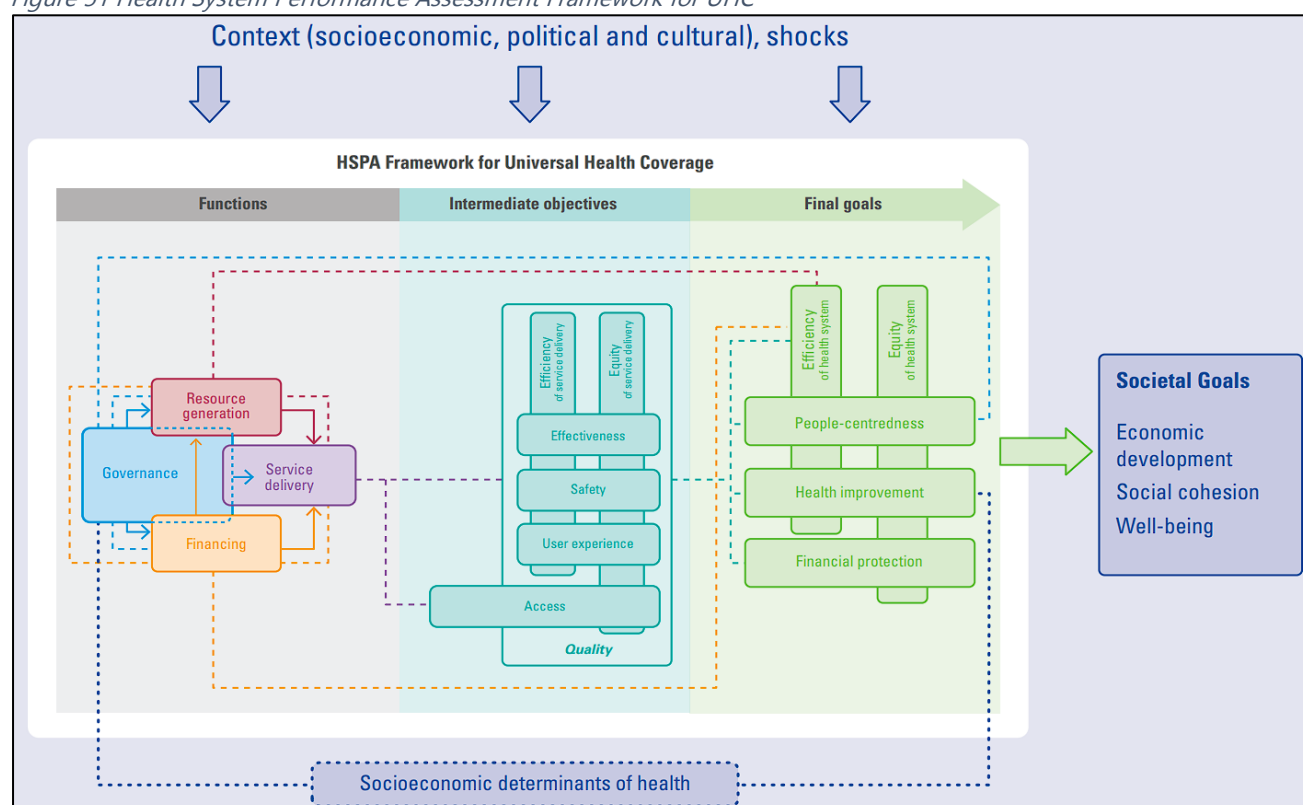
A health system consists of all the organizations, institutions, resources and people whose primary purpose is to improve health. Health systems have three fundamental objectives³⁸:

- Improving the health of the population they serve;
- Responding to people's expectations; and
- Providing financial protection against the costs of ill health.

The global COVID-19 pandemic serves as a stark reminder that a robust and resilient health system is one of the best defences against health threats. The WHO has developed a new **Health System Performance Assessment (HSPA) Framework for Universal Health Coverage (UHC)**⁴⁹ that can serve as a conceptual tool to better assess performance effectively. It builds on the traditional building blocks model, but focuses on four health system *functions* – **governance, resource generation, financing and service delivery** – and their sub-functions that matter most for the overall system performance.

The HSPA Framework for UHC (Figure 51) focuses on health actions within the traditional health care remit, but also includes public health and health promotion activities that contribute to health improvement.

Figure 51 Health System Performance Assessment Framework for UHC



Source: Papanicolas I et al. (2022).

⁴⁹ Papanicolas I et al. (2022). Health system performance assessment: A framework for policy analysis. Geneva, World Health Organization.

While it is beyond the scope of the framework, the **influence of the socioeconomic, political and cultural factors on health are explicitly acknowledged** by placing the health system within this broader context and by recognizing that the health system is meant to contribute to the wider societal goals.

The Governance function now includes health information systems as a sub-function, and both are covered under Chapter 2 Leadership and Governance. This chapter looks at:

- Health Financing (NHSP SD5, see 5.1 Health Financing);
- Resource Generation (NHSP SD5, see section 5.2 Resource Generation: Human Resources and 5.3 Resource Generation: Infrastructure and Supplies); and
- Service Delivery (NHSP SD2, see section 5.4 Service Delivery).

The COVID-19 pandemic put a strain on the health system and in 2022, effort was directed at fully restoring health services. This chapter provides data points and evidence to demonstrate an upward trend towards progress in restoring health services.

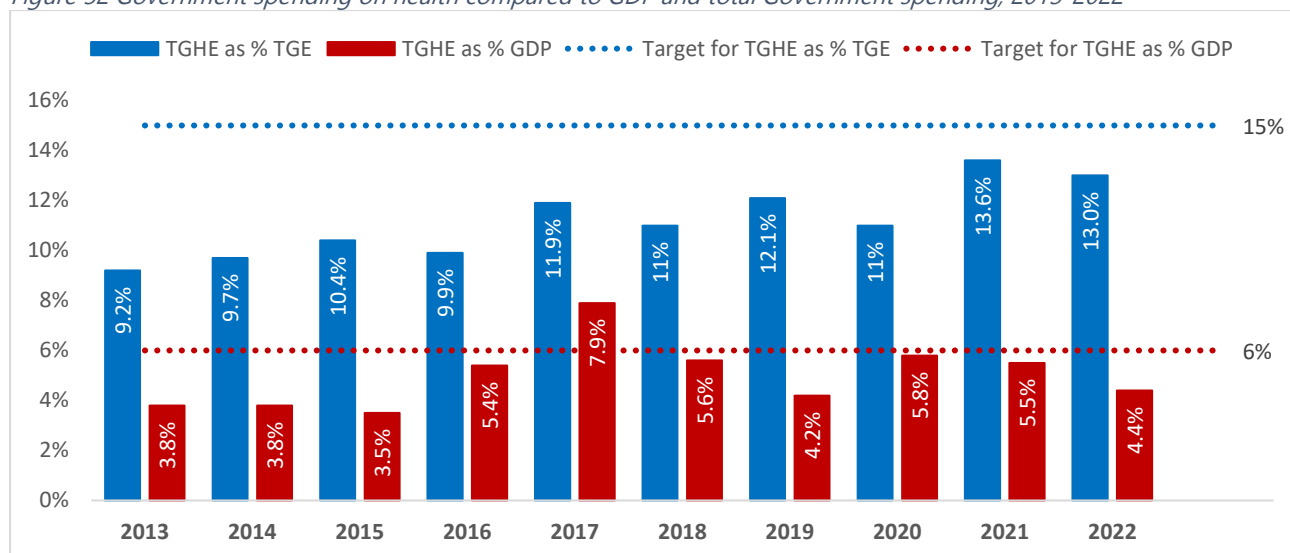
5.1 Health Financing

The NHSP identified the following **key priorities** for health financing, outlined under SD5 Invest for Results:

- Make PPBB work for the health sector.
- Address inefficiencies in healthcare management and service delivery.
- Improve revenue collection.
- Ensure sustainability of healthcare financing.
- Increase spending on preventive and promotive care.

Seychelles government remains the biggest financer of health. In 2022, the government dedicated **13% of the total budget allocation to the health sector** (target >15% per NHSP and the Abuja Declaration⁵⁰. Total government health expenditure as a percentage of the GDP was **4.4%** (NHSP target >6%, Figure 52)⁵¹.

Figure 52 Government spending on health compared to GDP and total Government spending, 2013-2022



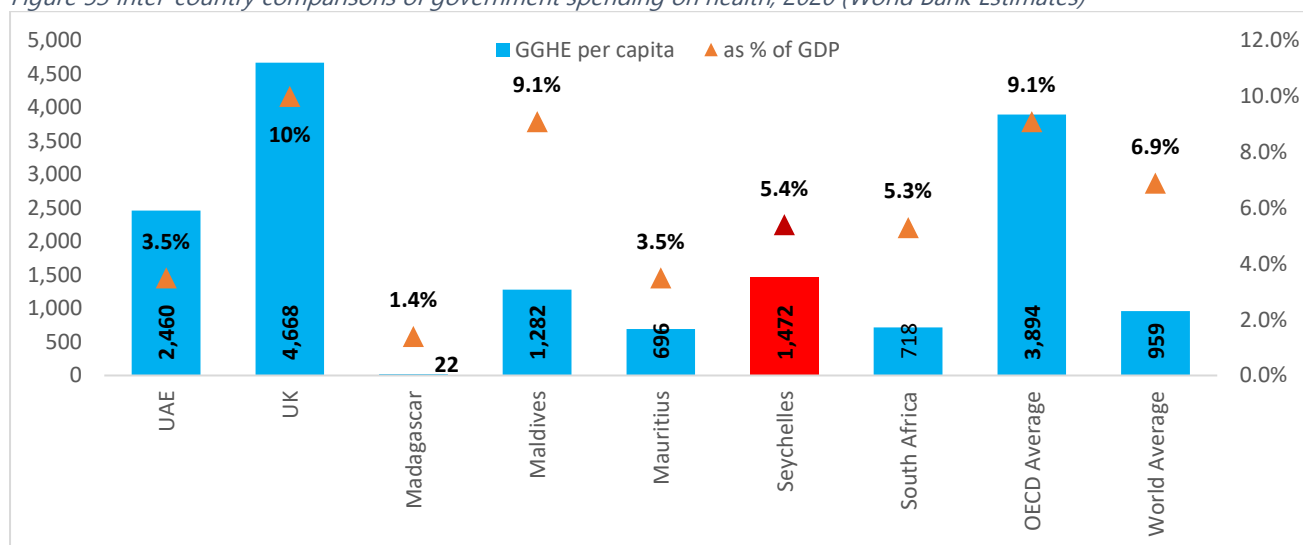
THE – Total Government Health Expenditure; TGE – Total Government Expenditure;
Data Source(s): Financial Controllers at MOH.

⁵⁰ Abuja Declaration on HIV/AIDS, Tuberculosis and other related infectious diseases (OAU, Abuja 2001); link: <https://au.int/sites/default/files/pages/32894-file-2001-abuja-declaration.pdf>

⁵¹ Central Bank of Seychelles Monetary Policy Report (Jan 2023).

Per-capita spending on health increased significantly in the past decade, catching up to, and surpassing, neighbouring island countries, but remained lower compared to other high-income countries (Figure 53). The per-capita spending figure would be *lower* if the revised resident population estimates were applied.

Figure 53 Inter-country comparisons of government spending on health, 2020 (World Bank Estimates)



Note: Chart compares domestic General Government Health Expenditure (GGHE) per capita, PPP (current international \$) and as % of GDP, of select countries with Seychelles, for 2020. Data Source(s); World Bank.

5.1.1 Budget allocations and spending by entities

In 2022, **total budget allocation to the health sector decreased by 7%** compared to the preceding year. HCA continued to receive the major share of Government health financing (86%), however, there was a decrease of 13% compared to the previous year (Table 19).

Table 19 Health sector budget allocation, 2019-2022

Entity	2019		2020		2021		2022	
	Budget	%	Budget	%	Budget	%	Budget	%
MOH	77,613,576	7.6%	48,012,000	4.3%	49,676,810	3.5%	109,704,243	8.4%
HCA	878,858,596	86%	1,024,780,000	90%	1,301,930,740	92%	1,133,839,050	86%
PHA	62,674,403	6.1%	59,078,000	5.3%	60,701,650	4.3%	65,656,000	5%
NAC	3,957,522	0.4%	4,064,770	0.4%	3,703,700	0.3%	3,846,000	0.3%
Total	1,023,104,097		1,135,934,770		1,412,309,200		1,313,045,293	

Data Source(s): MOH Financial Controllers.

Allocations for the MOH secretariat had doubled from 2021 to 2022 due to integration of the budget of DSAPTR (formerly-APDAR), and significant allocations for non-financial assets (for move of offices to new premises, which did not occur in 2022); **PHA allocations in 2022 were 8% higher** than in 2021. In 2022, **actual budget spending was 92% of the allocated budget**, an increase of 11% compared to 2021 (Table 20). **A significant portion (44%) of spending was towards compensation of employees.**

Table 20 Health sector actual budget spending, 2020-2022

Entity	2020		2021		2022	
	Actual Spent	% Allocated	Actual Spent	% Allocated	Actual Spent	% Allocated
MOH	37,528,890	78%	44,096,780	89%	74,877,263	68%
HCA	1,084,780,390	106%	1,139,341,000	80%	1,074,113,377	95%
PHA	63,830,510	108%	59,186,790	98%	60,759,000	93%
NAC	-	-	3,704,000	100%	3,403,000	88%
Total	-	-	1,246,328,570	81%	1,213,152,640	92%

Data Source(s): MOH Financial Controllers.

5.1.2 Donations

The MOH received **donations valued at SCR 34 million** in 2022 (2.6% of MOH budget), from international and local donors. This represented a decrease compared to 2021 (SCR 98 million). Among international partners, the WHO work plan for 2022-2023 amounted to almost USD 1 million; and contributions were received from the Japanese Embassy towards medical equipment. Local donations amounted to SCR 2.7 million, which included donations towards the replacement of the hospital lift (Table 21).

Table 21 Value of International and Local Donations to MOH in 2022

Currency	Amount	SCR equivalent
International Donations		
SCR	287,343.00	287,343.00
USD (USD 1 $\equiv$ SCR 13.83)	1,095,745.11	15,153,475.51
JPY (1 JPY $\equiv$ SCR 0.095)	165,811,000.00	15,713,870.49
GBP (1 GBP $\equiv$ SCR 17.24)	6,246.10	107,699.23
INR (1 INR $\equiv$ SCR 0.168)	1,000,000.00	167,540.45
Subtotal for international donations (SCR)		31,429,928.68
Local Donations		
SCR	1,664,266.75	1,664,266.75
USD	10,688.29	147,812.42
EUR (1 EUR $\equiv$ SCR 14.85)	60,911.52	904,396.36
Subtotal for local donations (SCR)		2,716,475.53
Grand Total (SCR)		34,146,404.21

Data Source(s): International Cooperation Unit, MOH.

5.1.3 Key messages on Health Financing

Although Seychelles has not reached commitments of the Abuja declaration, the **absolute financing for health is above peers** in the Indian Ocean region and well-above mainland Africa, and represents an almost three-fold increase compared to 2015⁵², in dollar-terms.

Prescription fees (introduced in 2019) were reversed in 2022 in view of costs of administering it, exceeding revenues. The MOH lacks capacities for conducting economic evaluations and efficiency studies.

The National Health Accounts reports were delayed, with the last released report (MOH, April 2023) covering data up to 2018 only. Updated reports will help understand changes in spending on preventive and promotive services and out-of-pocket spending.

5.2 Resource Generation: Human Resources

Human resources (HR) is one of key focus of *NHSP SD5 Invest for Result*. HR is the most essential resource for the healthcare system, enabling the delivery of a wide variety of healthcare services, and ultimately contributing to the achievement of national health goals.

The **availability, distribution, and competence** of healthcare professionals directly impact the overall performance of the healthcare system, as well as the wider societal development of Seychelles. Furthermore, investing in a robust healthcare workforce is crucial for the nation's progress towards SDG 3 - ensuring good health and wellbeing for all. The need for other professionals (e.g. biostatisticians, economists) is increasingly recognised in effective health system operations.

The key **priority areas for HR for Health** set-out in the NHSP 2022-2026 for the next five years are:

- Develop a National HR for Health Strategic Plan.

⁵² See Annual Health Sector Performance Report 2018 (MOH, 2019).

- Empower professional councils to become more effective regulators.
- Make careers in health more attractive.
- Enable HR to deliver on all core HR functions.
- Provide support and guidance for career planning.
- Care for the carers.
- Improve HR data collection, analysis and use in decision-making.

5.2.1 Regulatory Councils

Three **regulatory councils**, namely the Seychelles Medical and Dental Council (SMDC), Seychelles Nurses and Midwives Council (SNMC) and Health Professionals Council (HPC) have oversight role on the work of various health professionals across the relevant cadres. Councils publish updated registers of all health professionals under their purview registered to practice in Seychelles annually.

Analysing key performance indicators of relevance to regulatory councils (NHSP objective 5.1), the **HPC has introduced English-language proficiency requirements** for new and re-registrations of expatriate staff in 2022. Continuous Professional Development (CPD) is already compulsory for re-registration with the SNMC and HPC, but yet to be introduced by the SMDC.

5.2.2 Doctors and Dentists

As at end 2022, **514 doctors and 78 dentists were on the register of the SMDC**. Just under one third (32%) of all registrants are women. Only 22% are Seychellois nationals. Approximately of doctors (241) and of dentists (44) on register were professionally active in Seychelles in 2022. Table 22 summarises **new registrants** in 2022, and total on register for doctors and dentists as at end 2022.

Table 22 Doctors and dentists on register end-2022, and new registrants in 2022

		Doctors on Register		Dentists on Register	
		New	Total	New	Total
Seychellois National	Both Sexes	10	106	2	23
	Male	5	57		13
	Female	5	49	2	10
Foreign National	Both Sexes	102	408	6	55
	Male	81	294	3	36
	Female	21	114	3	19
Total	Both Sexes	110	514	8	78

Data Source(s): Human Resources for Health Report, 2022; Medical and Dental Practitioners (Office of the CMO, 2023).

Of the professionally active doctors and dentists at end-2022, 228 doctors (95%) and 41 dentists (93%) were in clinical practice; with others in differing roles (e.g. policy, regulatory, administrative, etc.). The distribution of doctors and dentists across public and private health sector are shown in Table 23.

Table 23 Professionally active doctors and dentists and distribution across public and private sectors, 2022

	Doctors	Dentists
Total Professionally Active	241	44
In Clinical practice	228 (95%)	41 (93%)
Administrative/Other	13 (5%)	3 (7%)
Density per 10,000 population (SDG 3.c)	20.1	3.7
Active in Public Health Sector	191 (79%)	30 (68%)
Seychellois	75	14
Non-Seychellois	116	16
Active in Private Health Sector	50 (21%)	14 (32%)
Seychellois	13	2
Non-Seychellois	37	12

Data Source(s): Human Resources for Health Report, 2022; Medical and Dental Practitioners (Office of the CMO, MOH 2023).

A few doctors working in Seychelles were not counted in the active head-count of doctors; these include ‘visiting doctors’ in the private sector (e.g. Hotel-based rotating doctors and at certain private clinics).

Age-breakdown of the workforce of doctors shows that 25% of active workforce were under age 35, 24% in range 25-44y, 23% 45-54y, 21% 55-64y, and 6% in the 65y+ range. The workforce is predominantly expatriate between ages 35 and 54. **The under 35y age group consisted predominantly of Seychellois, indicating an increase in number of returning Seychellois medical graduates.**

Overall, **37% of active doctors are women**, with the male-female ratio in the <35y age group almost 1:1, and an increasing fraction of men with increasing age. **Women constitute 43% (N = 19) of active dentists.** One fifth of all active dentists were aged <35y (nine, all female); older age groups were predominantly male.

Undergraduate training

All basic training for doctors and dentists, as well as most degree-level training, are undertaken with overseas training institutions, usually in-person or (more recently) through remote learning methods.

At the end of 2022, **56 medical students were undergoing undergraduate training**, the majority (27) in Slovakia. Ten new students enrolled in medicine in 2022 (2M, 8F). **No students were enrolled in dentistry** undergraduate training.

Five (4M, 1F) doctors graduated in 2022 and returned to Seychelles to join the HCA. Two dentists graduated, with one joining HCA, the other in private health sector.

Over the next five years, 48 new medical graduates are expected to complete their studies. This may provide an opportunity to reduce dependence on expatriate workforce, at least in primary care.

Post-graduate training

Budget constraints during the COVID-19 pandemic limited Government funding for post-graduate training. There was an increased uptake of remote learning options since 2020. Some doctors also sought alternative funding for their post-graduate training.

In 2022, **ten doctors (2M, 8F) started post-graduate training** (of whom seven are Government-funded, three self-funded). Thirty-one doctors were pursuing post-graduate training end 2022, including nine undertaking courses online. No new doctors completing post-graduate studies have returned to join the service in 2022. No dentists were enrolled in post-graduate training at end-2022.

Over the next five years, 29 doctors are expected to complete post-graduate training, with the largest group expected in 2023, eighteen in total, including five in family medicine.

5.2.3 Nursing and Midwifery

As at December 2022, **947 nurses were registered** with the Seychelles Nurses and Midwives Council (SNMC) to practice in Seychelles⁵³, **128 (14%) of these were expatriate** nurses. Of the total registered, **623 (66%) were practising; 453 in the public health sector, 170 in private sector.** Of those in public health sector, the vast majority are practising in the HCA (Figure 54).

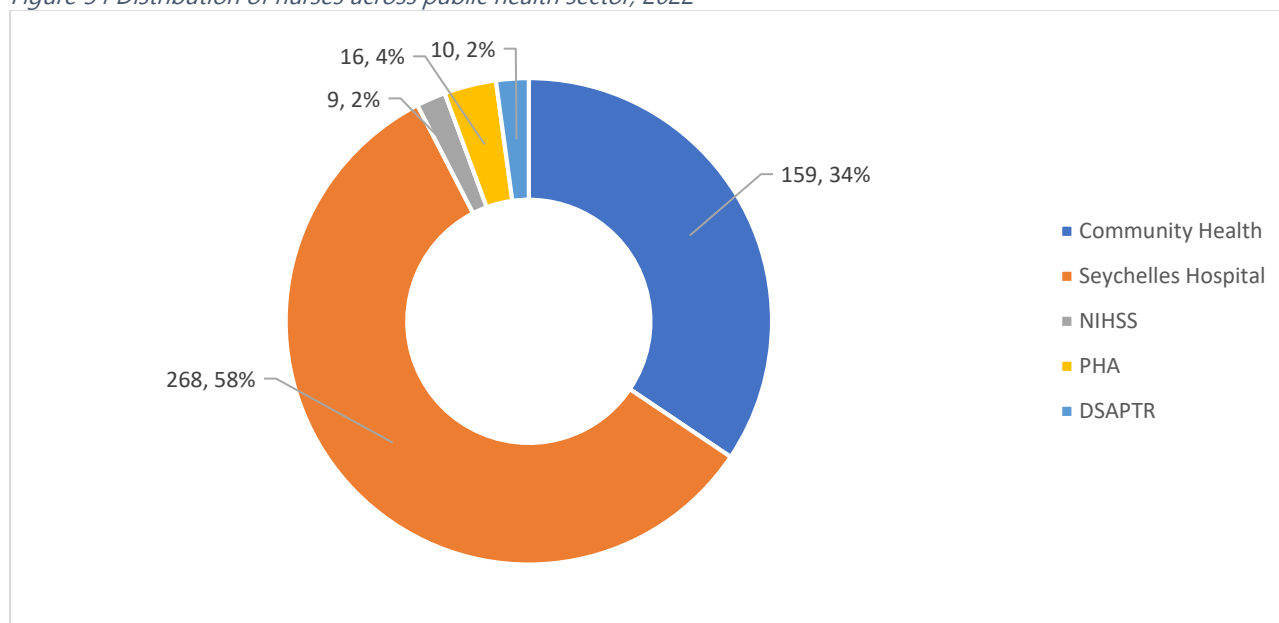
Among 310 midwives on the register, 145 were practising. The majority were nurses that subsequently completed training in midwifery (139 of 145 practising, or 96%).

⁵³ Draft CNOs Nurses Movement Report, 2022 (MOH, 2023).

Basic training for nurses, midwives and a number of allied health cadres is provided locally at the NIHSS. More advanced training is offered in partnership with universities abroad, or through remote/online courses. In-person training opportunities abroad for nurses remain limited (see 5.2.5 Workforce Production at the NIHSS).

Total attrition of nursing staff (17 retirements and 10 resignations) exceeded new recruits (18) in 2022.

Figure 54 Distribution of nurses across public health sector, 2022



Note: Seychelles Hospital includes Wellness centre, North East Point Hospital and Hospice. Data Source(s); Office of CNO (MOH, 2023).

5.2.4 Allied Health Workforce

The Health Professionals Council (HPC) reported **481 registered active allied health professionals** at end 2022. The exact numbers of each cadre are summarised in Table 24. There were 77 new registrants (41M, 36F), 43 in public sector and 34 in private.

Certain key cadres, such as pharmacists, may appear to be numerous in terms of numbers registered, but have a high turnover, are not necessarily the numbers actually practising, and are predominantly distributed to the private health sector. **Only seven of the registered pharmacists were practising in the public health sector** as at the end of 2022.

Table 24 Allied health professionals' counts by cadre, 2022

Allied Health Cadre	Total Active	Allied Health Cadre	Total Active
Dental Surgery Assistant	50	Optometrist	25
Physiotherapist	18	Speech Pathologist	3
Physiotherapy Technician	12	Dental Technologist	4
Occupational Therapist	3	Radiographer	9
Occupational Technician	7	Psychologist	25
Pharmacist	81	Emergency Medical Technician	21
Pharmacy Technician	61	Public Health Officers	28
Dental Hygienist	14	Nutritionist	2
Dental Therapist	19	Dietician	1
Biomedical Lab Technologist	19	Audiology	1
Biomedical Lab Technician	70	Prosthetic	1
Biomedical Scientist	5	Orthotic Technician	2

Data Source(s): Health Professionals Council, 2023.

5.2.5 Workforce Production at the NIHSS

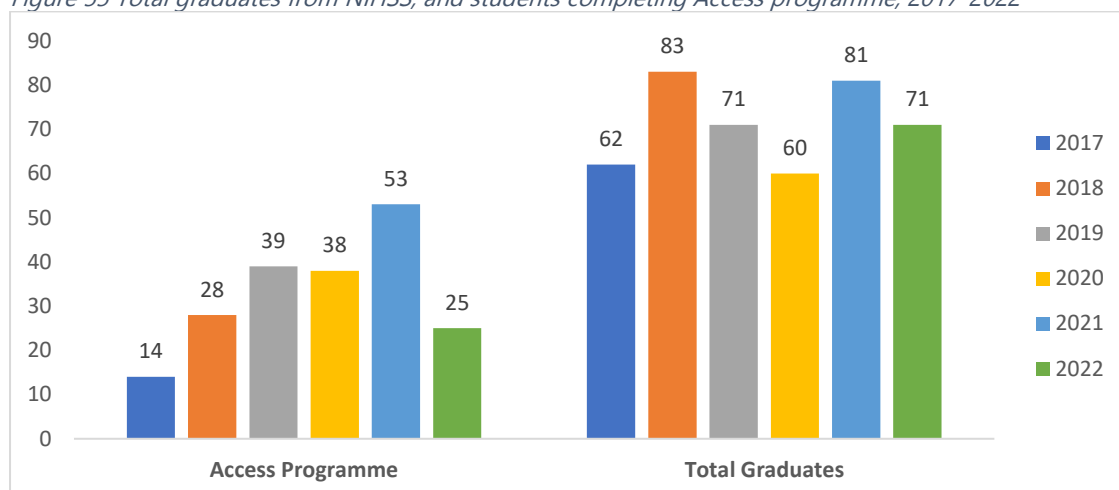
NIHSS is mandated to provide training for health and social workforce production. The key highlights of 2022 include:

- The development of a new **NIHSS strategy** in 2022 (pending approval by management at time of preparation of this report; contributes towards NHSP objectives 1.2 and 5.1).
- **Social work programme and nursing curriculum were fully accredited** by Seychelles Qualifications Authority (SQA) in 2022 (NHSP objective 5.1).
- The NIHSS developed an **emergency preparedness plan** for the institute.

An average of 70 students graduate from NIHSS each year, the majority receive a Certificate or Diploma-level qualifications. Degree level training are typically in cohorts. **In 2022, 71 students graduated** from the following training programmes (Figure 55):

- Diploma in Nursing – 22 graduates.
- Diploma in Social Work – 23.
- Diploma in Biomedical Laboratory Sciences – 16.
- Diploma in Pharmaceutical Sciences – 10.

Figure 55 Total graduates from NIHSS, and students completing Access programme, 2017-2022



Data Source(s): NIHSS Annual Report 2022 (MOH, 2023).

There were more nursing graduates in 2022 (22) compared to 2021 (14). The **Access programme has become a key pathway** for students who do not meet entry requirements to access courses at NIHSS. Twenty-five students completed the Access Programme for Health in 2022.

New infrastructure for NIHSS within the main hospital premises was completed in 2022 (NHSP objective 5.1), and will provide a more conducive learning environment in close proximity to the Seychelles Hospital.

5.2.6 Health Workforce Density

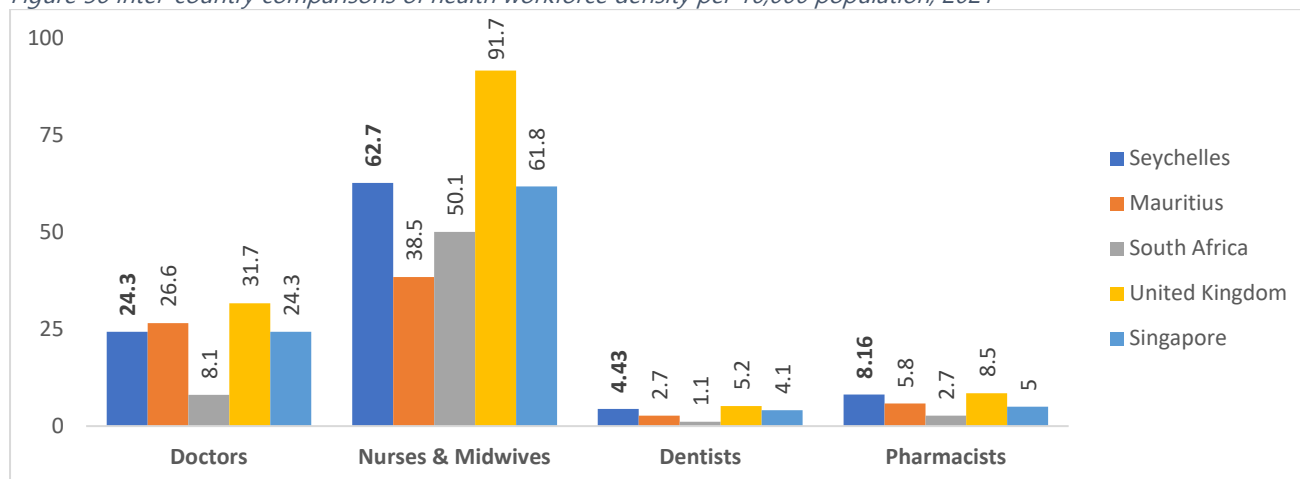
The workforce density refers to the number of health workers available in the country relative to total population, and typically expressed in HCWs by specific cadre(s) per 10,000 population. SDG 3.c indicators include workforce density of doctors, nurses and midwives, dentists and pharmacists, to measure progress towards achieving adequate health workforce in member states.

Figure 56 shows 2022 data for Seychelles, compared select other countries. The **doctor density in Seychelles was comparable to Mauritius and some high-income countries**, and significantly higher than in mainland African countries such as South Africa. **Dentist workforce density exceeded Mauritius, and is comparable to high-income countries.** **Nurses and midwives density were comparable to many high-income**

countries. Pharmacist density may be misleadingly high (see 5.2.4 Allied Health Workforce), and the majority work in the private health sector.

These workforce densities were above WHO recommendations, and above regional averages. Despite this, **effective distribution of the workforce across service areas in quantity, quality and skills-mix, remains a challenge.**

Figure 56 Inter-country comparisons of health workforce density per 10,000 population, 2021



Data Source(s): World Health Statistics Report, 2023 (WHO, 2023).

Key cadres of HR for Health have increased dramatically during the preceding 10 years (Table 25)⁵⁴. Whether this has improved quality, access or models of service delivery remains to be assessed.

Table 25 Historic trends in HR for Health, 2008-2022

	2008	2013	2018	2022	Δ 2013-2023
Doctors (practising, public and private)	85	134	188	241	+80%
Nurses (practising, public and private)	407	416	510	623	+50%
Dentists (practising, public and private)	12	21	-	44	+110%
Pharmacists (practising, public sector only)	4	4	-	7	+75%

Δ – Change in percentage. Data Source(s): Annual Report (MOH, 2015); Offices of the CMO and CNO; HPC Council.

5.2.7 Key points on HR for Health

Investing in the development and retention of human resources for health (HRH) is essential for Seychelles. A robust healthcare workforce ensures the efficient operation of the healthcare system, contributes to the achievement of NHSP goals and objectives, drives wider societal development, and propels Seychelles towards SDGs. There is need for a **National Human Resources for Health strategy** (NHSP objective 5.1.2) to clearly define workfare needs and define training, recruitment, skills-mix, distribution, motivation, capacity building, and retention of HCWs for providing integrated quality care.

Although work on the strategy had not started in 2022, progress was made towards preparing a **Workload Indicators of Staffing Needs (WISN)** report, which will inform preparation of a **National Health Workforce Accounts (NHWA)** report, followed by development of a HRH strategy for the health sector.

An **NIHSS strategic plan** was developed in 2022 (pending approval) and work progressed on preparation of a NHWA. Development of a HR for Health strategy remained pending. MOH participated in, or organised, multiple activities to market professions in health.

⁵⁴ Annual report of the MOH 2014 (MOH, 2015).

A key positive trend is the **steady inflow of Seychellois medical graduates** in recent years, which is expected to continue for the coming five years. However, none or very few Seychellois graduates are expected for other key cadres such as dentists and pharmacists. There were shortage of Seychellois doctors in the community health services, which hampers development of primary care. The need for regular continuous professional development is noted for doctors/dentists. There are **high number of pharmacists in the private sector**, indicating need to have greater collaboration and regulation.

The big increase in number of HCWs in key cadres in the past decade has likely **supported provision of more, and wider range of, health services; however, quality of health services were unmeasured.**

Multiple interruptions to service provision continue to persist due to shortage of key specialised doctors (e.g. cardiologists, physicians, vascular surgeon). Technicians in certain fields number just one (e.g. audiology), and are ageing, presenting a serious threat to continuity of the service.

Dependence on foreign workforce (particularly for high-skill jobs) and their high turnover are a risk to the resilience of the health system. The year saw the **Cuban Government scale down** the number of doctors supplied as part of the cooperation agreement between the two countries, placing acute staffing constraints on certain specialised fields of practice.

A **growing private health sector** focused on medicalised care, poaching staff from the public health sector, means more HCWs are typically diverted away from public health, primary care, and health promotion activities, and towards clinical care.

Although almost all **COVID-19-related staff re-deployments were reversed by end-2022**, certain lessons were learnt; in particular, the low agility of deployment in light of rapidly evolving needs during an emergency, and the need for continuity of essential services.

Freezes on basic salary, promotions and permission to take annual leave were eased in 2022, and all COVID-19-response related allowances discontinued.

A new system of **performance management** (under the results-based framework, NHSP objective 1.7) was rolled-out in 2022, linked with the reintroduction of a performance-based 13th month pay. The introduction involved significant burden in terms of paperwork to complete for supervisors. Induction packages and mentorship programmes are available for most cadres of staff.

Defining **integrated care teams** across cadres, including numbers, skills-mix, and task shifting, supported by clear **models for service delivery**, may help improve patient satisfaction with, quality of health services.

5.3 Resource Generation: Infrastructure and Supplies

The NHSP 2022-2026 sets-out key priorities for health technologies, infrastructure and information technology, all under SD5 (Table 26).

Table 26 Key Priorities for health technologies, infrastructure and IT, under NHSP SD 5

NHSP SD5 Components	Key Priorities
Health Technologies (NHSP objective 5.3)	- Strengthen procurement and stock management capacities. - Fully digitise stock management - Engage service providers in procurement process. - Improve service maintenance contracts. - Ensure donated equipment, medicines and supplies align with MOH standards and needs.
Infrastructure (NHSP objective 5.4)	- Needs-based infrastructure planning - Infrastructure maintenance - Addressing fire and other risks - Renovate, upgrade or replace infrastructure no longer fit for purpose - Disability-friendly facilities
Information Technology (NHSP objective 5.5)	- Use IT to support for new health care delivery models – appointment systems, reminders, treatment adherence support, health education, health promotion, self-management supports. - Provide health professionals with point of care access to information – guidelines, standards, decision-making tools. - Facilitate data collection – routine reporting, surveys, monitoring. - Institutionalise eLearning and CPD. - Utilise Telemedicine to improve access to services, including in crises.

5.3.1 Infrastructure and Equipment

The public health sector service infrastructure consists of a main tertiary hospital (Seychelles Hospital), four annexes (North East Point Hospital, Wellness Centre, Baie Sainte Anne Hospital and Logan Hospital), and 17 primary care facilities across the three main inhabited islands.

Key infrastructure projects in 2022 included the start of construction of a **new hospital on La Digue**⁵⁵, the construction of the **NIHSS facility** within the Seychelles Hospital premises, and **renovation work on parts of the North East Point Hospital** complex. A **lift for Seychelles Hospital** was purchased by HCA, and the PHA purchased three new vehicles in 2022.

There were **multiple new equipment** acquired for the hospital, including bedside portable X-ray machine, syringe pumps, intravenous fluid stands, wheelchairs, defibrillators, ultra-cold freezers and tablets. Vaccine fridges were procured in December 2022 for EPI (funded by WHO). New equipment and consumables were received by SPHL, including Multiplex SARS-CoV-2/Flu assays and two genomic sequencing machines (Illumina and Minion ONT, through a donation).

Major interruptions to supply chains in 2020 and 2021 eased in 2022; however, **goods and freight costs remained high** due to global and regional geopolitical issues (conflict and insecurity, fuel price hikes, geopolitical tensions) that kept fuel prices high, increased production costs, and created barriers to free trade.

⁵⁵ <https://www.nation.sc/articles/12758/construction-begins-on-la-digue-hospital#:~:text=Construction%20begins%20on%20La%20Digue%20hospital%20%7C10%20March%202022&text=The%20contractor%20commissioned%20to%20undertake,the%20hospital%20are%20being%20done.>

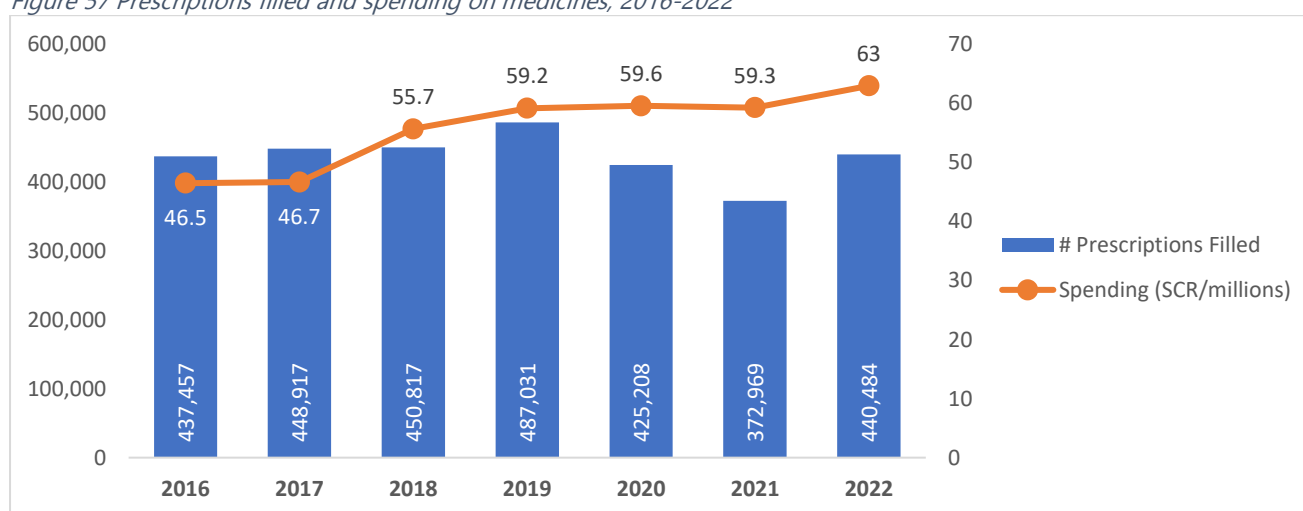
5.3.2 Medicines

A total of **440,484 prescriptions were filled and refilled** in 2022 in the public health sector, an increase of 19% over 2021 and 3.6% over 2020 (Figure 57). The counts appear high since it reflects a count of each (typically monthly) refill of prescriptions and not the actual number of prescriptions issued.

PHC clinics on Mahé filled 302,876 prescriptions (68.8%), out of which, the English River Health Centre filled the highest number (88,940, or 21% of total). The hospital (ward, SOPD and CDCU) pharmacies filled 90,772 (20.6%), and the inner islands, 46,836 (10.6%).

Spending on medicines increased by 6.24% in 2022 compared to 2021 (**SCR 63 million** vs SCR 59.3 million, see Figure 57). Prescriptions dispensed in community pharmacies accounted for SCR 27.4 million (43.5% of spending), and hospital pharmacies 36 million (57.1%). The expenditure for non-formulary medicines was approximately SCR 10.2 million (16%), most of which (88%) were for cancer medicines. The service developed and implemented SOPs for dispensing, to improve and standardise pharmacy practice.

Figure 57 Prescriptions filled and spending on medicines, 2016-2022



Data Source(s): Pharmaceutical Services Annual Report 2022 (HCA, 2023); Annual Health-sector Performance Report 2021.

The **Drug Quality Control Lab maintained its accreditation** of DQCL ISO/IEC 17025: 2017 and WHO pre-qualification of DQCL to be a nationally and regionally recognised reference laboratory. The lab was recertified in August 2022.

5.3.3 Discussion on Infrastructure and Supplies

There was **expansion in use of digital platforms for procurement and stock management**. New infrastructure for a Hospital on La Digue and the NIHSS were built in 2022; and maintenance plans exist for medical devices, but not for infrastructure.

Seychelles Hospital masterplan implementation had not begun ten years through. MOH facilities were not disability-friendly; and certain facilities are vulnerable to service interruptions and damage due to impacts of climate change. Assessment of health infrastructure for fire, flooding, chemical, radiological and biological risks is pending. Response plans for fire, flood, chemical, radiological or biologic risks have not been developed or updated.

New hardware has been distributed across health facilities as part of roll-out of the eHIS. Appointments and registration modules were rolled-out, with capacity-building for doctors and clerical staff. Use of electronic medical records remained limited (to some SOPD clinics, see 2.3.3 Health Information Systems). Internet or network outages, as well as persistent mind-sets and habits, present key challenges to optimal use of the eHIS.

5.4 Service Delivery

The core priorities and objectives of health service delivery are outlined under SD 2 of the NHSP 2022-2026: Protect and Improve UHC, including the following:

- Address health needs through a combination of life-course (RMNCAH, men's health, healthy ageing, etc.) and disease-specific (cancer, CVDs, COVID-19, HIV, etc.) approaches.
- Innovate and modernise care delivery – new service delivery models⁵⁶ aimed at improving outcomes of prevention and care.
- Make quality a top priority in service delivery.
- Increase spending on preventive and primary healthcare services.
- Strengthen and promote a preventive care agenda.
- Ensure no one is left behind: Measure and improve equity in access to health services, and health outcomes.
- Improve health system resilience.

Healthcare service delivery in the public health sector is organised into community health services and hospital based services. Preventive and promotive services are principally provided at community health level; long-term care is provided across both platforms; whereas most specialist services are provided at the Seychelles Hospital. See section 2.4 Legislation and Regulation for numbers of registered private health providers. Private providers are concentrated in and around Victoria, Beau Vallon and the East coast of Mahé, down to Anse Royale.

The NPHC 2022 (NBS, 2023, unpublished) findings show a large fraction of the population continue to use the public health services: **four in five households (79%) participating reported at least one visit to a health centre** in the preceding 12 months, 56% visited only Government health facilities, 8% only private, and 36% both Government and private health facilities. At least one person was **hospitalised in 16% of households** (about one in six) in the preceding 12 months, 95% of these to public hospital facilities, and 3.5% to private.

NHSP priorities for community health include **operationalisation of the PHC Package** of services (see section 2.2.2 Primary Health Care Package), underway in 2022, and creation and operationalisation of a PHC council (NHSP objective 2.1.1); regular coordination meetings between community health services and hospital-management were started in 2022 (NHSP objective 2.1.2).

Focus for secondary and tertiary care is on **strengthening clinical governance structures for improving quality-of-care**, and improving efficiency of service provision (NHSP objective 2.1.3).

5.4.1 Community Health Services

The community health services are organised into health regions with five regions on Mahé (North, Central, East, West and South), and Praslin and La Digue. Each region consists of a larger regional health centre providing a wider range of primary care and public health services, as well as specialist outpatient services, and a number of smaller health centres providing a more limited set of services. Regional centres on Praslin and La Digue provide inpatient services as well. The services of health programmes along the life-course (see 6.1.3 Programmes Across the Life Course) are mostly provided through the community health facilities.

⁵⁶ Service delivery models refer to "how health services are organised, delivered and financed". Link; https://extranet.who.int/kobe_centre/en/what_we_do/service-delivery.

Three health centres functioned as dedicated '**fever clinics**' in Q1-Q2 of 2022, focusing on testing for COVID-19 and outpatient care of patients. Additionally, the Music Stadium was used as a dedicated COVID-19 testing facility in Q1-Q2 of 2022. The Souvenir health centre in La Misère remained closed throughout 2022.

Doctors' consultations at community health centres were 271,437 (83% during working hours and 17% after working hours) in 2022, an increase of 18.5% compared to 2021 (Figure 58). **Nursing care**⁵⁷ attendance was 145,877 in 2022 (11% more than in 2021), with 121,912 (84%) during working hours and 23,965 (16%) after working hours. Apart from regional health centres, smaller **health centres account for less than 2% of doctor's consultations and/or nursing care services each**, indicating low utilisation.

There were 54,133 attendances for **dressings** in 2022 (a 7% increase from 2021). English River health centre recorded more attendance for dressing (9,076), but as a region, the highest number was in East Mahé (11,336).

Domiciliary care (home) visits in 2022 were 8,972, an increase of 87% from 2021 (4,796). The Central Mahé region recorded the most number of domiciliary care visits (2,598). These patterns suggest that the utilisation of PHC services were gradually returning to pre-pandemic levels.

5.4.2 Specialist Outpatient Services

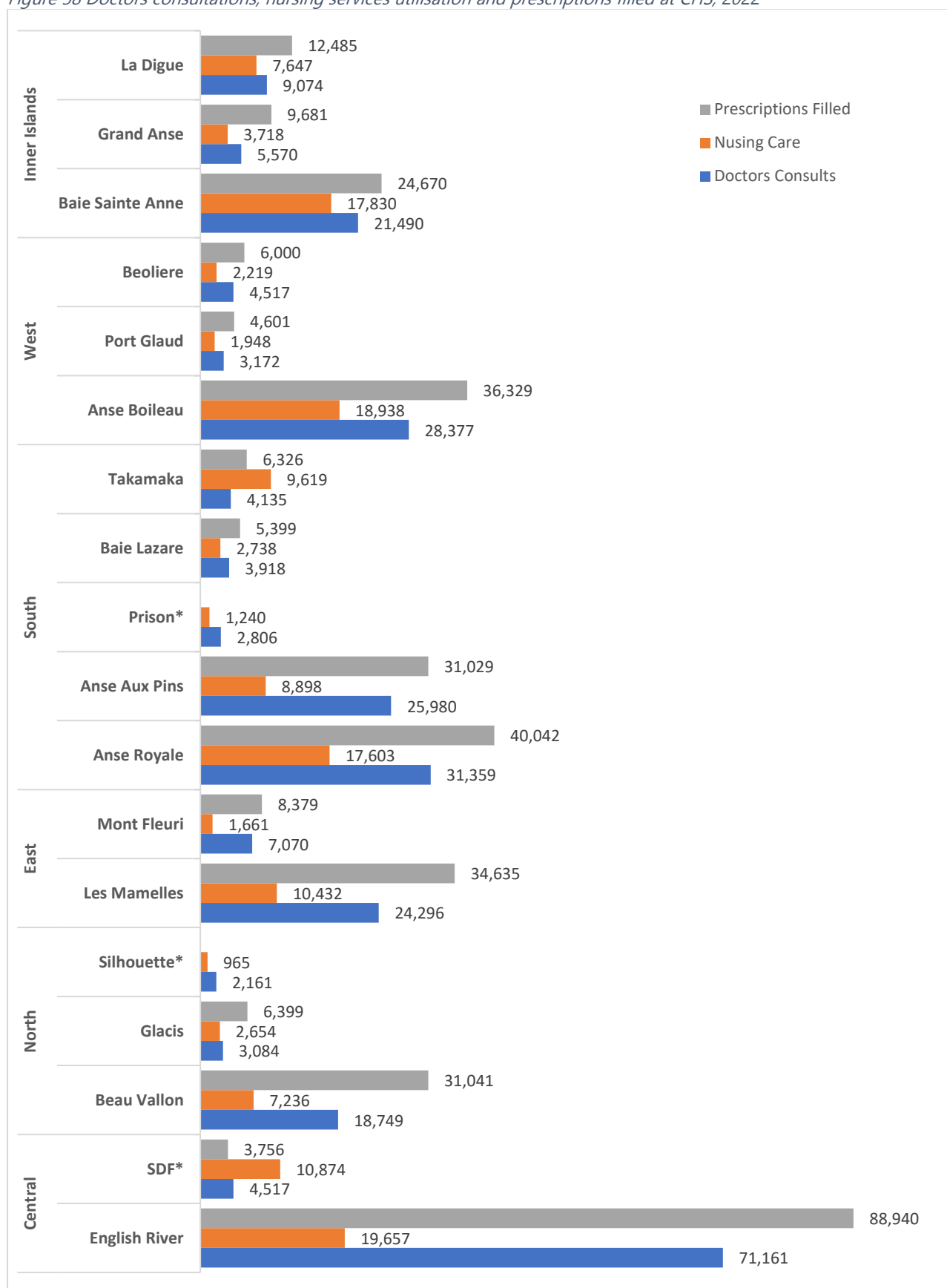
The specialist clinics are based at the Yellow Roof building of Seychelles Hospital, but services are also provided at most regional health centres on Mahé, Praslin and La Digue. Utilisation of specialist clinics increased in 2022 compared to 2021, reaching pre-pandemic levels (Figure 59). The SOPD was closed in Q1 and Q2 of 2021.

There were **39,068 visits to specialist outpatient clinics** in 2022, 28,595 (73.2%) at Yellow Roof and 10,473 (26.8%) at regional health centres. This count excludes visits to oncology and mental health clinics, but these are included in the graph (Figure 59).

The **Oncology unit administered 2,232 chemotherapy sessions in 2022** (see Annual Hospital Services Report 2022 for more information). Mental Health Services team worked to put into practice the Mental Health Care Act, 2020, and began development of a National Mental Health Policy in 2022.

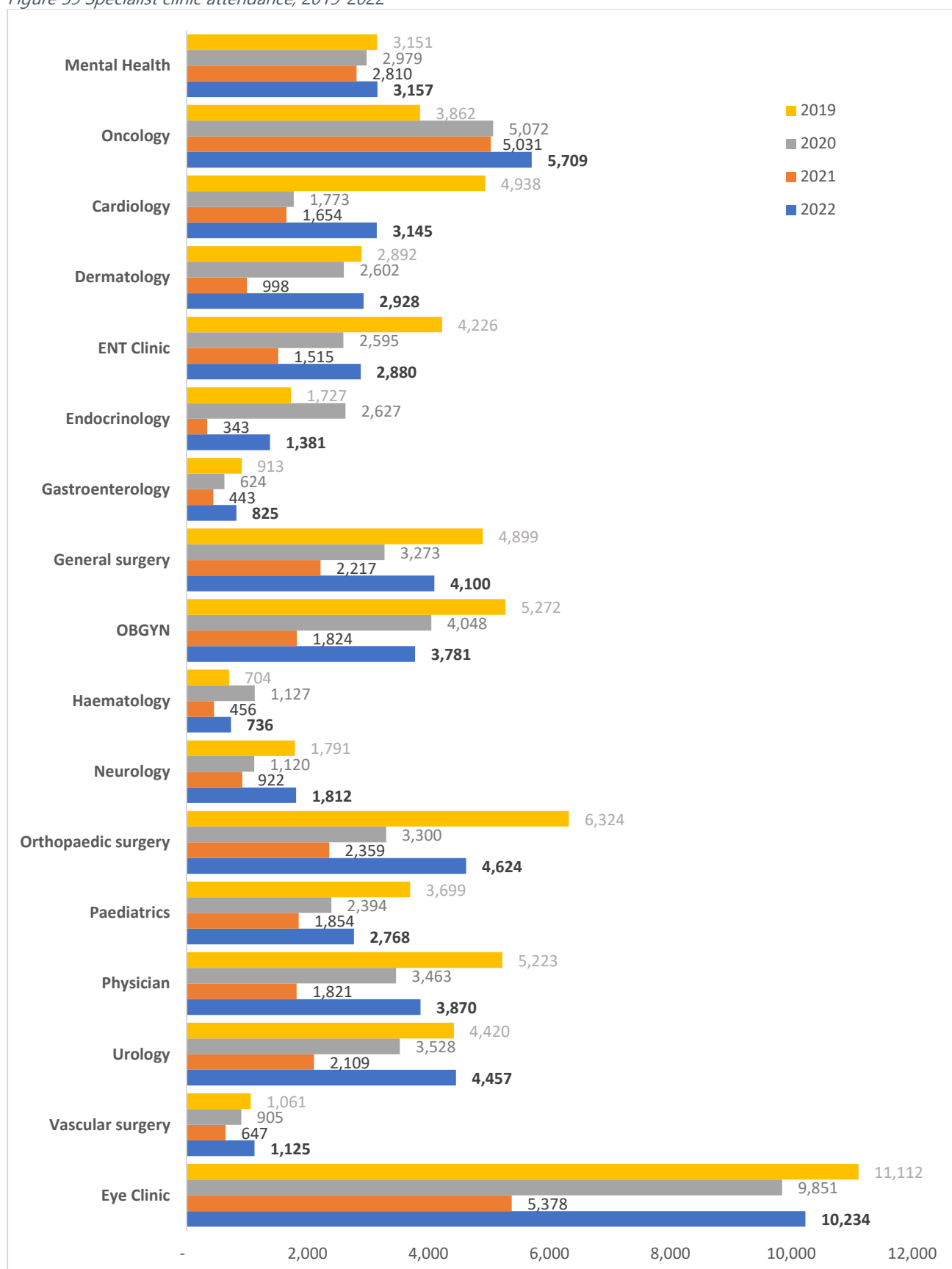
⁵⁷ Nursing care refers mainly to patients seen by a nurse for procedures requested by a doctor: example - blood pressure checks, nebulization, temperature checks, blood sugar tests and injections.

Figure 58 Doctors consultations, nursing services utilisation and prescriptions filled at CHS, 2022



Note: SPDF – Seychelles Defence Forces Clinic; SDF, Silhouette and Prison clinics are not operated directly by MOH; Prescriptions issued at Hospital or SOPD are mostly refilled in community health pharmacies. Data Source(s): Statistics Unit, MOH.

Figure 59 Specialist clinic attendance, 2019-2022



ENT – Ear, Nose Throat Specialist; OBGYN – Obstetrics and Gynaecology.

Data Source(s): Primary Health Care Annual Report 2021 and 2022, Statistics Unit, MOH.

Haemodialysis services

The **cohort of patients on long-term dialysis was 189 in 2022**, slight increase from 2021 (186). Total dialysis sessions increased by 6.4% (25,788 in 2022 vs 24,234 in 2021). **Thirty new patients started chronic dialysis**, while **18 patients on chronic dialysis died, and three discontinued** (Figure 60).

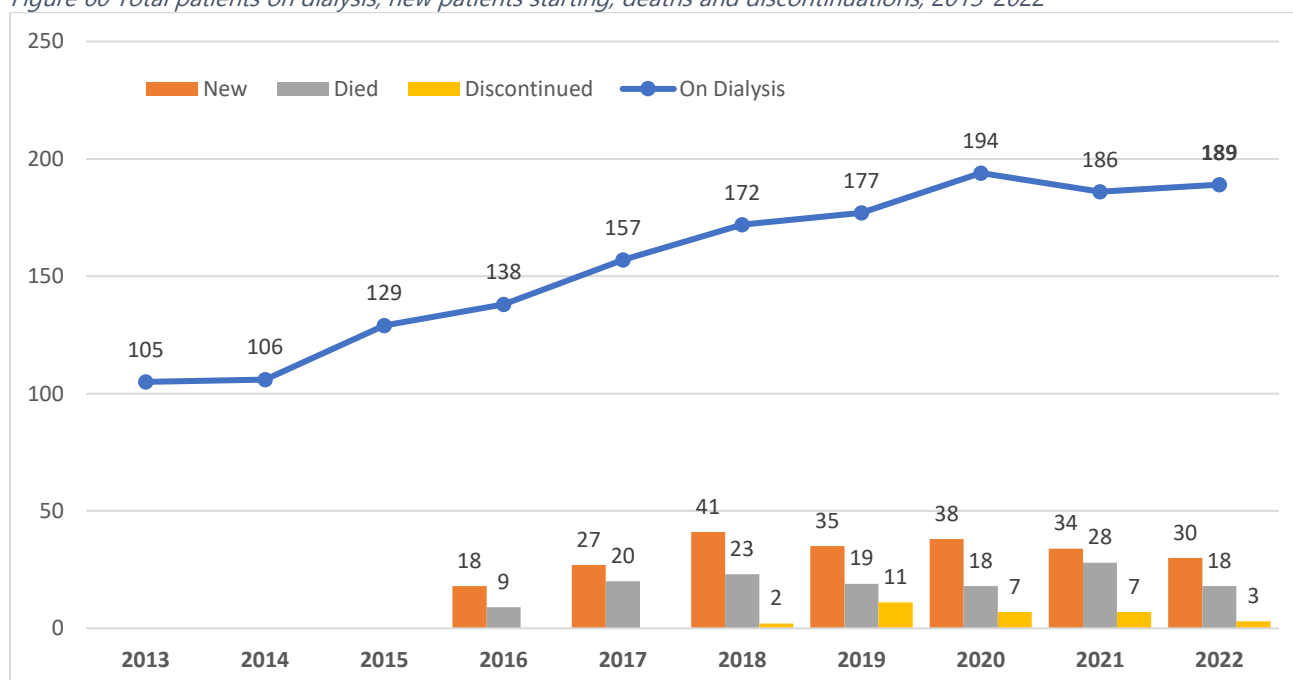
Dialysis adequacy indicators were reported to be above standards defined by the National Kidney Foundation Kidney Disease Outcomes Quality Initiative (NKF-KDOQI)⁵⁸, whereas certain patient health indicators such as prevalence of anaemia and bone metabolism have remained below NKF-KDOQI standards for the past four years. The **fraction of patients using dialysis catheters for routine dialysis remained high** (33.9% of patients) in 2022, this is not considered best practice and was likely due to inconsistent access to in-country expertise in arterio-venous fistula construction.

Using total sessions and taking costs at 200 USD per session, total estimated **costs for all dialysis sessions only are SCR 71.3 million, or 5.4% of MOH budget, in 2022**. About 136 dialysis session are conducted per year per person on long-term dialysis, taking note that this includes a small fraction of dialysis sessions which were for short-term emergency dialysis. The **equivalent cost per patient on long-term dialysis per year is SCR 377,000** (this calculation excludes other costs like for dialysis catheter insertion).

The MOH bears the costs of all blood tests, diagnostic imaging, inpatient care, most medications for dialysis patients, and management of all complications, in addition to payments AMSA Renal Care Services for dialysis services. The actual payments for 2022 were not made available at time of preparation of this report.

The Haemodialysis Unit procured additional equipment in 2022 – these included cardiac monitors, patient beds, vitals monitors and a computer.

Figure 60 Total patients on dialysis, new patients starting, deaths and discontinuations, 2013-2022



Note: Certain data points for past years have changed compared to previous annual reports.

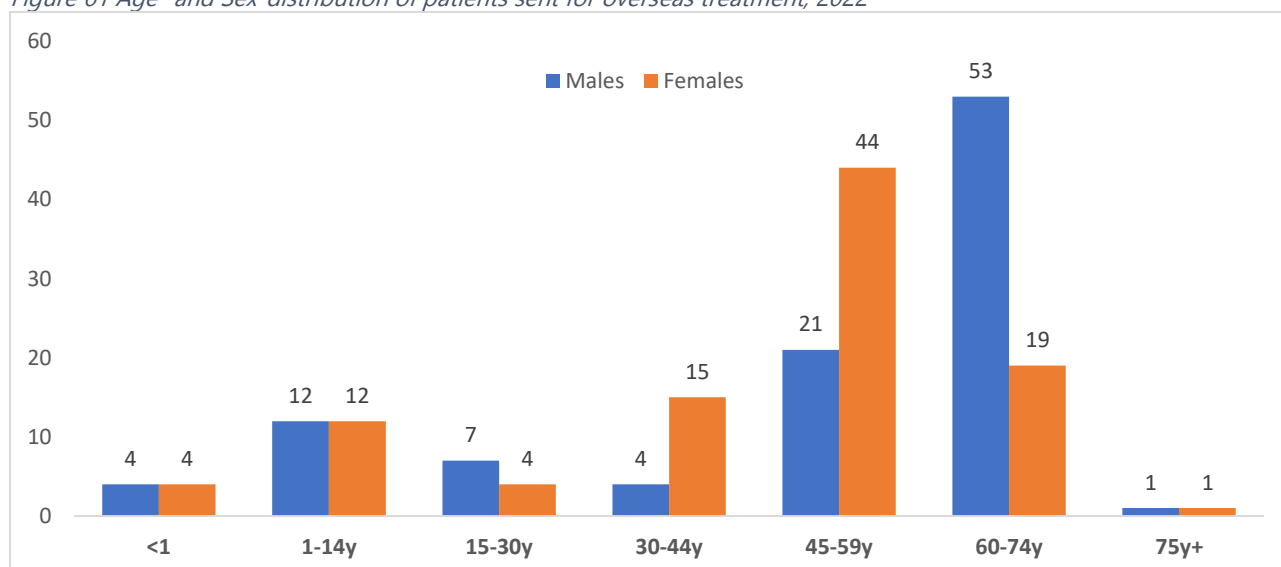
Data Source(s): AMSA Renal Care Annual Report, 2022; Annual health-sector performance report 2021 (MOH, 2022).

⁵⁸ National Kidney Foundation: KDOQI Clinical Practice Guideline for Hemodialysis: 2015 Update. Link: <https://www.kidney.org/professionals/guidelines/hemodialysis2015>.

Overseas Treatment

The MOH referred 201 patients (102 M, 99 F) for overseas treatment in 2022. Women in age-group 45-60y, and men in 60-74y, constitute the majority of those referred (Figure 61).

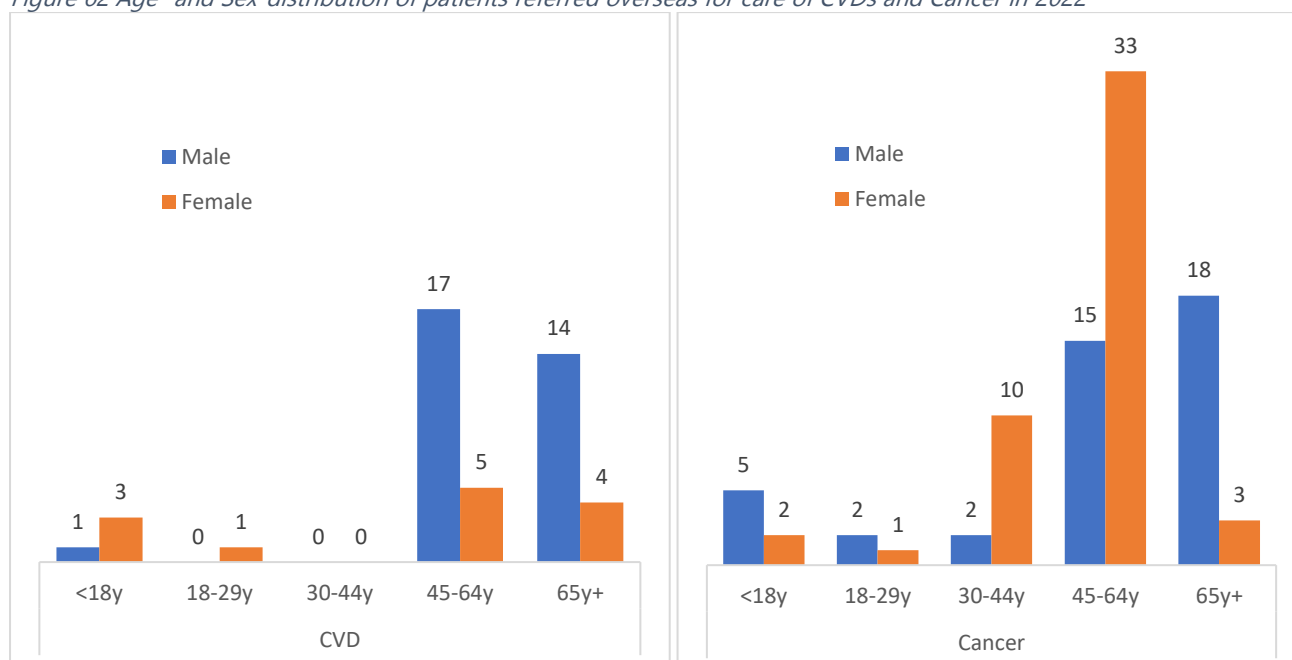
Figure 61 Age- and Sex-distribution of patients sent for overseas treatment, 2022



Data Source(s): Overseas Treatment Unit (HCA, 2023).

Cancer was the leading reason (45% of total) for referral, with breast cancer cases accounting for 20 of 49 women (40.8%) referred for cancer care. **CVD were the second leading reason** (22%), most referrals for CVD were amongst men (71% of CVD referrals), and predominantly for ischaemic heart disease (78% of CVD referrals, Figure 62).

Figure 62 Age- and Sex-distribution of patients referred overseas for care of CVDs and Cancer in 2022



Data Source(s): Overseas Treatment Unit (HCA, 2023).

The total costs borne by MOH for sending patients for **overseas treatment amounted to SCR 37.5 million** for 2022 (or 2.9% of MOH budget), **equivalent of SCR 187,000 on average per patient**. A breakdown of these costs are provided in Table 27; 91% of spending was for care costs and 9% travel-related.

Table 27 Cost estimate for Overseas Treatment, 2022

Cost Category	Sub-Total	Conversion Rate	Cost in SCR
Airfares	SCR 3,091,639	-	3,091,639
Revalidation of Airfares	SCR 122,192	-	122,192
Medical Escort	SCR 109,894	-	109,894
Treatment Costs	EUR 123,013.90	1 EUR = 17.24 SCR	2,120,759
	USD 2,086,041	1 USD = 13.83 SCR	28,849,952
	AED 670,334.10	1 AED = 3.63 SCR	2,433,313
	INR 4,631,112	1 INR = 0.168 SCR	778,028.50
Total (SCR)	-	-	37,505,777

Data Source(s): Overseas Treatment Unit (HCA, 2023).

Compared to 2020 and 2021, the number of patients referred and care costs appear to be approaching pre-COVID numbers in 2022 (Table 28), despite an increase number and variety of specialists working in the MOH in recent years. The duration of overseas stay for care in 2022 was similar to 2021 (Table 29).

Table 28 Estimated costs of overseas treatment, 2017-2022

	2018	2019	2020	2021	2022
# Patients Referred	212	210	162	134	201
Costs (millions of SCR)	41	43	54.5	32	37.5

Data Source(s): Overseas Treatment Unit (HCA, 2023); Annual health-sector performance report 2021 (MOH, 2022).

Table 29 Duration of stay for overseas treatment, 2021-2022.

	Total	1-10 days	11-30 days	31-60 days	61 days+	Not known
2021	# 134	5	52	56	14	7
	% -	4%	39%	42%	10%	5%
2022	# 201	5	89	75	25	7
	% -	2.5%	44.3%	37.3%	12.4%	3.5%

Data Source(s): Overseas Treatment Unit (HCA, 2023).

5.4.3 Emergency Services

Accident and Emergency (A&E) services are available 24x7 at Seychelles Hospital and four regional community health centres (Anse Royale and Anse Boileau Health Centres on Mahé, Baie Sainte Anne Hospital on Praslin, and Logan Hospital on La Digue).

The A&E Service of the Seychelles Hospital registered **41,153 visits** in 2022, 22% higher than 2021. One-quarter (25.7%) were during weekends, and just over half (53%) being after-hours (between 4pm and 8am). Amongst total attendees, about one-quarter (24.5%) were children (Table 30).

Table 30 Total A&E attendance 2018-2022

Year	2018	2019	2020	2021	2022
A&E Attendance	50,123	46,750	41,456	33,768	41,153

Data Source(s): Hospital Services Annual Report 2021 and 2022, Statistics Unit, MOH.

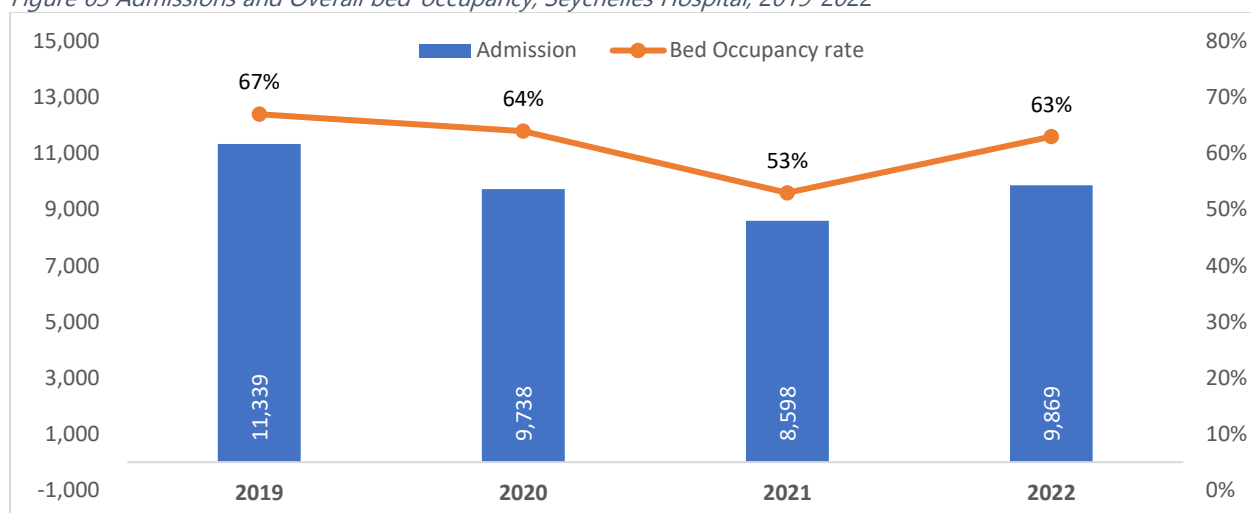
5.4.4 Inpatient Services

Ward Admissions

The **Seychelles Hospital recorded 9,869 admissions** in 2022, representing a 15% increase over 2021, giving an **average bed occupancy rate of 63%** in 2022, compared to 53% in 2021 (Figure 63).

Bed occupancy varied across wards (Table 31), **highest being in Male Medical Ward (90%) and Intensive Care Unit (ICU, 91%); lowest in Paediatric Ward (51%) and Neonatal ICU (37%)**. Shortest average length of stay was in Paediatric Ward, longest in Psychiatric Ward.

Figure 63 Admissions and Overall bed-occupancy, Seychelles Hospital, 2019-2022



Data Source(s): Statistics Unit, MOH.

Table 31 Admissions, average length of stay and occupancy by ward, 2022

Ward(s)	Admissions 2022	Average Length of Stay (days)	Bed Occupancy			
			2022	2021	2020	2019
Male Medical Ward	1,263	6.8	90%	79%	90%	89%
Female Medical Ward	1,094	6.2	62%	52%	54%	53%
Paediatric Ward	1,368	3.5	51%	35%	47%	66%
Neonatal ICU	114	12.0	37%	37%	35%	39%
Maternity Ward	1,758	4.2	67%	70%	69%	76%
Male Surgical Ward	1,722	4.9	56%	43%	57%	66%
Female Surgical Ward	1,982	4.4	55%	37%	53%	62%
Intensive Care Unit (ICU)	323	6.2	91%	68%	85%	75%
Psychiatric Ward	228	18.4	63%	71%	89%	87%
Total (Seychelles Hospital)	9,869	7.4	63%	53%	64%	67%

Data Source(s): Annual Hospital Services Report, 2022 (Statistics Unit, MOH, 2023).

The **Family Hospital (FH-ITC)** was repurposed into a **COVID-19 Isolation Centre** from mid-2020 to mid-2022. In 2022, it received **197 admissions** (Jan-Jun), with an average bed occupancy of 31%. Thereafter, maternal-child health related activities resumed gradually at this facility. Part of the facility was kept reserved in the latter half of 2022 to act as an isolation centre for Mpox cases, if any were detected locally.

Table 32 shows average occupancy by month, showing the **large month-to-month variations** across wards. ICU had over 90% average occupancy for 7 of 12 months in 2022, and over 100% on three of these. Male medical ward had over 90% occupancy on 8 of 12 months, and exceeded 100% on two of these.

Table 32 Average bed occupancy in each ward, by month, 2022

2022	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
ICU	103%	74%	60%	94%	97%	105%	75%	94%	123%	95%	79%	83%
HW	34%	44%	54%	55%	54%	59%	54%	61%	71%	62%	51%	49%
DW	39%	65%	60%	54%	58%	65%	48%	62%	64%	52%	62%	51%
MAT	64%	63%	80%	84%	68%	46%	64%	72%	73%	61%	69%	58%
FMW	50%	63%	64%	64%	60%	69%	75%	58%	61%	72%	60%	53%
MMW	72%	91%	77%	83%	92%	98%	91%	102%	105%	92%	95%	83%
PW	33%	41%	49%	50%	52%	56%	58%	60%	45%	55%	58%	51%
NICU	35%	49%	42%	36%	41%	34%	38%	35%	34%	39%	28%	55%
PSY	55%	72%	71%	46%	50%	66%	46%	51%	71%	85%	60%	68%
FH-ITC	46%	27%	23%	40%	35%	17%						

PSY – Psychiatric Ward; MAT – Maternity; Data Source(s): Statistics Unit, MOH.

Operating Theatre

The Seychelles Hospital houses six operating theatres, three for minor surgeries only, and three that can be used for minor and major surgical procedures. **In 2022, 5,211 surgical procedures were performed (56% F, 44% M), an increase of 54% over 2021, and above pre-pandemic levels** (Table 33). This increase could be explained by need to clear backlogs from periods where elective surgery was interrupted, especially in 2021.

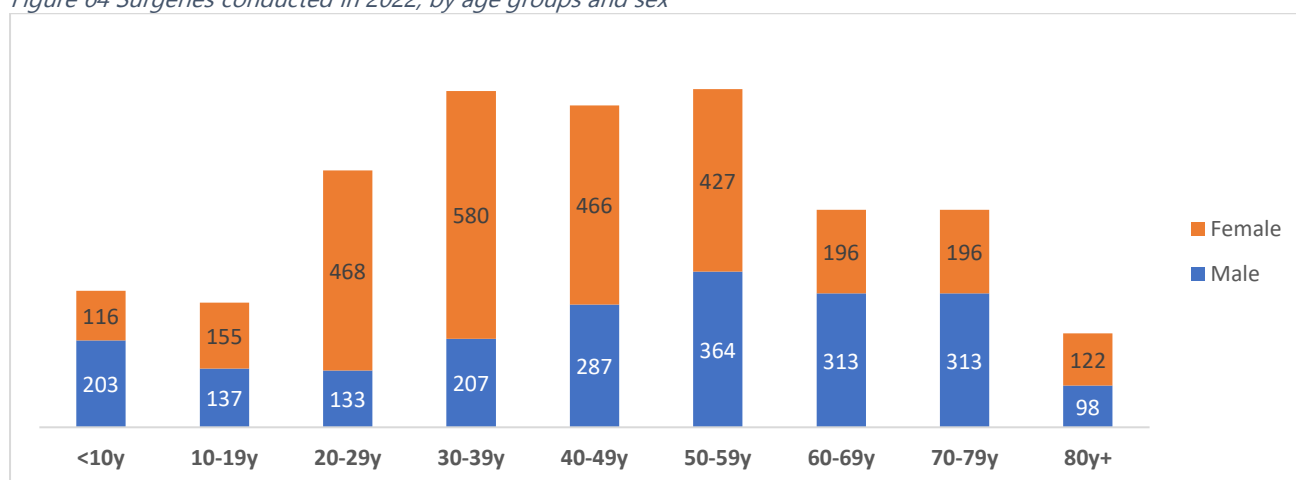
Table 33 Total number of surgeries conducted at Seychelles Hospital, 2019-2021

Year	2019	2020	2021	2022
# of Surgical procedures performed	4,242	3,559	3,337	5,211

Data Source(s): Hospital Services Annual Report, Statistics Unit, MOH.

Age- and sex- distribution of patients undergoing surgery are indicated in Figure 64. It is notable **that in the reproductive age-groups, many more women undergo surgeries, whereas in children under age 10 and adults 60y and above, more men undergo surgical procedures.**

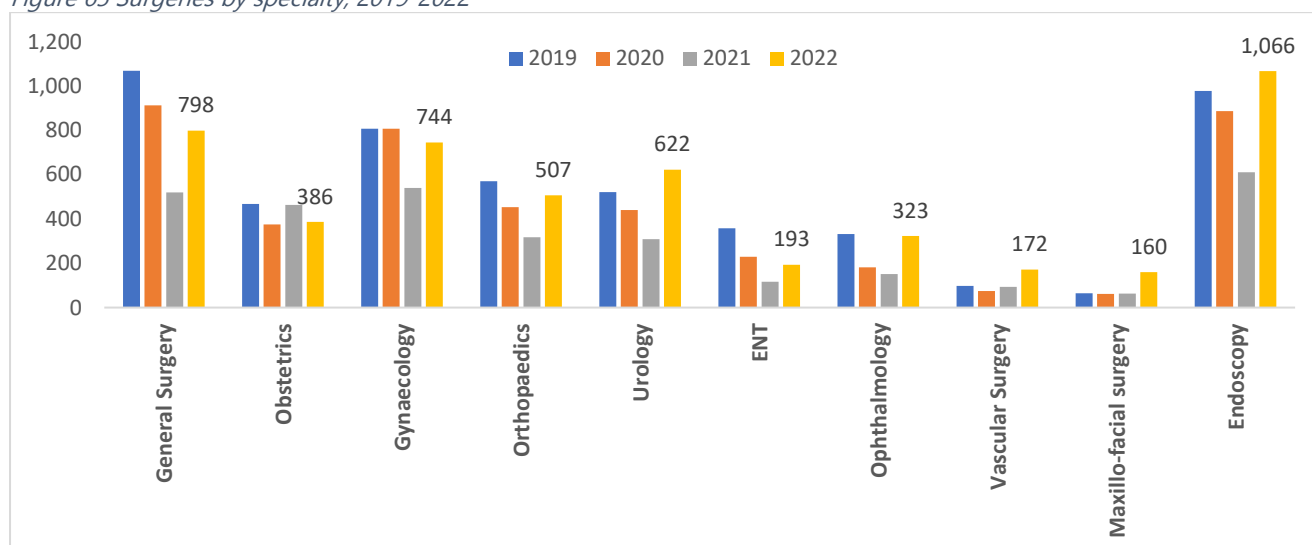
Figure 64 Surgeries conducted in 2022, by age groups and sex



Data Source(s): Hospital Services Annual Report, Statistics Unit, MOH.

Figure 65 shows number of **surgeries by specialty – the majority saw an increase** in 2022 compared to 2021, reaching pre-pandemic levels, and serving as evidence of full restoration of surgical services. For additional information, please refer to Annual Hospital Services Statistical Report 2022 (MOH, 2023).

Figure 65 Surgeries by specialty, 2019-2022



Data Source(s): Hospital Services Annual Report, Statistics Unit, MOH.

5.4.5 Other Services

Services for harmful use of substances

Substance abuse continues to take a heavy toll on the health and wellbeing of families and communities (NHSP objective 2.1.5). The Prevention of Drug Abuse and Rehabilitation Agency Act (Act 13 of 2017) was repealed in October 2021, followed by the dissolution of Agency for Prevention of Drug Abuse and Rehabilitation (APDAR). All functions of APDAR were formally transferred to the MOH in 2022, under a new Division for Substance Abuse Prevention, Treatment and Rehabilitation (DSAPTR), with 87 staff members.

Services are provided at four 'Methadone-Assisted Treatment' or MAT clinics, located at English River, Prison, Praslin and La Digue, as well as through mobile vehicles across all districts. Of the 4,391 cumulative clients (3,838M, 553F) registered with DSAPTR, **3,166 clients accessed services in 2022** (72%). The majority were for **Low-Threshold Methadone Maintenance Programme** (LTMMMP, 80% of total, Table 34). Across the three main islands, 2,676 (85%) of clients were on Mahé (excluding Prison), 112 (4%) at Mt Posée Prison, 249 (8%) on Praslin, and 93 (3%) on La Digue.

Nineteen percent of clients on LTMMMP (474 of 3,166) were lost to follow-up, with 483 new clients joining in the same period (19% of cohort). This indicates that one in five users dropped out of the programme, and an equivalent number re-entered, in 2022.

Table 34 New registrants and total active service users on register, 2018-2022

Year	2018 or earlier	2019	2020	2021	2022
New registrants	1,537	740	651	304	266
Active clients on Register	-	2,740	3,431	3,714	3,166
Of which for LTMMMP	-	2,435 (89%)	2,872 (84%)	3,126 (84%)	2,546 (80%)
Of which for other services	-	305 (11%)	559 (16%)	588 (16%)	620 (20%)

Data Source(s): DSAPTR Annual Report 2022 (May 2023); National Drug Observatory Report 2022 (MOH, 2022).

Active DSAPTR service users also accessed **High-Threshold Methadone Maintenance Programme** (HTMMMP, 232, 7%), treatment with buprenorphine (6), symptomatic detoxification (35) and services addressing relapse. Amongst clients on HTMMMP, 17% (38 of 232) completed treatment in 2022; whereas 34 (15%) were lost to follow-up.

Urine drug tests conducted on clients give insights into types of substances in use. Based on 1,292 tests conducted in 2022 on 990 clients. Of these, 63% of tests were done as part of client registration process, amongst these 80% of clients tested positive for opiates, 71% for cannabis, 32% for cocaine; **majority (93%) were poly-drug users** (using two or more illicit substances simultaneously). Twelve percent of clients reported injection drug use, of whom about one-quarter reported stopping injection drug use whilst on DSAPTR programmes (Figure 66).

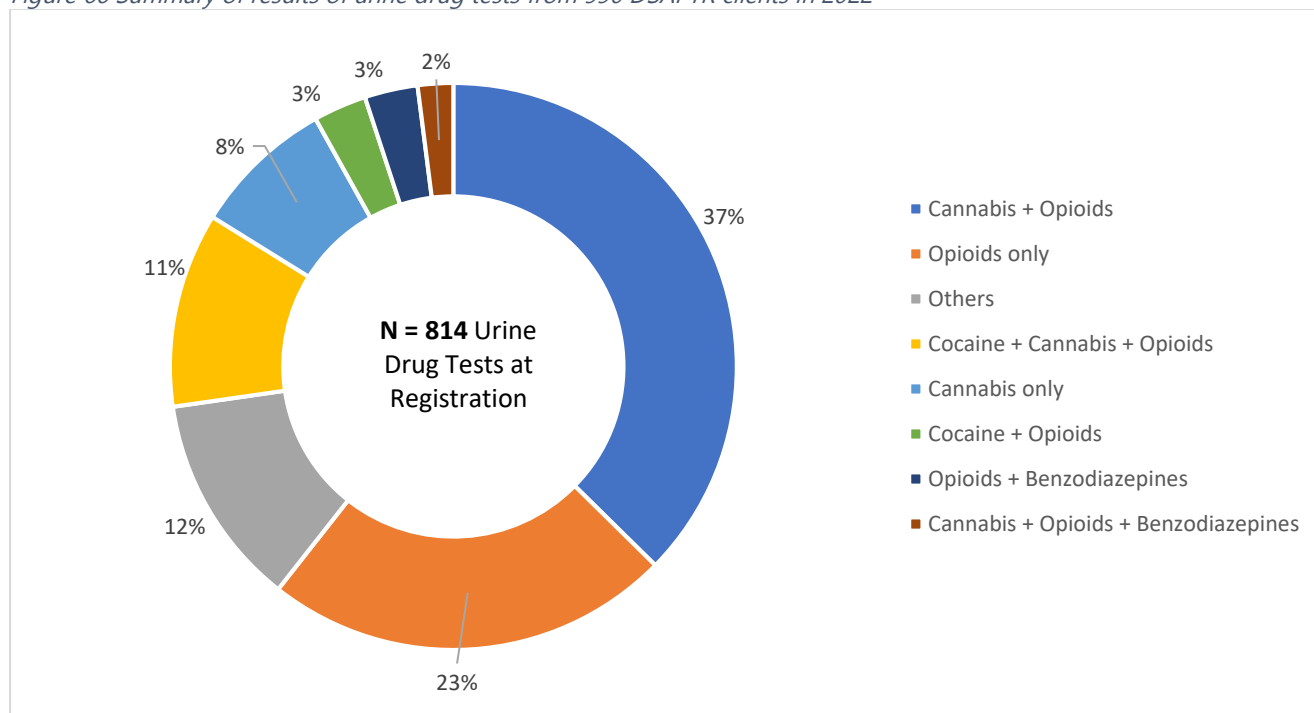
HIV and HCV testing conducted in 2022 shows 3% and 28% of clients tested (N=278) were reactive for HIV and HCV, respectively. HIV and viral hepatitis testing coverage was very low for this key population (only 8.8% of clients tested in 2022).

The **Needle-syringe exchange programme** offered by DSAPTR **distributed 11,038 needles and syringes** in 2022. Based on the 2017 survey⁵⁹ the injection users of Heroin was estimated at 2,144 – this means about

⁵⁹ Seychelles Biological and Behavioural Surveillance of Heroin Users 2017 (APDAR, 2018).

thirty clean needles were distributed per person who injects drugs (i.e. 12% of cohort), well-below the recommended >100/user/year minimum (ideally >300) to reduce complications of injection drug-use⁶⁰.

Figure 66 Summary of results of urine drug tests from 990 DSPATR clients in 2022



Data Source(s): DSPATR Annual report 2022 (MOH, 2023).

Only a few clients accessed services for other substance use disorders – 295 for alcohol-use disorders (9% of total), 17 for cannabis-use disorders (0.5%) and 26 for other reasons. Of those enrolled in services for alcohol-abuse, only three successfully completed their treatment in 2022. A total of 864 clients accessed psychosocial services in 2022.

Communicable Disease Control Unit

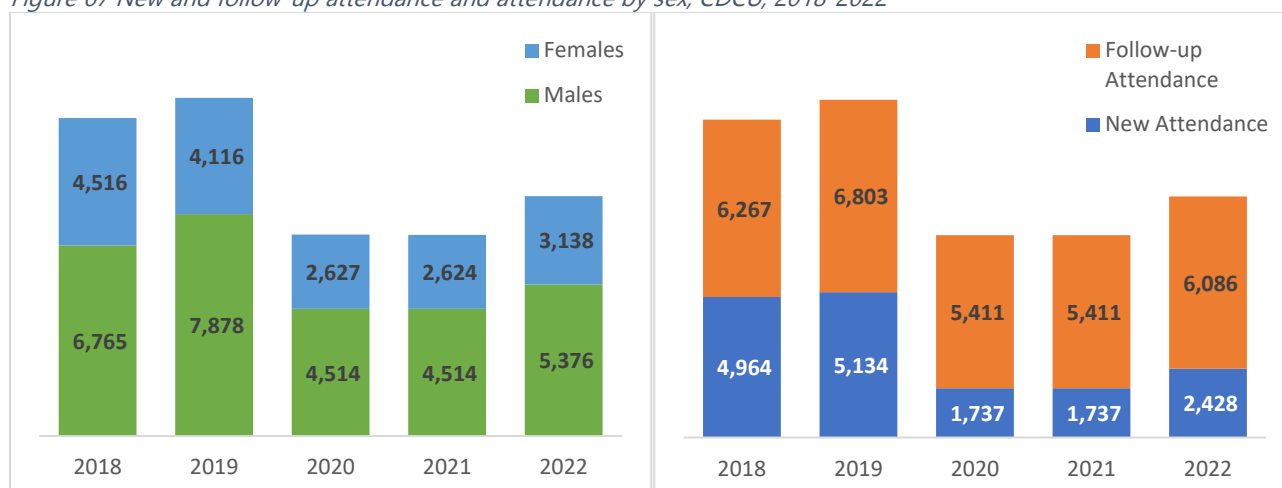
A total of **8,514 attendances were registered** at Communicable Disease Control Unit (CDCU). Of these, about **34% were for HIV-related care, and 26% for accessing the needle-syringe programme**. Figure 67 shows recent trends in CDCU attendance – utilisation of services remained below pre-pandemic levels in 2022.

There were 45 new attendance for Post-Exposure Prophylaxis for HIV (PEP) and five for Pre-Exposure Prophylaxis for HIV (PrEP). Needle syringe programme provided by the CDCU registered 27 new users in 2022, and 2,211 follow-up attendances (Figure 68), a significant increase compared to 2020 or 2021.

For cases and deaths from communicable diseases, see sections 3.2 Causes of Death, 3.3 Morbidity and 4.1 Key Disease Outbreaks.

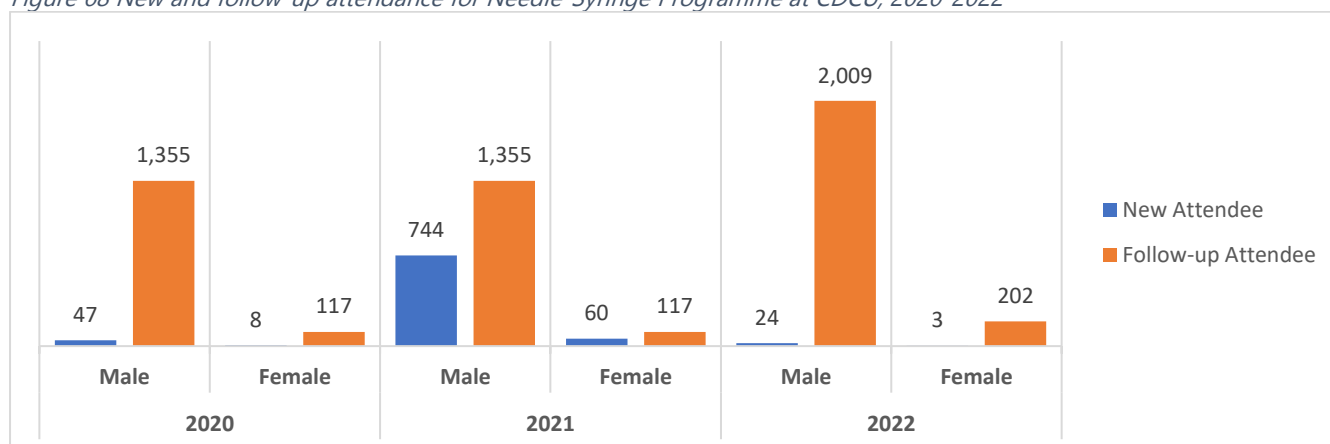
⁶⁰ UNAIDS (2023): Indicator Registry – Needles and syringes distributed per person who injects drugs. Link: <https://indicatorregistry.unaids.org/indicator/people-who-inject-drugs-prevention-programmes#:~:text=Medium%3A%20100%E2%80%93%20syringes%20per,who%20injects%20drugs%20per%20year> (Accessed 4 June 2023).

Figure 67 New and follow-up attendance and attendance by sex, CDCU, 2018-2022



Data Source(s): CDCU Annual Report 2022 (PHA, 2023); Statistics Unit, MOH.

Figure 68 New and follow-up attendance for Needle-Syringe Programme at CDCU, 2020-2022



Data Source(s): CDCU Annual Report 2022 (PHA, 2023); Statistics Unit, MOH.

Occupational Health Unit

The Occupational Health Unit (OHU) primarily provides relevant medical assessments and testing for work- or education-related requirements, and for social welfare support. In 2022, the unit registered **4,772 attendances (2,683, 58% M, 2,089, 42% F)**, recovering somewhat after a low in 2021 but still below 2019 figures (Table 35). Of these, 40% (1,910 attendees) were for tests and evaluations required for food handler's license, followed by 11.2% (534) for assessments related to welfare assistance. Apart from occupation-related and GOP-related medical assessments and assessments for social welfare assistance, other occupational health and safety activities were minimal.

Table 35 Occupational Health Unit total attendance 2018-2022

	2018	2019	2020	2021	2022
Attendance to OHU	6,632	6,521	4,873	3,978	4,772

Data Source(s): Annual Public Health Statistical Report 2022 (unpublished) (MOH, 2023).

Rehabilitation services

Rehabilitation is considered an integral part of Universal Health Coverage. Rehabilitation Services help individuals minimize the loss of physical and cognitive function resulting from illness or injury and therefore are essential for maximizing patient outcomes and ensure the population improves their quality of life.

Multidisciplinary teams (36 Physiotherapists, 17 occupational therapists, one speech pathologist, four Orthotic/Prosthetic technicians, one audiologist, and four acupuncturist) with rehabilitation expertise provide necessary management and care.

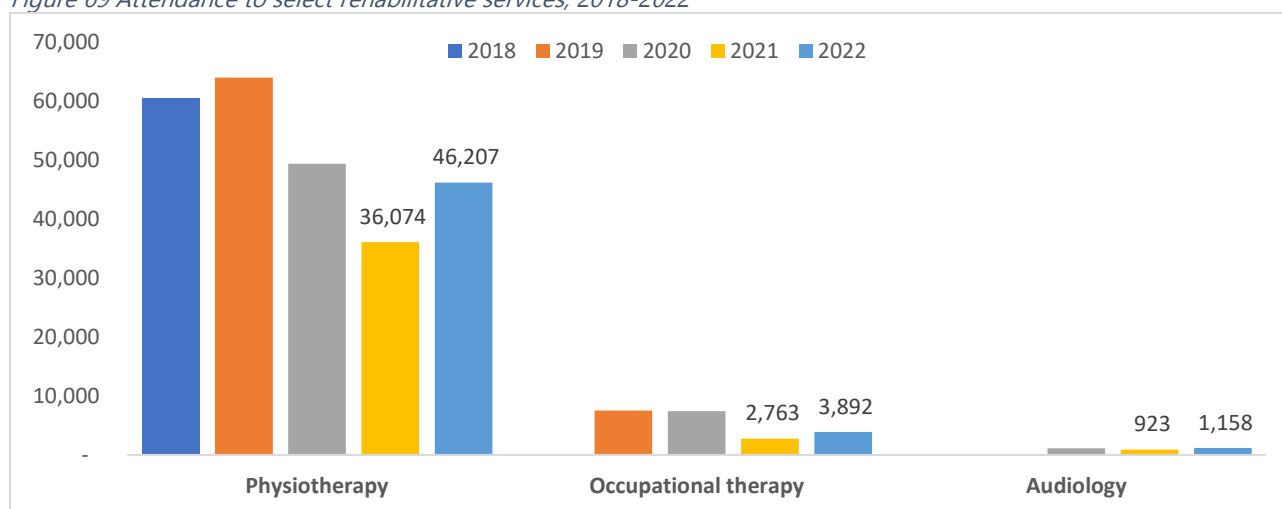
Available attendance for 2022 by rehabilitative service area is shown in Figure 69. Of the total attendance for the respective service, 5,879 (12.7% of total attendance) were new referrals to physiotherapy, and 376 (9.7%) new to occupational therapy. Refer to Annual Primary Care Services Report 2022 (MOH, 2023) for more details.

Data for attendance for speech pathology, orthoprosthesis and acupuncture services were not available.

Podiatry services were discontinued in 2022 (in consequence to scale-down of cooperation medical personnel from Cuba). There was an **increase in home visits** conducted – 1,046 in 2022 compared to 395 in 2021.

A **rehabilitative services strategy** is in late stages of development, developed with the support of WHO. Its implementation will improve guidance for rehabilitative care across the country (NHSP objectives 1.1 and 2.1).

Figure 69 Attendance to select rehabilitative services, 2018-2022



Data Source(s): Primary Healthcare Services Annual Statistical Report, 2022 (MOH, 2023).

Oral Health Services

The Oral Health Services (OHS) within the HCA provides preventive, restorative, and rehabilitative dental services free at the point of use to the population. OHS has a number of dental facilities that include seven school-based clinics, one at the North East Point hospital, 11 community dental clinics (housed within respective community health centres), and one at the Yellow Roof building of Seychelles hospital.

The utilisation of publicly provided oral health services remains high although there are a number of private dental service providers (Table 36). Total **attendance in dental clinics for 2022 was 51,505**, which includes cases seen in operating theatre (92 cases). There were 197 dental cases seen after working hours.

The **dental specialties expanded** in 2021 and 2022, from two specialist services in 2019, to five dental specialist services in 2022. **No oral health services were provided at school campuses** in 2022, or at the Montagne Posée prison. The service is one of a handful of clinical service areas that implements an effective appointment system for patient visits (NHSP objective 2.1.2).

Table 36 Attendance to Oral Health Services, 2019-2022

	2019	2020	2021	2022
Community Dental Services (Total)	65,248	49,945	29,856	51,505
Dental Hygienist	9,477	-	-	4,147
Dental Therapist	18,293	-	-	16,965
Dental Officer	37,478	-	-	30,393
Specialist Dental Services (Total)	13,565	9,209	6,912	13,949
Oro-Maxillo-Facial Surgery	6,536	-	-	5,517
Orthodontist	7,029	-	-	3,748
Periodontist	-	-	-	1,808
Endodontist	-	-	-	1,079
Paediatric Dentist	-	-	-	1,797

Data Source(s): Oral Health Services (HCA, 2023).

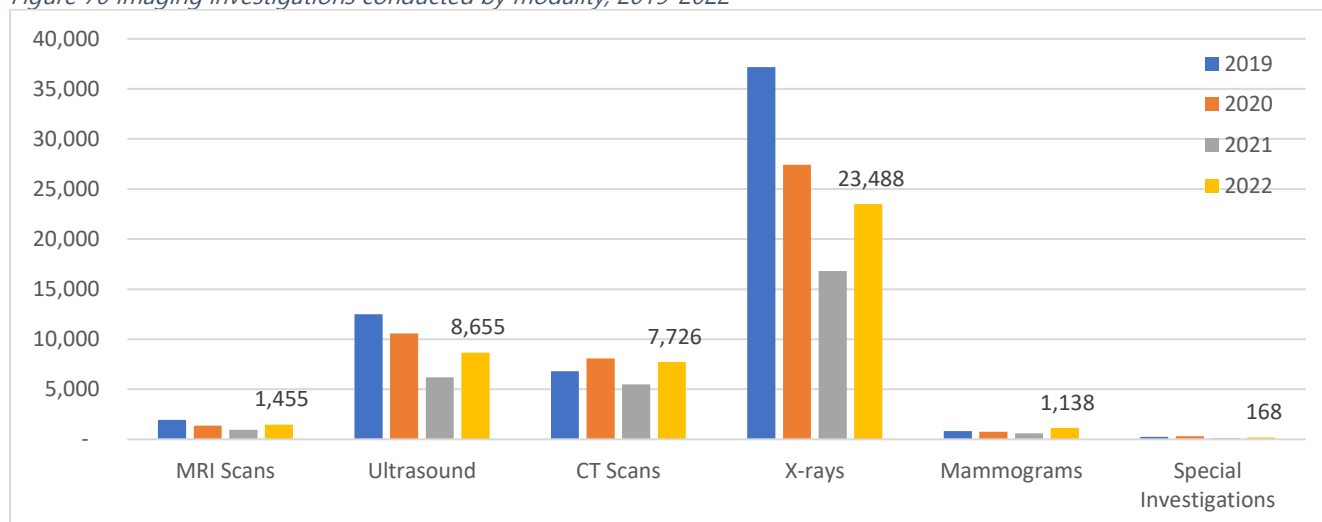
5.4.6 Clinical Support Services

Imaging

Imaging services are available 24x7 at Seychelles Hospital. There were **42,630 imaging investigations** conducted in 2022. Investigations by modality from 2019-2022 are shown in Figure 70; utilisation level appears to have surpassed 2019 levels for most modalities. About three-quarters of investigations were conducted during normal working hours, and 25% during after-hours and weekends.

More mammograms (1,138) were conducted in 2022 compared to 2019-2021, indicating progress towards increased screening for breast cancer, but it remains well-below the NHSP target (at 46% only) of at least 2,500/year.

Figure 70 Imaging investigations conducted by modality, 2019-2022



MRI – Magnetic Resonance Imaging; CT – Computer-assisted Tomography.

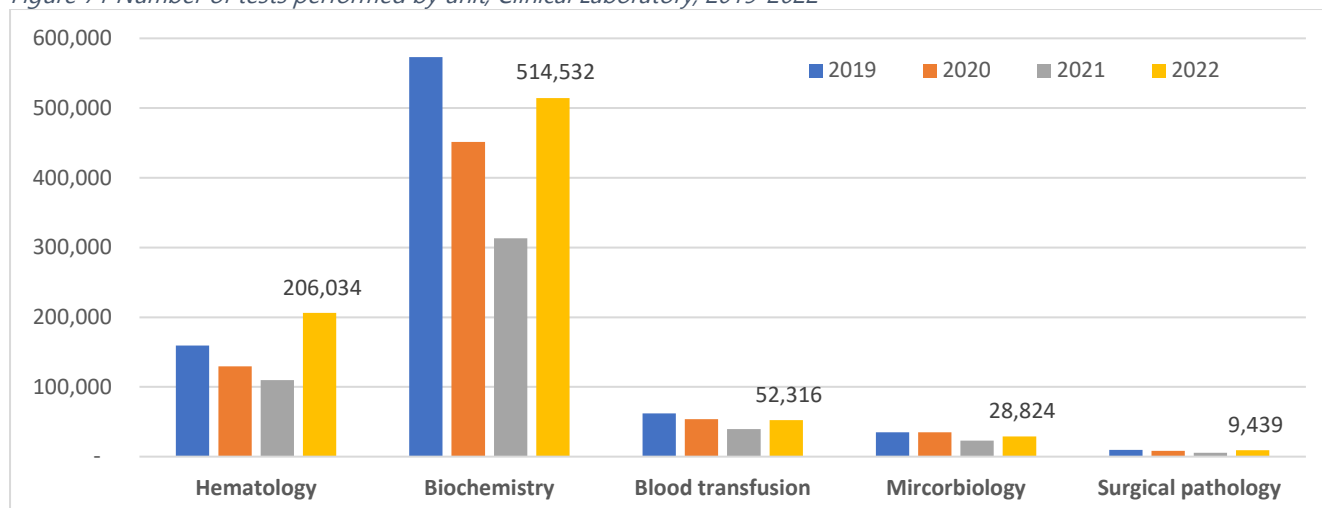
Data Source(s): Annual health-sector performance report 2021 (MOH, 2022); Annual hospital services report 2022 (MOH, 2023).

Clinical Laboratory

The number of **tests performed at the Clinical Laboratory in 2022 was 811,145, an increase of 65%** compared to 2021 (491,186), and comparable to 2019. As shown in Figure 71, the Biochemistry Unit performed 63% of the total tests, followed by the Haematology Unit (25%).

The **Pathology Unit conducted 172 (119 M, 53 F) post-mortem examinations** in 2022, a 19% increase from the 2021 (101 M, 44 F). More than half post-mortems conducted were police cases.

Figure 71 Number of tests performed by unit, Clinical Laboratory, 2019-2022

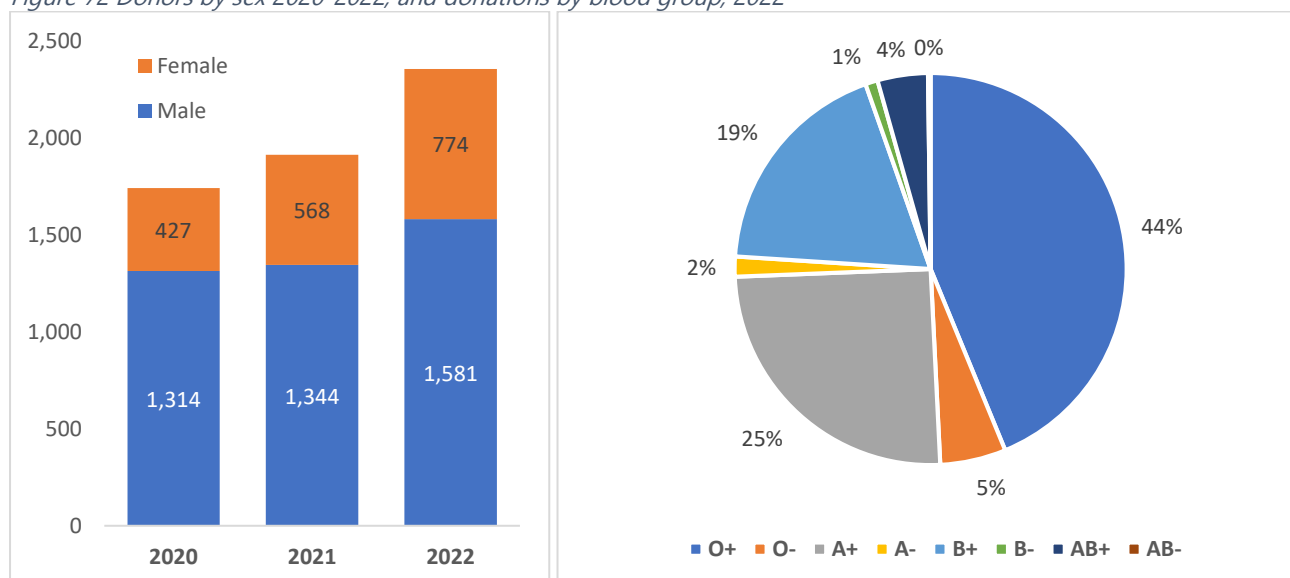


Data Source(s): Annual Hospital Services Report 2022 (MOH, 2023).

Blood Transfusion service noted a 23% increase in blood donations in 2022 over 2021 (Figure 72). Although the majority of donors are males (67.1%), the **proportion of women donating has increased** serially from 2020 to 2022. In terms of blood group, blood group O+ accounted for most (44%) of blood donated.

Amongst the 2,442 donors screened, 793 (33%) were regular donors, 516 (21%) were first-time (non-family-replacement) donors, and 1,133 (46%) were 'family-replacement donors' (referring to donors coming to give blood for specific family or friend in-need of blood). This pattern is similar to 2021.

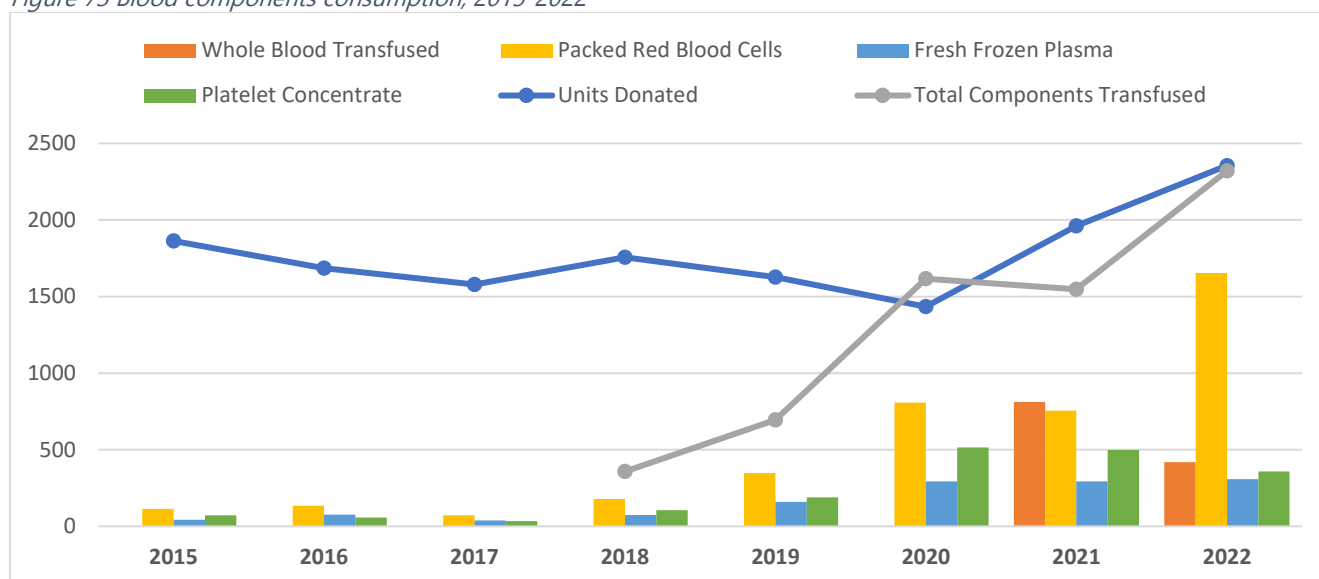
Figure 72 Donors by sex 2020-2022, and donations by blood group, 2022



Data Source(s): Annual Hospital Services Report 2022 (MOH, 2023).

Of collected blood, 418 units of donated blood were transfused as whole blood in 2022; **the majority of blood utilised was in the form of blood components**: 1,653 packed cells, 308 Fresh Frozen Plasma and 359 Platelet Concentrates. This is a pattern similar to preceding years (Figure 73). ICU remains the highest consumer of blood and blood products (23% of total used), followed by adult surgical and medical wards – see Annual Hospital Services Report 2022 (MOH, 2023).

Figure 73 Blood components consumption, 2015-2022



Data Source(s): Annual Hospital Services Reports, 2020 to 2022 (MOH, 2023); Annual health-sector performance report 2021 (MOH, 2022).

Infection Prevention and Control

The MOH has an active Infection Prevention and Control (IPC) programme, which continued to conduct related training for HCWs across the public and private health sectors in 2022. Work of the programme is guided by the **Infection Prevention and Control Strategic Framework 2021-2025** (MOH, 2021), with the IPC Committee providing oversight. The key objectives of this framework are:

- Institutionalize IPC programmes at national and facility levels with clearly defined functions.
- Promote enabling environment to support the implementation of IPC, including top leadership and health facility managers' commitment.
- Ensure the availability of evidence-based national IPC guidance documents for service delivery, including management of public health emergencies.
- Ensure that all HCWs are knowledgeable on IPC related to their job function.
- Harmonize system for IPC information management and ensure the availability of quality data and information to guide decisions.
- Continuously share lessons to help promote best practice.
- Improve Hospital Acquired Infection and Anti-microbial resistance surveillance.
- Optimize the built environment and ensure the continuous availability of essential infrastructure, materials and equipment needed to support implementation of effective IPC measures.

Assessment of healthcare facilities' adherence to IPC protocols was conducted across the MOH in 2022 (23 hospital service areas/wards/units, and 19 community health facilities), the overall score was rated as intermediate (51% average score for hospital-based service areas, 52% average for community health centres), leaving room for further improvement.

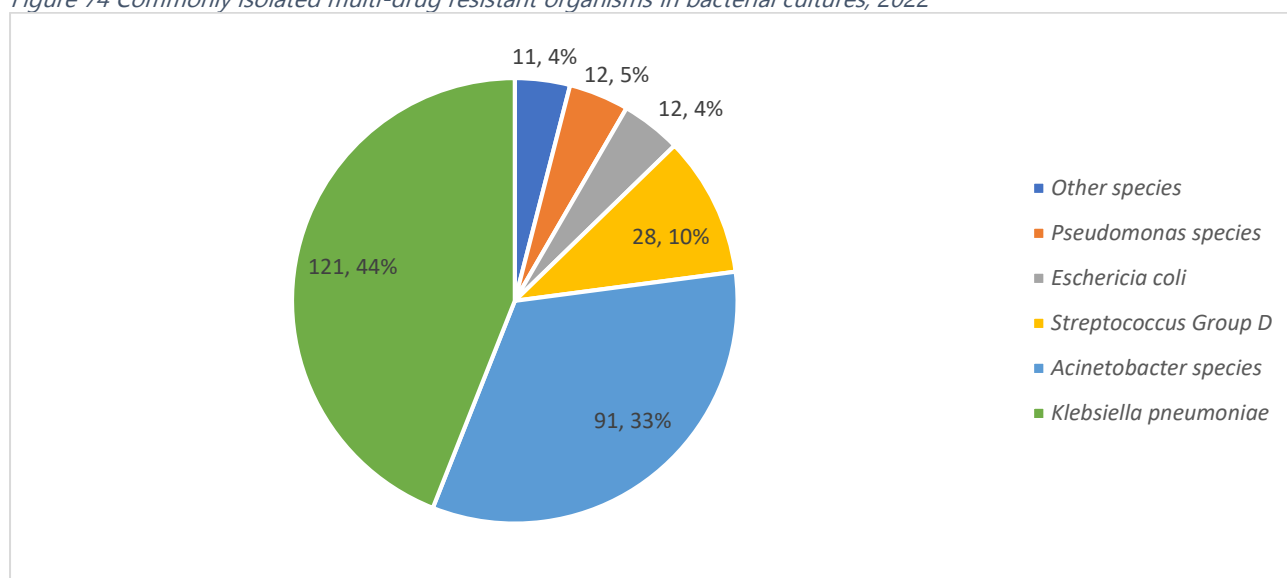
Hand hygiene audits conducted across 39 clinical areas and 20 cadres of staff (including non-health professionals), in partnership with WHO, show:

- Hand-rub (with sanitiser) score was 73 on average (range 43 to 100).
- Hand-wash (with soap and water) score was 78 on average (range 44 to 100).
- WHO IPC facilities score-card showed an overall score of 64 for facilities assessed.
- Results were disseminated to facility managers and IPC focal persons.

Surveillance of organisms associated with multi-drug resistance was conducted in conjunction with clinical laboratory and SEIS in 2022. It showed that out of 12,895 specimens sent for culture in 2022, 275 samples (2.2%) grew organisms that were multi-drug resistant (Figure 74).

Klebsiella pneumoniae were prominently found in urinary catheter tip samples whilst *Acinetobacter baumannii* were found predominantly in culture of endotracheal tubes or secretions. The majority of these cases were recorded in the ICU and Medical wards. See section 4.2.2 Surveillance and Epidemic Intelligence for additional details of anti-microbial resistance in Seychelles.

Figure 74 Commonly isolated multi-drug resistant organisms in bacterial cultures, 2022



Data Source(s): IPC Unit Annual Report 2022 (HCA, 2023); Clinical Laboratory –Microbiology Department (HCA, 2023).

5.4.7 Quality of Care Improvement

The NHSP (objectives 2.2.1 and 2.2.2) outlines key steps for quality improvement across the health system, including development of a quality framework, establishing standards of care and quality-of-care (QoC) indicators, promoting antibiotic stewardship and improving capacity to monitor efficiency and effectiveness of services.

The HCA set-up a **QoC Committee** in September 2022, tasked with developing a QoC strategy and operational plan (NHSP objectives 1.8 and 2.2.1). The committee is working in partnership with the WHO and University of Oxford to build capacity of the task force members, including through overseas training workshops for five members.

The committee identified three specific areas to support quality of healthcare delivery⁶¹: (1) **Colorectal Cancer Screening** (see Cancer), (2) **SEY-PEN** (see Diabetes) and (3) **improving A&E services**, where an audit was initiated.

Other notable initiatives across the MOH towards improving quality of services include:

- New treatment guidelines introduced for prevention of venous thromboembolism in hospitals.
- The Directly acting Oral Anti-Coagulant (DOAC) Rivaroxaban was introduced for treatment of Deep Vein Thrombosis and Pulmonary Embolism. Although these medications are more expensive than the

⁶¹ Internal communication from Chairperson of QoC committee.

traditionally used Warfarin, they do not require as close monitoring of the patient and the level of anti-coagulation is more stable, thus saving costs on other aspects of care⁶².

- The anti-diabetic medication Sitagliptin (a dipeptidyl peptidase 4 inhibitor) was also introduced as part of treatment options for certain patients with Diabetes Mellitus (Type 2).
- The Cancer Registry is upgrading its software systems.

5.4.8 Key points in Service Delivery

The MOH continues to provide an elaborate range of health services for the population, free at point of use and registering high utilisation. **Service utilisation levels have partially recovered in 2022**, with many facilities registering attendance similar to, or sometimes exceeding, pre-pandemic levels. The **very high utilisation of clinical laboratory diagnostic services requires further investigation**.

A full new management team for community health services was already spearheading the operationalisation of the PHC package, with initiation regular engagement with hospital services management to improve care coordination. Hospital services management posts were unfilled at end 2022.

MOH had begun to **operationalise new programmes** defined under the PHC package. **Use of harm-reduction services through the LTMMP remain high**, adequate for impact on population health and wellbeing. But utilisation of services for alcohol and other substances of abuse, and the needle-syringe programme, remained far too low for population-level impact.

Gaps exist in availability of **care standards and clinical guidelines**, including adherence to these guidelines and recommended medications; and they do not inform procurement of medicines and medical supplies.

Work of the QoC committee will contribute towards **improving patient experience and outcomes**, including through the development of a QoC framework.

Changes to our **service delivery models** can support improved integration of services, seamless referral mechanisms across the system, and improve clinical experience for both patients and staff, whilst contributing towards improving UHC. There is continued need for effective upstream interventions, including effective public health services (and laws) and evidence-based health promotion.

Not measuring equity parameters in health services and outcomes remains a barrier to identifying and understanding any specific population groups that are left behind.

⁶² Introduction of Direct acting oral anticoagulants (DOACs) in HCA: Cost-effectiveness and benefits versus warfarin (Medicines and Therapeutics Committee, HCA, 2022).

6 Progress towards NHSP and SDG3 Goals

WRITER(S): **ZACCHEAUS AYELEHIN**; **SANJEEV PUGAZHENDHI**; **AGNES CHETTY**;

The world is facing cascading and intersecting crises that are threatening progress towards the global goals. The COVID-19 pandemic has wreaked havoc across the SDGs and erased more than four years of progress towards reducing poverty⁶³. Together with Climate Change, conflict and insecurity, increasing National debt⁶⁴, and growing inequalities, they threaten food security and decades of progress on global health.

The need for accelerated action in health and across society is imperative to recover lost gains and resume the country's progress towards the SDGs. The health sector continues to plan and execute actions across the health system to fully recover from the pandemic challenges, and move towards increased resilience.

6.1 Universal Health Coverage

SDG 3.8 focuses on achieving universal health coverage (UHC), which means ensuring that all individuals and communities have access to essential health services without suffering from financial hardship. This goal aims to promote equitable and affordable healthcare, improve health outcomes, and provide protection against health emergencies. UHC underpins all other SDG 3 indicators.

Seychelles has made significant progress towards achieving SDG 3.8. People have access to a wide variety of preventive and curative services, free at point-of-care, through geographically de-centralised public health facilities. MOH is now prioritising improvement of quality-of-care.

6.1.1 The UHC Index

The UHC index is composed of two sub-indicators:

- **Average Service Coverage (ASC) index (SDG 3.8.1)**, an index derived as a composite score from set of 15 indicators across four domains.
- **Financial Hardship index (SDG 3.8.2)**, which estimates the fraction of population at risk for catastrophic health expenditure, typically at two thresholds – 10% and 25% of household income.

The latest global estimates show an increase in the **UHC-ASC index** for Seychelles, from 70% in 2019 to **75%** in 2021⁶⁵. The MOH undertakes the UHC-ASC calculation using primary data, 2022 estimate was 85% (i.e. 0.85), almost unchanged from 2021 (Table 37).

Financial risk protection continues to be estimated using 2013 household budget survey, and remains at **2.6%** risk at 10% threshold, and 1.3% at 25%, values that indicate very high financial risk protection (also see section 5.1 Health Financing). Past NHA data consistently estimated about one quarter of health spending was out-of-pocket (NHA 2016-2018).

The **UHC index has certain shortfalls** – it does not measure **quality** of services and **equity** of access to them.

⁶³ Foreword by Dr Tedros in World Health Statistics Report 2023 (WHO, 2023)

⁶⁴ IMF Country Report No. 22/374: IMF Executive Board Completes Third Review Under the Extended Fund Facility for Seychelles (Dec 2022).

⁶⁵ World Health Statistics Report 2023 (WHO, 2023)

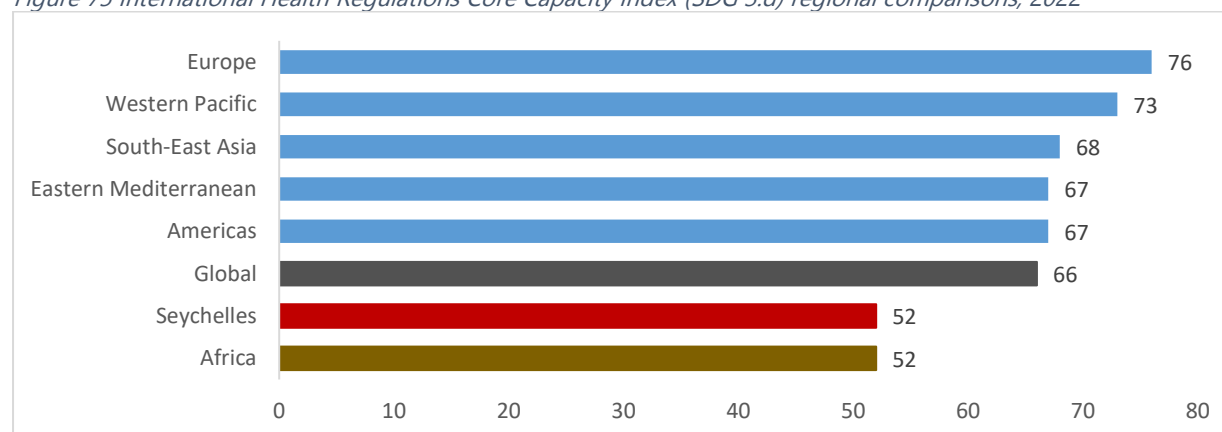
Table 37 Universal Health Coverage Index 2019-2022

UHC Components		2019	2020	2021	2022
Average Service Coverage Index (SDG 3.8.1)					
Maternal and Child Health		0.80	0.82	0.81	0.80
Family Planning Coverage	WHO Estimates (2019)	0.41	0.47	0.43	-
Antenatal Coverage (4+ visits)	Maternal Health Programme Data, 2018-2022	0.99	0.99	1.00	0.99
Immunization Coverage (DPT3) (SDG 3.b.1)	2019-2022, EPI Data	0.99	0.97	0.97	0.97
Care-Seeking for Pneumonia in <5y olds	Assumed to be 100%	1.00	-	-	1.00
Infectious Diseases		0.82	0.85	0.88	0.91
ART Coverage (of known PLHIV)	GAM Report 2022	0.81	0.82	0.87	0.90
Tuberculosis Treatment Coverage (% of known cases)	2019-2022, CDCU Data	0.70	0.78	0.82	0.87
WASH (Water Access, Sanitation and Hygiene) indicators (SDG 6.1/6.2)	Population & Housing Census 2022 (NBS, 2023)	0.97	-	-	0.97
Malaria Prevention.	Excluded	-	-	-	-
NCDs		0.73	0.73	0.73	0.73
Prevalence of normal Blood Pressure.	2013, Seychelles Heart Study IV	0.77	-	-	0.77
Prevalence of normal Fasting Plasma Glucose	-	0.64	-	-	0.64
Prevalence Tobacco Non-Smoking (SDG 3.a.1)	-	0.79	-	-	0.79
Service Capacity		0.96	0.96	0.96	0.96
Hospital Access (WHO target > 18 acute beds/10,000 pop)	2019-2022, MOH	1.00	-	-	1.00
Health Workers Density (WHO target of > 150 Doctors & Nurses/10,000 pop)	Offices of the CMO and CNO	1.00	-	-	1.00
IHR Core Capacity Index (SDG 3.d.1)	eSPAR State Parties Annual Report 2022 (WHO, 2023)	0.88	-	-	0.52
ASC Index (SDG 3.8.1)		0.82	0.82	0.84	0.85
Financial Hardship Index (SDG 3.8.2)					
Risk for Catastrophic Health Expenditure					
Proportion of population with spending on health compared to total household budget	WHO Comparable Estimates 2021 (World Health Statistics Report 2023)	-	-	-	-
At > 10% threshold	-	-	-	2.6%	-
At > 25% threshold	-	-	-	1.3%	-

6.1.2 International Health Regulations Core Capacities

Reported International Health Regulations (IHR) Core Capacity **Index for Seychelles was 52** for 2022 (eSPAR), a small increase from 48 in 2021 (NHSP objective 3.2). This is comparable to average for WHO-AFRO, and well-below the Global average and all other WHO regions (Figure 75).

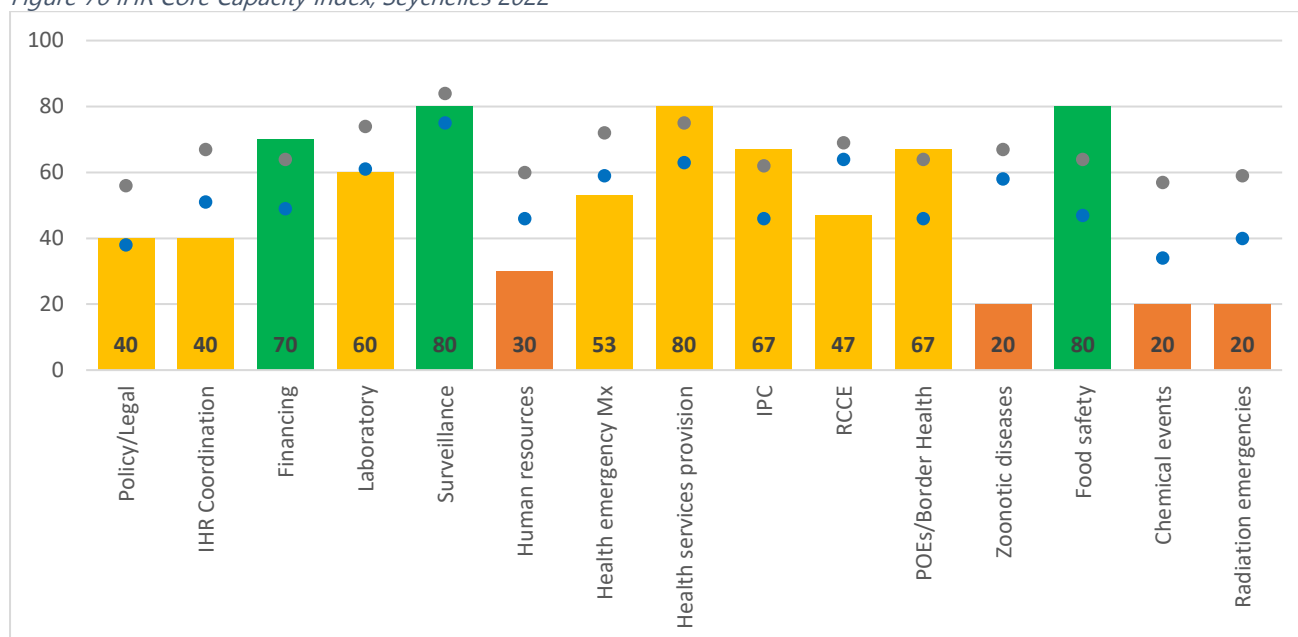
Figure 75 International Health Regulations Core Capacity Index (SDG 3.d) regional comparisons, 2022



Data Source(s): World Health Statistics Report, 2023: Monitoring Health for the SDGs (WHO, 2023).

The individual index components for 2022 are illustrated in Figure 76. The principal gains were in health service provision (+20%), ports of entry (+14%) and human resources (+10%); main loss of scores were under policy and legal frameworks (-10%) to implement IHR, compared to 2021.

Figure 76 IHR Core Capacity Index, Seychelles 2022



Note: (●) Africa regional average; (●) World average; Data Source(s): WHO (2022): eSPAR State Parties Annual Report, link: <https://extranet.who.int/e-spar/#submission-details>

6.1.3 Programmes Across the Life Course

Family health programmes focus on a series of health programmes across the life-course, with promotive and preventive services as the core mandate. The directorate saw major changes in leadership in 2022, including a new director and new programme managers leading the child health, school health and EPI programmes (NHSP objective 2.1.4).

Routine vaccination and child developmental screening were partially interrupted in 2021, with only partial catch-up in 2022; however, the country maintained high levels of early childhood vaccines coverage. Other health programmatic services such as family planning were also disrupted in 2020 and 2021.

A key challenge across all programmes remains the collection and analysis of relevant data, as well as its use in decision-making. In 2022, work was initiated to revisit how programme performance data is being collected across all community health centres.

Certain key guidance development and training were undertaken in 2022, including development of family planning manual, internal training for family planning and child health nurses. **Four programmes defined in the PHC package were still pending implementation (men's health, home-based care, healthy ageing, and palliative care).**

Sexual Reproductive Health

Family planning attendance and pap-smear uptake are shown in Table 38. An improvement in pap smear uptake is a notable positive trend. Family planning services offered in the private health sector are not captured in this dataset.

The NPHC 2022 (NBS, 2023, unpublished) findings showed that **overall level of contraceptive use amongst women aged 18-49y was 36%** in 2022; with prevalence of use being highest (about 45%) in the age range 20-34y. Note that this does not necessarily reflect SDG 3.8.1 index parameter (demand for family planning met

using modern contraceptive methods), since the country has not conducted relevant surveys to collect this information (in particular, estimate of population in need).

Table 38 Family Planning Attendance, 2019-2022

Indicators	2019	2020	2021	2022	% Change	NHSP Target
Family Planning Attendance	19,280	19,809	17,508	21,195	+21%	-
Family Planning – New acceptors	-	-	-	399	-	-
Surgical sterilisation performed	-	-	-	184	-	-
# Pap Smears done	6,068	4,796	3,257	5,925	+81%	>20,143

Data Source(s): Family Health Programmes Annual Report 2022 (HCA, 2023); APR 2021 for historic data.

The WHO's global strategy for elimination of cervical cancer as a public health threat⁶⁶ aims to **reduce cervical cancer incidence to <4 cases per 100,000 woman-years**, through meeting the **90-70-90 targets** by 2030:

- 90% of girls successfully vaccinated with HPV vaccine by age 15y.
- 70% of women are screened with a high-performance test by 35y of age, and again by 45y of age.
- 90% of women identified with cervical cancer disease receive treatment (90% of women with pre-cancer treated, and 90% of women with invasive cancer managed).

Maternal Health

Pregnant women in Seychelles were **seen an average of 11 times during pregnancy** in 2022, with **ANC coverage consistently above 99%** (Table 39).

Table 39 Antenatal care attendance and Post-natal home-visit Coverage 2019-2022

Indicators	2019	2020	2021	2022
Maternal-Child Health clinic attendance	27,973	26,896	17,077	23,539
ANC Clinic – booking visits	-	-	-	1,492
ANC Clinic – attendance	12,221	13,096	12,354	15,255
Post-natal visits coverage (%)*	53%	19%	24%	53%

* Refers to coverage of post-natal home-visit at 6 weeks post-partum; other post-natal care coverage was 97% in 2022.

Data Source(s): Family Health Programmes Annual Report 2022 (HCA, 2023); APR 2021 for historic data.

Certain key patterns of **co-morbidities amongst pregnant mothers** are summarised in Table 40; the patterns show a significant level of risk factors and illness even amongst women in the reproductive age group.

Table 40 Maternal risk factors during pregnancy and post-partum KPIs, 2022

Characteristics	#	%
Amongst 1,492 women booking for ANC care in 2022		
Pregnant women who reported using illicit drugs	46	3.1%
Pregnant women who reported smoking	54	3.6%
Pregnant women who reported using alcohol	18	1.2%
Anaemic at booking visit (<11g/dL) – SDG 2.2	336	22.5%
Amongst 1,512 deliveries in 2022		
Cases of gestational diabetes identified	53	3.5%
Cases of gestational hypertension identified	49	3.2%
Post-partum care indicators (of 1,522 live births)		
Fraction of new-borns breastfed within one-hour of birth	-	88.3%
Fraction of new-borns having skin-to-skin contact with mother within one-hour of birth	-	90.5%
Fraction of mothers breastfeeding at time of discharge from ward	-	96.9%
Post-natal home visits conducted	805	52.9%

Data Source(s): Family Health Programmes Annual Report 2022 (HCA, 2023).

⁶⁶ Global strategy to accelerate the elimination of cervical cancer as a public health problem. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO.

Expanded Programme on Immunisation

The Expanded Programme on Immunisation (EPI) is well established in Seychelles, and achieves consistently high coverage for critical childhood vaccines for many decades, including during the COVID-19 pandemic, demonstrating programmatic resilience. The programme works closely with the child health and school health programmes, and conduct shared delivery of relevant activities, through shared staff and resources.

Since 2019, a **National Immunisation Technical Advisory Group** (NITAG) was established, providing support to the MOH with strategic planning and decision-making around immunisation. Additionally, a **National Immunisation Technical Working Group** (NTWG) was established at the end of 2020, to coordinate the COVID-19 immunisation campaign. Both the NTWG and NITAG remained active throughout 2022.

The average coverage for all antigens in **routine childhood immunisation was 98%** in 2022 (same as 2021). A catch-up vaccination programme for students who missed doses of HPV vaccine in 2021 was conducted in 2022. Table 41 describes coverage of all key antigens in the childhood immunisation schedule.

For **COVID-19 vaccines** administered in 2022, see COVID-19 Vaccination; for **Rotavirus and Polio surveillance**, see 4.2.4 Surveillance of Vaccine-Preventable Diseases.

Table 41 Childhood vaccination: Coverage by Antigen, 2022

Age	Vaccine/Antigen	Target Pop.	# Doses Given	Coverage (%)	NHSP Target (%)
Birth	Bacille Calmette Guérin (BCG)	1,569	1,569	100%	
3 mo	Pentavalent Vaccine (Penta1) ⁶⁷ Oral Polio Vaccine Dose 1 (OPV1) Rotavirus Vaccine Dose 1 (Rota1) Conjugate Pneumococcal Vaccine Dose 1 (PCV1)	1,569	1,557	99.2%	
4 mo	Injectable Polio Vaccine Dose 1 (IPV1)	1,569	1,550	98.7%	
	Rota2	1,569	1,548	98.6%	
5 mo	Penta3 OPV2	1,569	1,520	96.8%	>95%
	PCV3	1,569	1,521	96.9% ⁶⁸	>95%
12 mo	Yellow Fever	1,665	1,638	98.3%	
15 mo	Measles, Mumps, Rubella Vaccine Dose 1 (MMR1)	1,665	1,422	90.2%	
18 mo	Diphtheria Pertussis Tetanus (DPT) Booster OPV4	1,665	1,422	90.2%	
6y	Diphtheria, Tetanus (DT) OPV5 MMR2	1,633	1,595	97.6%	
11y	Human Papilloma Virus Vaccine (HPVV) Dose 1 ⁶⁹	834	684	82%	>95%
15y	OPV6 Tetanus Toxoid (TT)	1,302	1,180	90.6%	
Pregnant >25y old	TT Booster	322	322	100%	

Data Source(s): EPI Programme (HCA, 2023).

⁶⁷ Vaccine injection containing five antigens – Diphtheria Pertussis and Tetanus Dose 1 (DPT1), Hepatitis B Dose 1 (HepB1), and Haemophilus influenzae B Dose 1 (HiB1) antigens.

⁶⁸ Some children vaccinated with initial doses overseas accessed their third dose locally.

⁶⁹ Administered only to girls in 2022.

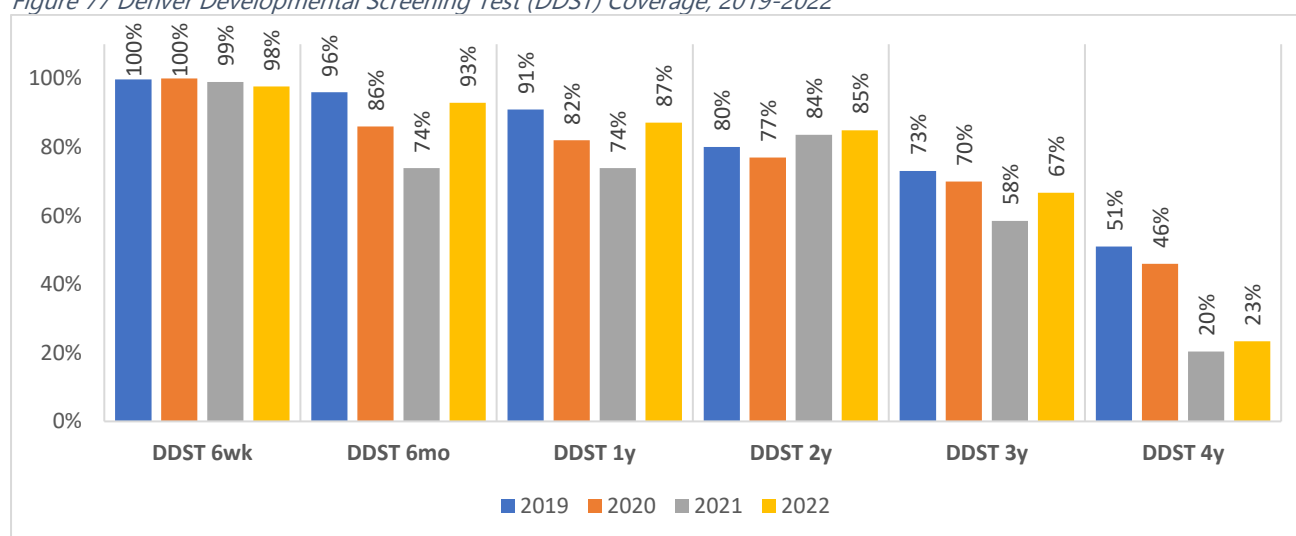
Child Health

Child Health programme targets children from birth to the start of school (approximately age 4). The key roles are for monitoring growth and development of children in this age group and make appropriate referrals to other health and social services as indicated.

A total of 23,578 attendances for child health services were recorded for 2022. The **Denver Developmental Screening Test (DDST) coverage rates are high at early ages, but reduces with age**. The reduction has been more drastic since the onset of the COVID-19 pandemic, in particular 2021, with partial recovery in 2022.

The **fraction of children failing DDST appears to increase with age**, from 0% at 6 weeks, to 0.4% at 6 months and 1 year, to 5.1% at 2 years, 7.2% at 3 years, and 4.5% at 4y (not shown in Figure 77)⁷⁰. Amongst mothers who attended the 6 weeks visit (N = 1,791), 488 breastfed exclusively (27%), 422 practiced mixed feeding (24%), and the breast-feeding status was not recorded for the remainder (49%).

Figure 77 Denver Developmental Screening Test (DDST) Coverage, 2019-2022



Data Source(s): Family Health Directorate Annual Report, 2022 (HCA, 2023).

In 2022, the child health programme referred 121 children for **developmental delays** to rehabilitative services; and 142 children for other abnormalities, with top three reasons being for **speech disorders, behavioural issues and nutritional issues**.

School Health

Conduct of school health screening was disrupted in Q1-Q2 of 2022, but caught up in Q3-Q4. Coverage was highest for screening in primary 4, lower in secondary 1 and 4, and lowest for Crèche. Screening of Secondary 5 students was not done in 2022 (Table 42). The screening focused on risk factors in schoolchildren including weight, physical activity, alcohol, and tobacco and substance use – see section 3.4 Risk Factors for Health.

The Early Childhood Care and Education (ECCE) technical team, in collaboration with Child Health programme, was conducting a re-evaluation implementation of the **Child Health Profile**, and will produce recommendations for strengthening the programme.

Staff re-deployment during the COVID-19 pandemic presented a major challenge to the full implementation of school health programmatic activities. With the increasing focus on tackling key risks at an early age,

⁷⁰ Family Health Programmes Annual Report 2022 (HCA, 2023).

schools, with a captured population, should serve as the critical platform for delivery of relevant preventive, promotive and screening services.

Table 42 School health screening coverage by level, 2022

Level	# Eligible	# Screened	Coverage	Comments
Crèche	1,107	222	20%	Data used for annual school health screening report, in partnership with UPCCD. Refer to section 3.4 Risk Factors for Health.
P4 (9y)	1,556	1,267	81%	
S1 (12y)	1,534	1,018	66%	
S4 (15y)	1,482	995	67%	
Note: P4 – Primary 4; S1 – Secondary 1; S4 – Secondary 4 (levels of schooling)				

Data Source(s): Family Health Directorate Annual Report, 2022 (HCA, 2023).

The top three health issues identified in 2022 school health screening were **dental caries, behavioural issues and speech impairment**. In addition to the schoolchildren, the programme also screened 341 school staff.

Adolescent and Youth Health

Responsibility to develop adolescent and youth health programme, as defined in the Primary Health Care Package (MOH, 2020), were added to school health programme manager portfolio as from September 2022. A draft plan is being developed in partnership with the Nurse Manager for Youth Health Centre, to expand youth health services (NHSP objective 2.1.1).

6.1.4 Disease-Specific Programmes

Cancer, HIV and CVDs prevention programmes are administered under the PHA. Of note, the post for programme manager for HIV/AIDS was vacant at the close of 2022. This section only outlines key programmatic achievements or new developments – please refer to relevant sections for disease incidence and mortality.

Cardiovascular Diseases

Key projects undertaken by the UPCCD in 2022 included (NHSP objective 3.6):

- Preparations for the next iteration of the Seychelles Heart Study series (due to be conducted in 2023).
- Conduct of school health survey for 2022 (see 3.4 Risk Factors for Health).
- Revision of the Tobacco Control Act, 2009 (see 2.4 Legislation and Regulation).
- Conduct of adherence surveys of hotels to anti-smoking regulations.
- NCD Screening services at workplaces.
- Conduct of Workplace Wellbeing Programme training workshops (see Health Promotion).

Diabetes

Staffing of diabetes programme was reduced in 2022. Activities in 2022 principally focused health promotion and education, review of implementation of the **Seychelles Package of Essential NCD interventions (SEY-PEN)** in 2019-2020, and resumption of the SEY-PEN initiative in Q3-Q4 of 2022. Successful implementation of SEY-PEN will contribute towards NHSP objective 2.1.1 – implementation of new service delivery models for chronic diseases, and help decrease disease burden associated with diabetes mellitus and hypertension.

SEY-PEN tools and communication materials were revised based on experience and feedback, and screening activities resumed as from September 2022. Repeat training was also provided for all community health staff. Reporting burden remains a key barrier to high screening coverage.

The diabetes programme re-established NCDs registers in community health facilities in 2022, and received reports of 267 new diabetes cases diagnosed; the total patients on the Diabetes register was 4,284 at end 2022⁷¹. Although these are likely underestimates, they indicate a key step towards monitoring diabetes care.

Cancer

The Cancer-Control Programme launched a **pilot screening campaign for colorectal cancer using Faecal Occult Blood Tests (FOBTs)** on La Digue, targeting population aged 50-64y. Of the estimated 350 people in this age group on La Digue, 184 (53% turnout) attended for screening over a ten-day period. Screening continues to be offered on La Digue. Of these, 30 FOBTs returned positive (16.8% or one in six), and 28 were referred for colonoscopy and further management as indicated; one case of colorectal cancer (Stage I) was detected, and a high number of false positive tests were noted.

In 2022, 407 women were referred for screening through the programme (accounting of 35.8% of all Mammograms conducted in 2022, see Imaging)⁷². Screening levels for all key cancers remain far below the scale required to realise any population level health impacts.

The **cancer registry** falls under the purview of the cancer programme. The registry is upgrading its software environment from CanReg4 to CanReg5, with the help of Africa Cancer Registry Network (AFCRN)⁷³. Some challenges were noted during adoption of new software package (NHSP objective 5.5.2).

Registry staff conducted visits to all clinics and units in 2022, to strengthen data collection and transmission, including additional training for nurses maintaining cancer registers in health facilities (NHSP objective 6.4). These activities constitute a quality-improvement initiative at unit/service level, and will contribute to strengthening cancer information systems in Seychelles.

Mental Health

Relevant activities were principally conducted by the mental health services team in 2022. It included screening programmes for mental health wellbeing and conditions, testing for autism-spectrum disorders and integration of mental health screening and care in maternal health services⁷⁴.

HIV-AIDS

The post of programme manager for HIV-AIDS programme remained vacant in 2022 – this constitutes a serious barrier to achieving stronger public health programmes (NHSP objectives 3.6 and 2.1.5). Activities related to HIV-AIDS were undertaken by CDCU and NAC teams in 2022, with related NGOs.

The estimated population of People Living with HIV/AIDS (PLHIV) in Seychelles is not known; **988 individuals were known to be living with HIV/AIDS** (11 children under age 15, 630 males aged 15y+ and 347 females aged 15y+), of which 893 were on anti-retroviral treatment (ART). Of these, 112 started ARTs in 2022, and 709 were reported to be virally suppressed⁷⁵.

Based on above data, the fraction of PLHIV who know their status is unknown (the first 95). Of known PLHIV, 90% were on Anti-retroviral Treatment (ARTs), of these (the second 95), 79% were virally suppressed (the third 95). Thus **Seychelles has attained one of the 90-90-90 by 2020 UNAIDS goals** in 2022. However, this is below the average globally and in the African region⁷⁶. Seychelles has some way to go to reach UNAIDS global targets of 95-95-95 by 2030, for ending the HIV/AIDS pandemic (Figure 78).

⁷¹ Family Health Programmes Directorate Report 2022 (HCA, 2023)

⁷² Cancer-control programme (PHA 2023)

⁷³ Link: <https://afcrn.org/>.

⁷⁴ Mental Health Services Annual Report 2022 (HCA, 2023).

⁷⁵ Global AIDS Monitoring Report – Seychelles 2022 (NAC, April 2023).

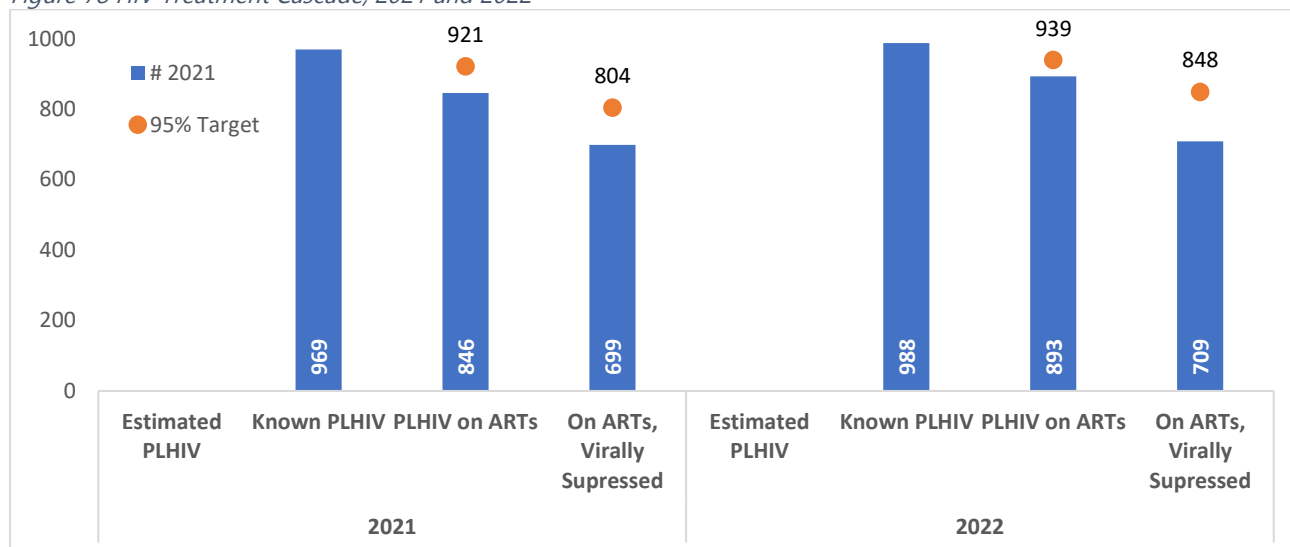
⁷⁶ World Health Statistics Report 2023 (WHO, 2023).

Just over one-in-four (93 of 347) women living with HIV were referred for cervical cancer screening in 2022.

All pregnant women are screened for HIV at booking ANC visit and again in early third trimester; 2,713 HIV tests were done through ANC clinics in 2022, and 16 pregnant women were positive. **Ten babies were born to HIV-positive mothers in 2022, of these, one case (female) of Mother-To-Child-Transmission of HIV/AIDS (MTCT) was recorded**, i.e. 10% MTCT rate.

Free condom distribution is principally in the public health sector; in 2022, 343,422 male condoms (309,131 in public health sector, 31,659 private health sector, 2,632 by NGOs), and 3,148 female condoms (all public health sector), were distributed by the MOH⁷⁷.

Figure 78 HIV Treatment Cascade, 2021 and 2022



Data Source(s): MOH contributions to Global AIDS Monitoring Report, 2021 and 2022 (NAC, 2023).

One in nine prison inmates (38 of 344, or 11.4%) were known PLHIV in 2022, with 19 on ARTs (50% of known PLHIV). Twelve prisoners were co-infected with HIV and HCV (3.6% of total), 18 with only HCV (5.4% of total). About one third (112 of 334) of the prison population were on opioid maintenance therapy⁷⁸.

The **low treatment coverage for HIV among prison inmates living with HIV, combined with the lack of access to the needle-syringe programme in prison setting in 2022, constitute a serious missed opportunity** to reduce HIV and Viral Hepatitis transmission, and provide effective care for these chronic infections in this key population (NHSP objective 2.1.5).

6.1.5 Other Programmatic Services

Nutrition

The nutrition team participates in a range of activities aimed at sensitising the population on healthy living (and eating) and also provide nutrition support and counselling for those in-need.

The Nutrition Unit reported 715 patient visits (159 new, 556 repeat) in 2022, with administration of medical nutrition therapy to 516 patients in-need. This is an increase from 2021 (569 attendance), but less than in 2020 (897). A total of 139 patients were given dietary advice through NCD clinics, and just over 400 expectant

⁷⁷ Global AIDS Monitoring Report – Seychelles 2022 (NAC, April 2023).

⁷⁸ Global AIDS Monitoring Report – Seychelles 2022 (NAC, April 2023).

mothers attended antenatal nutrition sessions (estimated coverage 26.5%). Additional nutrition advice is also provided during ANC attendance⁷⁹.

The unit conducted an assessment of hospital meals quality, with subsequent engagement with service provider with recommendations; and another assessment of nutrition standards at 18 day-care service providers, collaboration with IECD.

In collaboration with WHO, the team is facilitating adoption of WHO growth charts for monitoring early childhood growth and development through development of the tools and conduct of training.

Health Promotion

The health promotion team supported a range of programmatic areas with production of infographics, information products and audio-visual content. Active public engagement through social media and mass media continued throughout 2022.

The **MOH Facebook page had 165,632 followers** at the time of preparation of this report (June 2023), slightly less than in 2021 (168,488 followers). Post reach was 57,307 in Jun 2023 (56% M, 44% F), compared to an average of 50,000 in 2021.

The **Workplace Wellbeing Programme** is housed within the health promotion unit, but is cross-cutting capacity-building initiative for workplaces to improve services aimed at staff health and wellbeing. Health Promotion and Occupational Health lead the initiative jointly, with support from UPCCD, psychology services, nutrition unit and external partners (e.g. SCAA, National Sports Council).

After a hiatus from Q2 of 2020 to Q2 of 2022, five cohorts were trained in Q3-Q4 of 2022, spanning seven organisations, including the MOH (totalling 113 staff). The programme is yet to be fully institutionalised, and does not have dedicated human resources and finance.

6.1.6 Progress towards Promoting Healthy Populations

In addition to contributing towards SDG3, the various programmatic services contribute towards key priorities under the NHSP 2022-2026 SD4 Promote Health Populations:

- Promote healthy living for different age groups.
- Address key risk factors for health.
- Revitalize the health in all policies approach.
- Promote prevention and management of substance abuse and mental health.
- Transform Health Promotion.
- Promote/ advocate for Effective Public Health.

Multiple sectors contribute towards promotion of healthy populations. Inter-sectoral collaboration continued in 2022, both at programmatic level, and at policy level. Enforcement of standards for water, food and sanitation continued in 2022, with strong health promotion and community engagement through mass and social media. A **National Health Promotion policy** was approved in January 2023.

With a global drive to accelerate progress towards tackling the obesity pandemic, high-level momentum was seen in the MOH. Meetings of the policy network rallied support for action on obesity, with terms of reference defined for a technical working group to drive the development of a **National Action Plan to Stop Obesity**.

⁷⁹ Nutrition Unit Annual Report 2022 (HCA, 2023).

6.2 Progress towards SDG3

In general, there has been an **improvement in SDG indicators related to maternal-child health, and a deterioration in indicators reflecting mental health** (reflected by deaths related to homicides and suicides). Infectious-diseases-related indicators show a mixed picture. No new data is available for NCD-related indicators, although a new STEPS survey ⁸⁰ is planned for 2023, to provide data on NCD risks.

Programmatic service coverage has improved overall from 2021 to 2022, with most coverage indicators reaching pre-pandemic levels. Compared to global and NHSP targets:

- Maternal and Child health indicators show good performance, with almost universal skilled birth attendance, high vaccination coverage and child developmental assessment coverage. Coverage of post-natal visits remains low.
- Family planning coverage is estimated by WHO to be 43%; this is above global average (estimated at 20%)⁸¹, but below NHSP target of >60%.
- Screening for cancer (e.g. through pap-smears, mammograms and FOBTs) and other key NCDs (e.g. Diabetes and Hypertension) needs improvement, falling well-below NHSP targets.
- Targeted screening of key populations for infectious diseases such as HIV/Viral Hepatitis is low (e.g. amongst clients enrolled in services offered by DSAPTR, prison inmates, etc.).
- An acute increase in suicide-, homicide-, and RTA-related mortality occurred in 2022.
- The needle-syringe programme, despite being a proven effective harm-reduction method, is offered at a scale too low to have any impact on communicable disease transmission.
- Service coverage as reflected by facility-, bed- and workforce-densities are well-above WHO recommendations, indicating good access; but quality of services, and equity in access and outcomes across various population groups, are poorly measured.
- The increase in domiciliary care services provided in 2022 is a positive development.

The PHC package (see section 2.2.2 Primary Health Care Package) makes provision of bigger scope of services across the life course, but **new programmes require full institutionalisation**, including clearer definition of structure and guidelines, and subsequent implementation. The MOH is also working on improving quality of services and reducing low-value care.

All SDGs are inter-related and interconnected; to continue to improve SDGs, there is need for progress on other SDGs, most of which influence the social determinants of health.

Table 43 summarises trends in key SDG 3 indicators, and certain non-SDG3 indicators of relevance.

⁸⁰ WHO: STEPwise approach to NCD risk factor surveillance (STEPS); Link: <https://www.who.int/teams/noncommunicable-diseases/surveillance/systems-tools/steps>.

⁸¹ UN SDG3 Goal Profile (2021); Link: https://www.unescap.org/sites/default/d8files/event-documents/SDG%203%20Goal%20Profile_Final%20Web.pdf.

Table 43 Health-related SDG indicator trends, 2019-2022

SDG	Indicator & Data Source(s)	2019	2020	2021	2022	Trend
3.1.1	Maternal mortality ratio (per 100,000 live births, target <70) Source: Vital Statistics Report 2022 (MOH, 2023)	62.3	64.4	120.1	0.0	↑
3.1.2	Proportion of births attended by skilled health personnel Source: Vital Statistics Report 2022 (MOH, 2023)	98.8%	99.5%	99.0%	99.8%	→
3.2.1	Under-five mortality rate per 1,000 live births (target <25) Source: Vital Statistics Report 2022 (MOH, 2023)	17.4	14.2	13.2	18.5	→
3.2.2	Neonatal mortality rate per 1,000 live births (target <12) Source: Vital Statistics Report 2022 (MOH, 2023)	8.7	9.0	8.4	8.3	→
3.3.1	HIV incidence per 1,000 populations. Source: CDCU Annual Report 2022 (PHA, 2023)	1.12	0.9	0.59	0.78	↘
3.3.2	Tuberculosis incidence (per 100,000 population) Source: CDCU Annual Report 2022 (PHA, 2023)	13.3	9.1	11.1	14.3	↘
3.3.4	Hepatitis B incidence (per 1,000 population). Source: CDCU Annual Report 2022 (PHA, 2023)	0.45	0.35	0.22	0.13	↗
3.4.1	Probability of dying in the age 30-70 years from any of the 4 key NCDs per 100,000 populations. Source: World Health Statistics Report 2023 (WHO, 2023)	21.1%	-	-	-	■
3.4.2	Suicide mortality rate per 100,000 population Source: Vital Statistics Report 2022 (MOH, 2023)	6.1	3.0	8.1	13.1	↓
3.5	Total alcohol per capita (>15y age) consumption (litres of pure alcohol) Source: World Health Statistics Report 2023 (WHO, 2023)	12	-	-	-	■
3.6.1	Death rate due to RTAs (per 100,000 population) Source: Vital Statistics Report 2022 (MOH, 2023)	11.3	11.2	6.0	19.1	↓
3.7.2	Adolescent fertility rate (aged 15-19 years) per 1,000 women in age group Source: Population Bulletin 2022 (NBS, 2023)	89	69	63	46	→
3.8.1	Coverage of essential health services (UHC-ASC) Source: Internal Calculation (MOH, 2023)	70	80	84	85	↗
3.9.3	Mortality rate attributed to unintentional poisoning per 100,000 population Source: Statistics Unit, MOH.	0.0	0.0	0.0	0.0	→
3.b.1	Surviving infants who received 2 WHO-recommended vaccines (%) (Using DPT3 as proxy). Source: EPI Programme (HCA, 2023)	98.8%	97%	99%	98%	→
2.2.1	Prevalence of stunting in children <5y (%) Source: World Health Statistics Report 2023 (WHO, 2023)	-	-	-	7.2%	■
2.2.2	Prevalence of overweight in children <5y (%) Source: World Health Statistics Report 2023 (WHO, 2023)	-	-	-	5.2%	■
2.2.3	Prevalence of anaemia (%) in women aged 15-49y Source: World Health Statistics Report 2023 (WHO, 2023); Family Health Directorate Report, 2022 (HCA, 2023)	25.1%	-	-	22.5%	↑
6.1	Proportion of population using safely managed drinking water services (%) Source: National Population & Housing Census, 2022, NBS (2023)	97% (2010)	-	-	>95%	→
6.2	Proportion of population using safely managed sanitation services (%) Source: National Population & Housing Census, 2022, NBS (2023)	95.6% (2010)	-	-	99%	→
11.6	Annual mean concentrations of fine particulate matter (PM2.5) in urban areas (µg/m3). Source: World Health Statistics Report 2023 (WHO, 2023)	17.4	-	-	-	■
16.1	Mortality rate due to homicide (per 100,000 pop.) Source: Vital Statistics Report 2022 (MOH, 2023)	2.0	1.0	4.0	6.0	↓
Note	↓ Worsening → Unchanged ↑ Improving					↗ Improving slowly ↘ Worsening slowly ■ Trend not ascertained
						■ Green - Good ■ Amber - Neutral ■ Red - Bad

7 Conclusions

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The NHSP 2022-2026 was launched in April 2022, on occasion of World Health Day; thus **2022 was the first year of implementation of the new NHSP**. Overall, the guidance has been used extensively in development of operational plans across the MOH, with some milestones and results already in-reach.

There are **significant changes to the institutional structure of the MOH since 2020** (repeal of APDAR Act and discussion on repeal of the NAC Act) causing significant uncertainty amongst staff that may be impacted. A new MOH structure will evolve from this process.

The **MOH staffing counts and budget allocations have increased drastically in the past decade**; it remains to translate this into effective, efficient, equitable, cost-effective health services of high quality.

The trends and analyses of the mortality and available morbidity data have demonstrated the **continued increase in NCD burden** (especially cancer amongst men), with a **re-emergence of infectious and communicable diseases** (e.g. COVID-19 and Syphilis), and a sharp **increase in deaths from accidents and injuries** (RTAs, homicides and suicides).

Risk factors for health such as obesity (from school health screening), substance use (based on substance use service utilisation and psychiatric ward admission patterns) and unsafe and early sex (based on incidence of teenage pregnancy and communicable diseases) are **at alarming levels even amongst children**.

The **utilisation of health services (from service utilisation data) and coverage for key health programmatic interventions (from programmatic reports) have recovered** to almost pre-pandemic levels. However, equity in access and utilisation of services, health outcomes, and discrimination within the health service, across different population groups remain unmeasured.

Haemodialysis services, overseas-specialised care and oncology care stand out as key cost drivers relative to number of beneficiaries. The ideal mechanism to address this should be through upstream intervention to prevent cancer, NCDs and kidney diseases, and not through scaling down of services provided to patients in-need.

New COVID-19 cases reduced sharply from Q2 of 2022; other infectious and communicable disease threats continue to emerge (e.g. Mpox and Syphilis), emphasising the need to maintain strong prevention and preparedness services, and health system readiness to respond effectively.

The first half of 2022 required strong focus on the COVID-19 response, and MOH entities began implementation of the new NHSP only in the latter half of the year. Going forward, entities will work to address all key priorities outlined in the NHSP, with the ultimate aim of attaining the key sector goals.

It is essential to note that this health report has certain limitations. The data used for analysis may have certain gaps and inherent biases, and the findings might not be representative of the entire population.

Despite this, this annual health-sector performance report provides valuable insights into the current health status, key health issues, and potential areas for improvement within sector. These findings can serve as a foundation for evidence-based policymaking, healthcare planning, and targeted interventions aimed at improving the overall health and well-being of the population.

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